

Algebra 3

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1 Rings

§1.1 Rings

§1.1.i Definitions and Examples

Definition 1.1.1. A **monoid** $(M, \cdot)$ is a set M equipped with a binary operation $\cdot: M \times M \rightarrow M$ and an element $1_M \in M$ (called the multiplicative identity) such that:

- $\cdot$ is associative.
- $1_M \cdot m = m \cdot 1_M = m, \forall m \in M$.

Example 1.1.2

Any group is a monoid. (a group without the existence of an inverse is a monoid)

Example 1.1.3

The natural numbers $\mathbb{N} = \{1, 2, \dots\}$ forms a commutative monoid under multiplication, with multiplicative identity $1_{\mathbb{N}} = 1$.

Example 1.1.4

The nonnegative integers similarly forms a commutative monoid under multiplication.

Example 1.1.5

The set of $n \times n$ real matrices $M_n(\mathbb{R})$ forms a monoid under matrix multiplication, with multiplicative identity the unit matrix I . For $n > 1$, this is not a commutative monoid.

Definition 1.1.6. A **ring** is a set R together with functions $+: R \times R \rightarrow R$ and $\cdot: R \times R \rightarrow R$ and given elements $0_R, 1_R \in R$, such that:

- $(R, +)$ is an abelian group with identity 0_R .
- $(R, \cdot)$ is a monoid with multiplicative identity 1_R .
- The distributive property holds.

Remark 1.1.7 — A ring R is commutative if $(R, \cdot)$ is a commutative monoid.

Remark 1.1.8 — We refer to $+$ as **addition** and $\cdot$ as **multiplication**.

Remark 1.1.9 — Since $+$ is a function, it makes sense to write $+(x, y)$, but we write $x + y$ for sanity.

Definition 1.1.10. A subset $S \subseteq R$ is a **subring**, denoted $S \leq R$, if $0_R, 1_R \in S$ and for all $r, s \in S$, $-r, r + s, rs \in S$ too. This implies that S itself is a ring with the operations $+, \cdot$.

Example 1.1.11

The usual sets of numbers are all rings:

$$\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}.$$

However, $\mathbb{N} \subseteq \mathbb{Z}$ isn't a subring as $-1 \notin \mathbb{N}$.

Example 1.1.12

$\mathbb{Z}/m\mathbb{Z}$, for $m \in \mathbb{N}$, are rings.

Example 1.1.13

The set of **Gaussian integers** $\mathbb{Z}[i] = \{a + bi \in \mathbb{C} \mid a, b \in \mathbb{Z}\}$ is a subring of $\mathbb{C}$.

Example 1.1.14

We also have the subring $\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$ of $\mathbb{R}$.

Example 1.1.15

$M_n(\mathbb{R})$ is a ring.

Proposition 1.1.16

Let R be a ring and $r \in R$. Then, $r \cdot 0_R = 0_R \cdot r = 0$.

Proof. 0_R is the identity element in the group $(R, +)$, so $0_R + 0_R = 0_R$. Then,

$$r \cdot 0_R = r \cdot (0_R + 0_R) = r \cdot 0_R + r \cdot 0_R.$$

Then, adding $-r \cdot 0_R$ gives the desired result. $\square$

Example 1.1.17

The **trivial ring** $R = \{0\}$ has $0 \cdot 0 := 0$ and $0 + 0 = 0$. It is the unique ring with a single element.

Proposition 1.1.18

Let R be a ring. Then, $1_R = 0_R$ if and only if $R = \{0\}$ is the trivial ring.

Proof. If $R = \{0\}$, we have a singular element, so $1_R = 0_R$. On the other hand, suppose $1_R = 0_R$. Then, for $r \in R$, we have

$$r = r \cdot 1_r = r \cdot 0_R = 0.$$

□

Throughout the rest of the course, we will assume any rings encountered are nontrivial rings.

Definition 1.1.19. An element $u \in R$ is a **unit** if there is another element $v \in R$ such that $u \cdot v = v \cdot u = 1_R$. We write $R^\times \subseteq R$ for the set of units in R .

Definition 1.1.20. A **division ring** is a nontrivial ring where every nonzero element is a unit. That is, $R^\times = R \setminus \{0\}$.

Definition 1.1.21. A **field** is a commutative division ring.

Example 1.1.22

$\mathbb{Z}$ isn't a field, as 2 isn't invertible. In fact, $\mathbb{Z}^\times = \{\pm 1\}$.

Example 1.1.23

$\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are all fields.

Example 1.1.24

Similarly, $\mathbb{Z}[i]$ isn't a field, and $\mathbb{Z}[i]^\times = \{\pm 1, \pm i\}$.

Example 1.1.25

$\mathbb{Q}[\sqrt{2}]$ is a field.

Example 1.1.26 (Non-Commutative Division Ring)

The **quaternions** $\mathbb{H}$ is the abelian group $\mathbb{R}^4$, with standard basis vectors labeled $\{1, i, j, k\}$. Then, $\mathbb{H}$ is a ring with 1 the unit, and multiplication determined by $ij = -ji = k$, $i^2 = j^2 = -1$, and $(r \cdot 1) \cdot s = rs$ for any $r \in \mathbb{R}$ and $s \in \mathbb{H}$. This is not a commutative ring, since $ij \neq ji$. It is a division ring since

$$(a1 + bi + ci + dk)(a - bi - cj - dk) = a^2 + b^2 + c^2 + d^2,$$

so if $a1 + bi + ci + dk \neq 0$, it has an inverse:

$$\frac{1}{a^2 + b^2 + c^2 + d^2} \cdot (a - bi - cj - dk) \in \mathbb{H}.$$

Remark 1.1.27 — Note that the definition above for quaternions determines all other products. For instance,

$$k^2 = (ij)(-ji) = -ijji = i^2 = -1.$$

We can similarly attain that $jk = i$.

Example 1.1.28 (Non-Commutative Domain that is not a Division Ring)

The **Weyl algebra**, the ring of differential operators of polynomials. Given a polynomial ring $k[x]$, we can define this as $k[x, \partial]$. It satisfies $\partial x - x\partial = 1$.

Proposition 1.1.29 (Uniqueness of Inverses)

If $r \in R$ has inverses u and v , then $u = v$.

Proof. We have that $1 = ru = rv$, so $r(u - v) = 0$. $ur = 1$, so $0 = ur(u - v) = u - v$, so $u = v$. $\square$

Proposition 1.1.30

For any ring R , $R^\times$ is a group under multiplication.

Proof. First, $R^\times$ is closed under multiplication. If $a, b \in R^\times$, there are $c, d \in R^\times$ such that $ac = ca = bd = db = 1$. Then, $(ab)(cd) = a(bd)c = ac = 1$. Similarly, $(dc)(ab) = 1$, so $ab \in R^\times$.

Next, $1 \in R^\times$ is a unit, and inverses exist. As in the above, the inverse of a is c , which is also in $R^\times$. $\square$

§1.1.ii Construction of Rings

Example 1.1.31

Let R, S be rings. Then, their **product** $R \times S$ is a ring with addition and multiplication, defined as follows.

- $(r, s) + (r', s') = (r + r', s + s')$
- $(r, s) \cdot (r', s') = (r \cdot r', s \cdot s')$

The zero is $(0, 0)$ and the multiplicative unit is $(1, 1)$. Note that this is commutative if and only if R and S are.

Definition 1.1.32. Let R be a ring. Then, a **polynomial** f with coefficients in R is a sequence $f = (a_0, a_1, \dots)$ in R which is eventually 0. That is, $a_i = 0$ for i sufficiently large.

If $a_i = 0$ for $i > N$, we write the polynomial as

$$f = a_0 + a_1X + \dots + a_Nx^N.$$

Note that this is equivalent to $f = a_0 + a_1X + \cdots + a_Nx^N + 0x^{N+1}$. Note that if there's some i such that $a_i \neq 0$, we define the degree of f , $\deg(f)$, to be the maximal i such that $a_i \neq 0$ (this exists by assumption). If $a_i = 0$ for all i , then $f = 0$, and we define $\deg(0)$ to be $-\infty$.

Definition 1.1.33. Let R be a ring. Then, its **polynomial ring**, $R[X]$, is defined to be the set of polynomials with coefficients in R , with operations defined as below.

Given two polynomials $f = a_0 + a_1X + \cdots + a_nX^n$ and $g = b_0 + b_1X + \cdots + b_NX^m$, we define addition and multiplication as follows:

$$f + g = \sum_{i=0}^n (a_i + b_i)X^i$$

$$f \cdot g = \sum_{i=0}^{n+m} \left(\sum_{j=0}^i a_j b_{i-j} \right) X^i$$

Note that in the above, we can assume $n = m$ by adding $0X^i$'s if necessary. The subset of polynomials of constant polynomials, that is those of the form $f = a_0$ with $a_i = 0$ for $i > 0$, is a subring, which we identify with R .

Remark 1.1.34 — We can check that $R[X]$ is a ring; further, it is commutative if and only if R is.

Definition 1.1.35. If $f = a_0 + a_1X + \cdots + a_nX^n$ is a polynomial of degree n , we say f is **monic** if $a_n = 1$.

Example 1.1.36

$1 + X^2$ is monic, whereas 2 isn't.

Remark 1.1.37 — A polynomial $f = a_0 + a_1X + \cdots \in R[X]$ determines a function $R \rightarrow R$, sending r to $f(r) := a_0 + a_1r + \cdots$. The polynomial f itself is just a sequence of elements of R , and X some formal symbol: we don't identify the polynomial with the function, and in fact different polynomials can have the same function.

Example 1.1.38

An example to highlight the above is when $R = \mathbb{Z}/2\mathbb{Z}$, $f = X \in R[X]$, $g = X^2 \in R[X]$. Then, f, g are distinct polynomials, but $f(r) = g(r)$, for all $r \in R$.

Remark 1.1.39 — We can take the polynomial ring of a polynomial ring; that is, $(R[X])[Y]$, which we'll write as $R[X, Y]$. We can iterate this however many times:

$$R[X_1, \dots, X_n] := (\dots (R[X_1]) \dots) [X_n]$$

Definition 1.1.40. A **Laurent polynomial** f in R consists of a sequence $f = (\dots, a_{-1}, a_0, a_1, \dots)$, with only finitely many a_i nonzero. We write this in the form

$$f = \sum_{i \in \mathbb{Z}} a_i X^i.$$

We define $R[X, X^{-1}]$ to be set of Laurent polynomials on R . This has a ring structure defined the same as in $R[X]$.

Definition 1.1.41. A **power series** f in R consists of a sequence $f = (a_0, a_1, \dots)$ in R , where we don't assume only finitely many a_i are non-zero. We define $R[[X]]$ to be the **ring of power series** in R .

Definition 1.1.42. Let M be a monoid, and R a ring. The **monoid ring** of M over R , $R[M]$, is the set of tuples $f = \{a_m\}_{m \in M}$, where each $a_m \in R$, and only finitely many are non-zero. We write such a tuple f in the form

$$f = \sum_{m \in M} a_m m.$$

We define addition and multiplication similarly to before (and is listed below).

Given $f = \sum_{m \in M} a_m m$ and $g = \sum_{m \in M} b_m m$ be in $R[M]$, we define addition and multiplication as follows:

$$f + g = \sum_{m \in M} (a_m + b_m) m$$

$$f \cdot g = \sum_{m \in M} \left(\sum_{\substack{k, l \in M \\ k \cdot l = m}} a_k b_l \right) m.$$

We can check that $R[M]$ is commutative if and only if both R and M are. If M is a group (i.e. it contains inverses), we call this the **group ring** of M over R .

Example 1.1.43

Let M be the monoid of nonnegative integers, with addition as the binary operation. Then, we can identify $R[X]$ with $R[M]$, by identifying aX^n with $a \cdot n$, where $a \in R$ and $n \in M$. If we take $M = \mathbb{Z}$ with addition, we can identify $R[X, X^{-1}]$ with $R[M]$ in a similar way.

Example 1.1.44

If R is a ring and $n \geq 1$, the set of $M_n(R)$ forms a ring under the usual rules for matrix addition and multiplication. $M_n(R)$ is never commutative for $n > 1$, and $M_1(R)$ is commutative if and only if R is.

§1.1.iii Homomorphisms and Ideals

Definition 1.1.45. Let R, S be rings. A function $\phi: R \rightarrow S$ is a **ring homomorphism** if for all $a, b \in R$, the following hold:

- $\phi(a + b) = \phi(a) + \phi(b)$
- $\phi(ab) = \phi(a)\phi(b)$
- $\phi(0) = 0$
- $\phi(1) = 1$

Remark 1.1.46 — A bijective homomorphism is an isomorphism, and the inverse of an isomorphism is also an isomorphism.

Definition 1.1.47. Let $\phi: R \rightarrow S$ be a homomorphism.

- Its **kernel** is $\ker(\phi) := \{r \in R \mid \phi(r) = 0\} \subseteq R$
- Its **image** is $\text{Im}(\phi) := \{s \in S \mid s = \phi(r), \text{ for some } r \in R\} \subseteq S$.

Proposition 1.1.48

Let $\phi: R \rightarrow S$ be a homomorphism. Then, $\text{Im}(\phi) \subseteq S$ is a subring.

Proof. We just need to check the conditions. First, $0 = \phi(0)$, $1 = \phi(1)$, so $0, 1 \in \text{Im}(\phi)$. Next, suppose $s, s' \in \text{Im}(\phi)$ so that $s = \phi(r)$ and $s' = \phi(r')$, for some $r, r' \in R$. Then, $s + s' = \phi(r + r') \in \text{Im}(\phi)$. Similarly, $0 = s - s$, but also $0 = \phi(0) = \phi(r - r) = s + \phi(-r)$, implying $-s = \phi(-r) \in \text{Im}(\phi)$. $\square$

Remark 1.1.49 — Note that $\ker(\phi)$ is not a subring, since $1 \notin \ker(\phi)$.

Proposition 1.1.50

A homomorphism $\phi: R \rightarrow S$ is injective if and only if $\ker(\phi) = \{0\}$.

Proof. If ϕ is injective, there's no other $r \in R$ such that $\phi(r) = 0$, so $\ker(\phi) = \{0\}$.

On the contrary, suppose that for $r, r' \in R$, we have $\phi(r) = \phi(r')$. Then, $\phi(r - r') = 0$ so $r - r' \in \ker(\phi) = \{0\}$, implying $r = r'$. $\square$

Definition 1.1.51. Let $I \subseteq R$.

- I is a **left ideal** if it's an additive subgroup of R , and if $i \in I$ and $r \in R$, then $ri \in I$.
- I is a **right ideal** if it's an additive subgroup of R , and if $i \in I$ and $r \in R$, then $ir \in I$.
- I is a **two-sided ideal** if it's an additive subgroup of R , and if $i \in I$ and $r \in R$, then $ri \in I$ and $ir \in I$.

Remark 1.1.52 — For convention, we normally just say an ideal to mean left ideal. If R is commutative, then all definitions coincide.

Example 1.1.53

For any ring R , $\{0\}, R \subseteq R$ are both ideals.

Example 1.1.54

Suppose that $I \subseteq R$ is an ideal, and that $1 \in R$. Then, for any $r \in R$, $r \cdot 1 = r$ also lies in I . Therefore, $I = R$.

Particularly, the only subring which is also an ideal is the whole of R .

Definition 1.1.55. An ideal $I \subseteq R$ such that $I \neq R$ is called a **proper ideal**.

Remark 1.1.56 — By the above, if $1 \in I$, then I is not a proper ideal.

Example 1.1.57

We can generalise this: suppose $I \subseteq R$ is an ideal containing some unit u with inverse v , then I contains $vu = 1$ and hence $I = R$.

Lemma 1.1.58

Let $\phi: R \rightarrow S$ be a homomorphism. Then, $\ker(\phi) \subseteq R$ is a two-sided ideal.

Proof. Since ϕ is a homomorphism of abelian groups, $\ker(\phi) \subseteq R$ is a subgroup of R . Now suppose $i \in \ker(\phi)$ and $r \in R$. Then,

$$\phi(ri) = \phi(r)\phi(i) = \phi(r) \cdot 0 = 0.$$

Thus, $ri \in \ker(\phi)$, and, analogously, we get $ir \in \ker(\phi)$. □

Example 1.1.59

Let $R = \mathbb{Z}$ and $n \in \mathbb{Z}$. Then, $n\mathbb{Z} \subseteq \mathbb{Z}$ is an ideal.

To prove this, we claim all ideals are of this form. Let $I \subseteq \mathbb{Z}$ be an ideal. If $I = \{0\}$, then $I = 0\mathbb{Z}$, so suppose it's not zero. Let $n \in I$ be its smallest positive element. Then, $rn \in I$ for all $r \in \mathbb{Z}$, so $n\mathbb{Z} \subseteq I$. Now, let $i \in I$. The Euclidean algorithm then says

$$i = an + b,$$

with $0 \leq b < n$, and since $n \in I$ also $an \in I$, so $b = i - an \in I$. Since n was the smallest positive element of I , we must have $b = 0$, and so $i \in n\mathbb{Z}$. So $I = n\mathbb{Z}$.

Remark 1.1.60 — Note that we can do this in any ring that has a *Euclidean algorithm*.

Similar to group theory, we have the following lemma:

Lemma 1.1.61

Let $\phi: R \rightarrow S$ and $\psi: S \rightarrow T$ be homomorphisms of rings. Then their composition $\psi \circ \phi$ is also a homomorphism.

Definition 1.1.62. For an element $a \in R$, the **ideal generated by a** is

$$(a) := R \cdot a = \{ra \mid r \in R\} \subseteq R.$$

An ideal $I \subseteq R$ is **principal** if $I = (a)$ for some a .

Definition 1.1.63. If $S \subseteq R$ is any subset, the ideal generated by S is

$$(S) := R \cdot S = \left\{ \sum_{s \in S} r_s \cdot s \mid r_s \in R, \text{ and finitely many } r_s \text{ are non-zero} \right\}.$$

Example 1.1.64

In $\mathbb{Z}$, $n\mathbb{Z} = (n)$, so we have shown that all ideals in R are principal.

Example 1.1.65

The following subset

$$\{f \in R[X] \mid \text{the constant coefficient of } f \text{ is zero}\}$$

is an ideal. Particularly, it's a principal ideal generated by the polynomial X .

Definition 1.1.66. Let $I \subseteq R$ be a two-sided ideal. The **quotient ring R/I** consists of additive cosets of the form $r + I$, with zero given by $0 + I$ and one given by $1 + I$, and sum and product given by the following:

$$\begin{aligned} (r + I) + (s + I) &= (r + s) + I \\ (r + I) \cdot (s + I) &= (rs) + I \end{aligned}$$

We can then define an equivalence relation as follows:

$$r_1 \equiv r_2 \iff \exists a \in I \text{ such that } r_1 + a = r_2 \iff r_1 - r_2 \in I.$$

The equivalence classes are

$$[r] = r + I = \{r + a \mid a \in I\}.$$

Proposition 1.1.67

Let $I \subseteq R$ be a two-sided ideal. The quotient ring R/I is a ring, and the function $R \rightarrow R/I$ sending r to $r + I$ is a (surjective) ring homomorphism.

Proof. We know that addition is well-defined, since $(R/I, +)$ is the quotient of $(R, +)$ by a normal subgroup, so we need to check multiplication is well-defined. Now let $r, s \in R$ and $i \in I$. Thus, $r + I = r + i + I$. Then,

$$(r + i + I) \cdot (s + I) = rs + is + I = rs + I,$$

since $is \in I$. Similarly $(r + I)(s + i + I) = (rs + I)$. So multiplication is well-defined. Finally, associativity and distributivity follow from those of R , and one can check similarly that $0 + I$ and $1 + I$ are the zero and one respectively. $\square$

Example 1.1.68

Consider the ideal $n\mathbb{Z} \subseteq \mathbb{Z}$. Elements of the quotient ring $\mathbb{Z}/n\mathbb{Z}$ are cosets of the form $r + n\mathbb{Z}$ for $r = 0, \dots, n-1$, and addition and multiplication are the usual addition and multiplication and addition modulo n .

Example 1.1.69

Let R be a ring, and $\phi: G \rightarrow H$ a group homomorphism. Then ϕ induces a ring homomorphism $\phi_*: R[G] \rightarrow R[H]$, defined by

$$\phi_* \left(\sum_{g \in G} a_g g \right) := \sum_{g \in G} a_g \phi(g).$$

We now ask what $\ker(\phi_*)$ is. $N = \ker(\phi) \trianglelefteq G$. Thus, consider the (left) ideal $(N-1) \subseteq R[G]$ generated by elements of $R[G]$ of the form $g-1$ for $g \in N$. Then $\phi_*(g-1) = 0$ for all $g \in N$, so $(N-1) \subseteq \ker(\phi_*)$. In fact, $(N-1)$ is a two-sided ideal, and that $\ker(\phi_*) = (N-1)$.

Example 1.1.70

Let R be a commutative ring. Consider the ideal $(X) \subseteq R[X]$. Let's compute $R[X]/(X)$. Elements of this quotient ring are of the form

$$a_0 + a_1X + \dots + a_nX_n + (X).$$

Note that every term but the first is in (X) . Therefore, any such element is equivalent to $a_0 + (X)$, and this representation is in fact unique. Thus, there is an isomorphism $R \rightarrow R[X]/(X)$ sending a to $a + (X)$.

Proposition 1.1.71 (Euclidean Algorithm for Polynomials over a Field)

Let F be a field and $f, g \in F[X]$, with $g \neq 0$. Then there are some $r, q \in F[X]$ such that

$$f = gq + r,$$

with $\deg(r) < \deg(g)$.

With this, we can consider examples such as $\mathbb{R}[X]/(X^2 + 1)$.

Example 1.1.72

Let $R = \mathbb{R}[X]/(X^2 + 1)$. Elements of R are of the form

$$a_0 + a_1X + \cdots + a_nX^n + (X^2 + 1).$$

If we let $f = a_0 + \cdots + a_nX^n$, we can then apply the Euclidean algorithm for polynomials, and find $q, r \in R$ such that

$$f = q(X^2 + 1) + r,$$

with $\deg(r) < 2$, implying that $r = b_0 + b_1X$ for some $b_0, b_1 \in \mathbb{R}$. Therefore, any element of R is of the form $a + bX + (X^2 + 1)$, for some $a, b \in \mathbb{R}$. We note that this representation is unique: If $a + bX + (X^2 + 1) = a' + b'X + (X^2 + 1)$, then $(a - a') + (b - b')X = g(X^2 + 1)$, for some $g \in R$. However, if $g \neq 0$, so we get $\deg(g(X^2 + 1)) > 1$. Therefore, g must be zero, implying $a = a', b = b'$.

We conclude that every element of R is of the form $a + bX + (X^2 + 1)$, and the element $X + (X^2 + 1)$ squares to -1 . This looks a lot like the complex numbers. Define $\phi: R \rightarrow \mathbb{C}$ such that

$$\phi(a + bX + (X^2 + 1)) =: a + bi \in \mathbb{C}.$$

This is a well-defined, bijective, ring homomorphism.

This was a lot of work, but the following theorem will make similar problems much easier:

Theorem 1.1.73 (First Isomorphism Theorem for Rings)

Let $\phi: R \rightarrow S$ be a ring homomorphism. Then, there is an isomorphism of rings

$$R/\ker(\phi) \cong \text{Im}(\phi).$$

Proof. Define $\psi: R/\ker(\phi) \rightarrow \text{Im}(\phi)$ by $\psi(r + \ker(\phi)) =: \phi(r)$. This is a well defined map, since if $r + \ker \phi = r' + \ker \phi$, then $r - r' \in \ker \phi$, so $\phi(r) = \phi(r')$. Note that it is a ring homomorphism as well.

It is also an isomorphism of abelian groups, by the first isomorphism theorem for groups. $\square$

This quotient construction allows us to make many rings:

Example 1.1.74

Let k be a field. An important class of rings is finitely generated k -algebras; that is, quotients of a polynomial ring over k , $k[x_1, \dots, x_n]$, by a finitely generated ideal $I = (f_1, \dots, f_r)$. It turns out that for all $I \subseteq k[x_1, \dots, x_n]$, there exists finitely many elements $f_1, \dots, f_r \in I$ such that $I = (f_1, \dots, f_r)$. This is called the **Hilbert's Basis theorem**.

Definition 1.1.75. Let R be a commutative ring and k be a field. $(R, +)$ is a k -**algebra** if R is a vector space over k . Particularly, $k \subset R: \lambda \in k \mapsto \lambda \cdot 1$.

Definition 1.1.76. A k -algebra is **finitely generated** if $\exists a_1, \dots, a_n \in R$ such that the obvious ring homomorphism $\phi: k[x_1, \dots, x_n] \rightarrow R$ is $\phi(x_i) = a_i$, for all i , is surjective.

Example 1.1.77

We can define a homomorphism $\phi: \mathbb{R}[X] \rightarrow \mathbb{C}$ by setting $\phi(f) := f(i)$. We can then check that ϕ is surjective and that $\ker(\phi) = (X^2 + 1)$. Thus, $\mathbb{C} \cong \mathbb{R}(X)/(X^2 + 1)$.

Theorem 1.1.78 (Second Isomorphism Theorem)

Let S be a ring and $R \leq S$ a subring, and $I \subseteq S$ a two-sided ideal. Then,

1. $R + I := \{r + i \mid r \in R, i \in I\} \leq S$ is a subring.
2. $I \subseteq R + I$ and $R \cap I \subseteq R$ are both two-sided ideals.
3. $(R + I)/I = \{r + I \mid r \in R\} \leq S/I$ is a subring, and

$$R/(R \cap I) \cong (R + I)/I.$$

Proof. 1. Note that since $R \subseteq R + I$, $R + I$ contains 0 and 1. Now, let $r, s \in R$ and $i, j \in I$ so $r + i, s + j \in R + I$. Then,

$$(r + i) + (s + j) = (r + s) + (i + j) \in R + I$$

and

$$(r + i) \cdot (s + j) = (rs) + (is + rj + ij) \in R + I.$$

Note that the second term is in I since I is a two-sided ideal.

2. $I \subseteq R + I$ is a two-sided ideal since $I \subseteq S$ is one. Next, we define a homomorphism $\phi: R \rightarrow S/I$ by $\phi(r) := r + I$. Then, $\ker(\phi)$ consists of elements $r \in R$ such that $r + I = I$, i.e. $r \in I$. Thus, $\ker(\phi) = R \cap I$ which is therefore a two-sided ideal in R .

3. Let ϕ be the same map as in the proof for the second statement. Then, $\text{Im}(\phi) = \{r + I \mid r \in R\} = (R + I)/I$, so this is a subring of S/I . However, by the first isomorphism theorem (1.1.73), $\text{Im}(\phi) \cong R/(R \cap I)$. Therefore,

$$R/(R \cap I) \cong (R + I)/I.$$

□

Theorem 1.1.79 (Third Isomorphism Theorem)

Let R be a ring, and $I, J \subseteq R$ be two-sided ideals such that $I \subseteq J$. Then, $J/I \subseteq R/I$ is a two-sided ideal and

$$(R/I)/(J/I) \cong R/J.$$

Proof. We define a homomorphism $\phi: R/I \rightarrow R/J$ defined by

$$\phi(r + I) := r + J.$$

This is well-defined, and is a surjective ring homomorphism.

$\ker(\phi)$ consists of those $r + I \in R/I$ such that $r + J = 0$, i.e. $r \in J$. Therefore, $\ker(\phi) = J/I$. The result then follows from the first isomorphism theorem. $\square$

There's a similar important result, which doesn't have a standard name:

Proposition 1.1.80

Let R be a ring, and $I \subseteq R$ a two-sided ideal. Then there is a natural bijection

$$\begin{aligned} \{\text{two-sided ideals of } R/I\} &\longleftrightarrow \{\text{two-sided ideals of } R \text{ containing } I\} \\ \alpha: J \subseteq R/I &\longmapsto \{r \in R \mid r + I \in J\} \\ K/I \subseteq R/I &\longleftarrow I \subseteq K \subseteq R : \beta \end{aligned}$$

Proof. Let α be the map going right and β be the map going to the left. One can check that $\alpha(J) \subseteq R$ and $\beta(K) \subseteq R/I$ are two-sided ideals, so α and β are well-defined functions.

We now need to check that these are inverses to each other. First, let $K \subseteq R$ be a two-sided ideal containing I . Then,

$$\alpha(\beta(K)) = \alpha(K/I) = \{r \in R \mid r + I \in K/I\} = K,$$

where the last equality comes from $I \subseteq K$. For the other direction, let $J \subseteq R/I$ be a two-sided ideal. Then,

$$\beta(\alpha(J)) = \beta(\{r \in R \mid r + I \in J\}) = \{r + I \mid r + I \in J\} = J.$$

Therefore, α, β must be bijections. $\square$

Let R be any ring. Then, there's a unique homomorphism $\iota: \mathbb{Z} \rightarrow R$, where $\iota(n) = 1_R + \cdots + 1_R$ (n times) if $n \geq 0$, and $\iota(n) = -(1_R + \cdots + 1_R)$ ($-n$ times) if $n < 0$. This is a homomorphism by distributivity. Note that any homomorphism $\mathbb{Z} \rightarrow R$ must be equal to ι , since it must send 1 to 1_R . Further, note that $\ker(\iota) \subseteq \mathbb{Z}$ is an ideal, so we must have $\ker(\iota) = n\mathbb{Z}$, for some $n \in \mathbb{Z}$.

Definition 1.1.81. The **characteristic** of R is the unique $n \geq 0$ such that $\ker(\iota) = n\mathbb{Z}$, where ι is the unique ring homomorphism $\mathbb{Z} \rightarrow R$.

Example 1.1.82

$\mathbb{Q}$ has characteristic 0 since $\iota: \mathbb{Z} \rightarrow \mathbb{Q}$ is injective, so its kernel is 0. Similarly, $\mathbb{R}, \mathbb{C}, \mathbb{Z}[i]$ have characteristic 0.

Example 1.1.83

The homomorphism $\iota: \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ sends r to $r + n\mathbb{Z}$ and has kernel $n\mathbb{Z}$, so $\mathbb{Z}/n\mathbb{Z}$ has characteristic n , for any $n \geq 1$. Particularly, any $n \geq 0$ is the characteristic of some commutative ring.

§1.2 Integral Domains

Remark 1.2.1 — In this section, we will assume all rings are commutative and non-trivial.

§1.2.i Integral Domains and Ideals

Definition 1.2.2. Let R be a (commutative) ring. An element $r \in R$ is a **zero divisor** if $r \neq 0$ and there is some $s \neq 0$ such that $rs = 0$.

Definition 1.2.3. R is an **integral domain** if for all $u, v \in R$, $u \cdot v = 0 \implies u = 0$ or $v = 0$. That is, there are no non-zero zero-divisors.

Abuse of Notation 1.2.4. We may just say domain, instead of integral domain.

Definition 1.2.5. Given an integral domain R , an element $u \in R$ is a **unit** if $\exists r \in R$ such that $ur = 1$. We denote by $R^\times = \{u \in R \mid u \text{ is a unit}\}$.

Remark 1.2.6 — $R^\times$ is a group.

Example 1.2.7

The ring of integers $\mathbb{Z}$ is an integral domain.

Example 1.2.8

Any field is an integral domain, since if $rs = 0$ with $r \neq 0$, then if we let r^{-1} be an inverse of r , then

$$0 = r^{-1}rs = s.$$

Example 1.2.9

$\mathbb{Z}/6$ isn't an integral domain, since $2 + 6\mathbb{Z}$ and $3 + 6\mathbb{Z}$ are non-zero, but their product is 0.

More generally, we have the following.

Lemma 1.2.10

For $n \geq 1$, $\mathbb{Z}/n\mathbb{Z}$ is an integral domain if and only if n is prime.

Proof. Suppose $\mathbb{Z}/n$ is an integral domain, and suppose $n \mid rs$, for some $r, s \in \mathbb{Z}$. This means $(r + n\mathbb{Z})(s + n\mathbb{Z}) = 0$, so since $\mathbb{Z}/n$ is an integral domain, $r + n\mathbb{Z} = 0$ or $s + n\mathbb{Z} = 0$. So $n \mid r$ or $n \mid s$, so n is prime.

Now suppose n is prime, and suppose $(r + n\mathbb{Z})(s + n\mathbb{Z}) = 0$. Then $rs + n\mathbb{Z} = 0$ so $n \mid rs$. So $n \mid r$ or $n \mid s$, so $r + n\mathbb{Z} = 0$ or $s + n\mathbb{Z} = 0$. $\square$

Example 1.2.11

If R is an integral domain and $S \leq R$ is a subring, then S is also an integral domain, since a zero divisor in S would also be one in R .

Example 1.2.12

$\mathbb{C}$ is an integral domain since it's a field. So $\mathbb{Z}[i] \leq \mathbb{C}$ is an integral domain.

Proposition 1.2.13

A finite integral domain is a field.

Proof. Let R be our integral domain and let $|R| = N$, for some $N > 0$. Then, consider $a \in R$ and look at the set $\{a, a^2, a^3, \dots\}$. By finiteness, we must have that $a^m = a^n$ for some $n > m$, and since we have cancellation in integral domains, we attain that $a^{n-m} = 1$. In other words, a has an inverse. $\square$

Lemma 1.2.14

Let R be an integral domain. Then $R[X]$ is also an integral domain.

Proof. Let $f, g \in R[X]$ be non-zero, and write them as

$$\begin{aligned} f &= a_0 + \dots + a_n X^n \\ g &= b_0 + \dots + b_m X^m \end{aligned}$$

where a_n and b_m are non-zero (so $n = \deg(f)$ and $m = \deg(g)$). Then, the coefficient of X^{n+m} in fg is $a_n b_m$, which is non-zero since R is an integral domain. So $fg \neq 0$. $\square$

Corollary 1.2.15

If R is an integral domain, then $R[x_1, \dots, x_n]$ is an integral domain for all n .

Proof. We iterate lemma 1.2.14. $\square$

Example 1.2.16

In particular $\mathbb{Z}[X]$ is an integral domain.

Lemma 1.2.17

If R is an integral domain, then $R[X]^\times = R^\times$.

Example 1.2.18

The above lemma doesn't hold for non domains R . For instance, take $R = \mathbb{Z}/4\mathbb{Z}$. Then, $1 + 2X$ is a unit.

Lemma 1.2.19

A (non-trivial, commutative) ring R is a field if and only if its only ideals are $\{0\}$ and R .

Proof. Suppose R is a field and $I \subseteq R$ is a non-zero ideal, so it contains some non-zero element r . Then since R is a field, r is a unit, so $I = R$.

Now for the other direction. Let $r \in R$ be non-zero. Then $(r) \subseteq R$ is a non-zero ideal, and so must be R by assumption. So there is some s such that $rs = 1$, so r is a unit. So every non-zero $r \in R$ is a unit, therefore R is a field. $\square$

Definition 1.2.20. An ideal $I \subseteq R$ is **maximal** if $I \neq R$ and any proper ideal $J \subseteq R$ containing I is equal to I .

Lemma 1.2.21

An ideal $I \subseteq R$ is maximal if and only if R/I is a field.

Proof. Suppose I is maximal. Then the only ideals in R containing I are I and R , by the bijection between such ideals and ideals in R/I , we see that the only ideals in R/I are 0 and R/I , so R/I is a field.

Suppose R/I is a field, and suppose $J \subseteq R$ is a proper ideal containing I . Then, the only ideals of R/I are itself and 0 . Now, the correspondence of ideals tells us that the ideals of R/I are in bijection with ideals containing I . Therefore, 0 corresponds to R and R/I corresponds to I , and there is no in between. Thus, I is maximal. $\square$

We give a similar description for integral domains.

Definition 1.2.22. An ideal $I \subseteq R$ is **prime** if $I \neq R$ and whenever there are $r, s \in R$ such that $rs \in I$, we have that $r \in I$ or $s \in I$.

Example 1.2.23

Let $n \in \mathbb{Z}$ be non-zero. Then the ideal $n\mathbb{Z} \subseteq \mathbb{Z}$ is prime if and only if n is prime.

Suppose n is prime. If $rs \in n\mathbb{Z}$, then $n \mid rs$. So $n \mid r$ or $n \mid s$ i.e. $r \in n\mathbb{Z}$ or $s \in n\mathbb{Z}$.

Conversely suppose $n = uv$ is composite (where $u, v \neq 0, \pm 1$). Then $n = uv \in n\mathbb{Z}$ but $u, v \notin n\mathbb{Z}$, (since $0 < |u|, |v| < |n|$).

Lemma 1.2.24

An ideal $I \subseteq R$ is prime if and only if R/I is an integral domain.

Proof. ($\implies$): First suppose R/I is prime. Suppose we have elements $r + I, s + I \in R/I$ such that $(r + I)(s + I) = 0 + I$. This means $rs \in I$. Since I is prime, one of r and s is in I , so one of $r + I$ and $s + I$ is 0.

($\impliedby$): Suppose R/I is an integral domain, and suppose we have $r, s \in R$ such that $rs \in I$. Then $(r + I)(s + I) = 0 + I$, so one of $r + I, s + I$ is 0, so one of r, s is in I . $\square$

Example 1.2.25

Let R be a ring. Then $R[X]/(X) \cong R$, so the ideal $(X) \subseteq R[X]$ is prime if and only if R is an integral domain, and it's maximal if and only if R is a field.

Corollary 1.2.26

If $I \subseteq R$ is a maximal ideal, it is a prime ideal.

Proof. Since I is maximal, R/I is a field, so it is an integral domain, so I is a prime ideal. $\square$

The converse isn't true, as is shown in the following example.

Example 1.2.27

$(0) \subseteq \mathbb{Z}$ is a prime ideal but isn't maximal.

Lemma 1.2.28

Let R be an integral domain. Then its characteristic is either 0 or prime.

Proof. Let $\iota: \mathbb{Z} \rightarrow R$ be the unique ring homomorphism, so $\ker(\iota) = n\mathbb{Z}$, where $n \geq 0$ is the characteristic of R . Then, by the first isomorphism theorem, $\text{Im}(\iota) \leq R$ is isomorphic to $\mathbb{Z}/n\mathbb{Z}$. Since R is an integral domain, $\mathbb{Z}/n\mathbb{Z}$ is one too. We proved earlier that if $n > 0$ and $\mathbb{Z}/n\mathbb{Z}$ is an integral domain, n must be prime. $\square$

Visualizing Polynomials and Rings

Here are some remarks to help us see the pictures of polynomials and rings. That is, to understand the *geometry* of what is going on.

1. Polynomials give rise to functions, since they can be evaluated. $f \in k[x, y]$ can give rise to $f: k^2 \rightarrow k$ with $a, b \mapsto f(a, b)$.
2. We can consider the zero set

$$V = V(f) = \{(a, b) \in k^2 \mid f(a, b) = 0\}$$

3. We can restrict functions. That is, for $g \in k[x, y]$, we can restrict so that $g|_V : V \rightarrow k \subset \{\phi : V \rightarrow k\}$ (the latter of which is a ring. This ring homomorphism

$$k[x, y] \xrightarrow{\psi} \{\text{ring of functions } \phi : V \rightarrow k\}.$$

Note that $(f) \subset \ker(\psi)$. Sometimes, we can show that $(f) = \ker(\psi)$.

§1.2.ii Factorization

Remark 1.2.29 — Throughout this subsection, we’ll only work with rings that are integral domains.

We’ll explore some number-theoretic properties of certain classes of rings.

Definition 1.2.30. Let R be an integral domain, and $r, s \in R$.

- We say that r **divides** s , written $r \mid s$, if there is some $u \in R$ such that $s = ru$. This is equivalent to having $(s) \subseteq (r)$.
- We say r and s are **associates** if there is some unit $u \in R^\times$ such that $s = ru$. This is equivalent to having $(r) = (s)$, or that both $r \mid s$ and $s \mid r$.

Example 1.2.31

In $\mathbb{Z}$, r and s are associates if and only if $r = \pm s$.

This isn’t true in general: in $\mathbb{Z}[i]$, $2i$ and 2 are associates.

Definition 1.2.32. Let R be an integral domain. Then, $m \in R$ is **irreducible** if $m = m_1 m_2$ in R , then either $m_1 \in R^\times$ or $m_2 \in R^\times$.

Definition 1.2.33. Given an integral domain R , $n \in R$ is **prime** if $\forall a, b \in R, n \mid ab \implies n \mid a \text{ or } n \mid b$.

Example 1.2.34

These properties depend on the ring, not just the element. For example, 2 is prime in $\mathbb{Z}$, but not in $\mathbb{Q}$ (since it’s a unit in $\mathbb{Q}$). Similarly, $2X$ is irreducible in $\mathbb{Q}[X]$, but not in $\mathbb{Z}[X]$.

In $\mathbb{Z}$, primality and irreducibility coincide, but not in general. Furthermore, another important number-theoretic property of $\mathbb{Z}$ is the fact that any $n \in \mathbb{Z}$ has an essentially unique prime factorisation (unique up to reordering and signs), but in the following example, we see an example of factorisation into irreducibles being non-unique.

Example 1.2.35

The ring $R = \mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$ is a subring of $\mathbb{C}$ so it's an integral domain.

We claim 3 is irreducible but not prime in R . 2 and 3 both divide $6 = (1 - \sqrt{-5})(1 + \sqrt{-5})$ but neither 2 nor 3 divides either factor on the right, so 2 and 3 both aren't prime. Suppose $3 = (a + b\sqrt{-5})(u + v\sqrt{-5})$, $a, b, u, v \in \mathbb{Z}$. Applying $|\cdot|^2$ to both sides, we see that

$$9 = (a^2 + 5b^2)(u^2 + 5v^2).$$

The only solutions are $a + b\sqrt{-5} = \pm 3$ and $u + v\sqrt{-5} = \pm 1$, and the other way around. So 3 is irreducible. A similar argument shows 2 and $1 \pm \sqrt{-5}$ are all irreducible. So 6 can be factored into irreducibles in 2 distinct ways.

Lemma 1.2.36

A principal ideal (r) in an integral domain R is a prime ideal if and only if $r = 0$ or r is prime.

Proof. ($\implies$): Suppose (r) isn't a prime ideal. If $r = 0$, we're done, so assume $r \neq 0$. Since prime ideals aren't the whole ring, r isn't a unit. Now suppose that $r \mid uv$. Then, $uv \in (r)$. Since (r) is prime, it follows then that $u \in (r)$ or $v \in (r)$. Consequently, $r \mid u$ or $r \mid v$, implying that r is prime.

($\impliedby$): Conversely, if $r = 0$, then the ideal $(r) = \{0\}$ is prime since R is an integral domain. Suppose $r \neq 0$ is prime. If $uv \in (r)$, this means $r \mid uv$, so $r \mid u$ or $r \mid v$. Thus, $u \in (r)$ or $v \in (r)$. Thus, (r) is prime. $\square$

Lemma 1.2.37

If $r \in R$ is prime, then r is irreducible.

Proof. Let $r \in R$ be prime, and suppose $r = uv$. Thus, $r \mid uv$, so since r is prime, $r \mid u$ or $r \mid v$. WLOG, assume $r \mid u$. Thus, there is some $s \in R$ such that $rs = u$. Therefore, $r = uv = r(sv)$. Since R is an integral domain, this implies $sv = 1$, so v was a unit. $\square$

Remark 1.2.38 — We have seen that $3 \in \mathbb{Z}[\sqrt{-5}]$ is irreducible, but not prime, so the converse isn't true over integral domains.

Definition 1.2.39. A **Euclidean domain** is an integral domain R , with a function $\theta: R \setminus \{0\} \rightarrow \mathbb{N}$ such that:

- $\theta(rs) \geq \theta(r)$, for all $r, s \neq 0 \in R$.
- For all $a, b \in R$ with $b \neq 0$, there are $q, r \in R$ such that

$$a = bq + r,$$

and $r = 0$ or $\theta(r) < \theta(b)$.

Remark 1.2.40 — θ is often called the *Euclidean function* or the *degree function*.

Example 1.2.41

$\mathbb{Z}$ is a Euclidean domain with $\theta(n) = |n|$.

Example 1.2.42

If F is a field, then F is a Euclidean domain, with $\theta(r) = 0$, for all r .

Example 1.2.43

If K is a field, then $K[X]$ is a Euclidean domain with the usual degree function $\deg$. Note that there is an analogy between $\mathbb{Z}$ and $K[X]$.

Example 1.2.44

The ring of Gaussian Integers $\mathbb{Z}[i]$ is a Euclidean domain with the complex norm function $|a + bi| = a^2 + b^2$. We need to show that this function is indeed a Euclidean function.

$$\begin{aligned} \frac{a + bi}{c + di} &= \frac{(a + bi)(c - di)}{c^2 + d^2} \\ &= \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i. \end{aligned}$$

We then declare the following:

$$q = \text{Round}\left(\frac{ac + bd}{c^2 + d^2}\right) + \text{Round}\left(\frac{bc - ad}{c^2 + d^2}\right)i.$$

We also use the fact that the complex norm is multiplicative, and that if we round, the *remainder* will be of the form $\alpha + \beta i \in \mathbb{Q}[i]$, where $0 \leq \alpha, \beta < \frac{1}{2}$. This implies that $\alpha^2 + \beta^2 < \frac{1}{2}$, which allows our argument to work.

Definition 1.2.45. The ring of **Hurwitz quaternions** is the ring $R = \mathbb{Z}[i, j, k] + \frac{1+i+j+k}{2}\mathbb{Z}$.

Definition 1.2.46. For $\alpha = x + iy + jz + kt \in \mathbb{R}[i, j, k]$, define

$$N(\alpha) = \alpha\bar{\alpha} = x^2 + y^2 + z^2 + t^2.$$

Note that $\bar{\alpha} = x - iy - jz - kt$.

Remark 1.2.47 — Note that the norm on the real quaternions is multiplicative.

Example 1.2.48

$\forall a, b \in R$, the ring of Hurwitz quaternions, it has the Euclidean property on the right and left. That is,

- $\exists \gamma, \rho \in R$ such that $\alpha = \beta\gamma + \rho$ and $N(\rho) < N(\beta)$.
- $\exists \gamma', \rho' \in R$ such that $\alpha = \gamma'\beta + \rho'$ and $N(\rho') < N(\beta)$.

To prove this, the key point is the following lemma:

Lemma 1.2.49

$\forall \epsilon \in \mathbb{R}[i, j, k]$, $\exists \alpha \in R$ such that

$$N(\epsilon - \alpha) < 1,$$

where N is the norm introduced above.

As a consequence of the lemma above, we get that every right ideal (left ideal) is principal.

Remark 1.2.50 — Note that in the example above, the lemma would not be true in $\mathbb{Z}[i, j, k]$. For instance, $\epsilon = \frac{1+i+j+k}{2}$ has norm 1.

Definition 1.2.51. Let R be an integral domain. Then, R is a **principal ideal domain** if every ideal is principal. That is, for any ideal $I \subseteq R$, $I = (r)$ for some $r \in R$.

Example 1.2.52

$\mathbb{Z}$ is a PID.

Theorem 1.2.53 (ED $\implies$ PID)

Let R be a Euclidean domain. Then, R is a principal ideal domain.

Proof. Let θ be the Euclidean function for R . Let $I \subseteq R$ be an ideal, which we can assume to be non-zero. Choose $b \neq 0 \in I$, with $\theta(b)$ minimized, and let $a \in I$ be another element. Then, there are some $q, r \in R$ with $a = qb + r$, with $r = 0$ or $\theta(r) < \theta(b)$.

Since I is an ideal and $a, b \in I$, $r = a - qb \in I$ as well. This means we can't have that $\theta(r) < \theta(b)$ and $r \neq 0$, due to our selection of b , implying that $r = 0$. Therefore, $a = qb$, so $a \in (b)$. Since $a \in I$ was arbitrary, this implies that $I = (b)$ is principal. $\square$

Example 1.2.54

For any field F , $F[X]$ is a Euclidean domain and hence a principal ideal domain. Similarly, $\mathbb{Z}[i]$ is a principal ideal domain.

Proposition 1.2.55

The ideal $(2, X) \subset \mathbb{Z}[X]$ is not generated by a single element.

Proof. Suppose there is some $f \in \mathbb{Z}[X]$ such that $(2, X) = (f)$. $2 \in (f)$, so there is some $g \in \mathbb{Z}[X]$ such that $fg = 2$. Thus, $\deg(f) = 0$, and f must be ± 1 or ± 2 . $f \neq \pm 1$, since $1 \notin (2, X)$. Similarly, $f \neq \pm 2$, since $2 \nmid X$. Thus, $(2, X)$ can't be a principal ideal. $\square$

Corollary 1.2.56

$\mathbb{Z}[X]$ is not a principal ideal domain (and hence not a Euclidean domain either).

Proof. This follows by proposition 1.2.55. $\square$

Example 1.2.57

Let F be a field and $A \in M_n(F)$. Consider the set

$$I := \{f \in F[X] \mid f(A) = 0\} \subseteq F[X].$$

This is an ideal: If $f, g \in I$, $(f + g)(A) = f(A) + g(A) = 0$. If $f \in I$ and $h \in F[X]$, $(fh)(A) = f(A)h(A) = 0$.

Since $F[X]$ is a PID, then $I = (m)$, for some $m \in F[X]$. Then $m(A) = 0$ and for any $f \in F[X]$ such that $f(A) = 0$, $m \mid f$. m is called the **minimal polynomial** of A .

Remark 1.2.58 — Existence of minimal polynomials can be proven much more directly but would be much longer.

Example 1.2.59

Let R be a PID and $(a), (b)$ be two nonzero ideals for $a, b \in R$. Then,

$$(a) + (b) = (\gcd(a, b))$$

$$(a) \cap (b) = (\text{lcm}(a, b))$$

Definition 1.2.60. An integral domain R is a **unique factorization domain** (UFD) if

- Every non-zero non-unit $r \in R$ is a product of irreducibles.
- If $p_1 \dots p_n = q_1 \dots q_m$ with each $p_i, q_j \in R$ irreducible, then $n = m$, and they can be reordered such that each p_i is an associate of q_i (i.e. they're related by multiplication by a unit).

Our next goal will be to prove that principal ideal domains are unique factorization domains, but we need a few preliminaries first.

Lemma 1.2.61

Let R be a principal ideal domain. Then a non-zero principal ideal $(r) \subseteq R$ is maximal if and only if r is irreducible or ($r = 0$ and R is a field).

Proof. ($\implies$): Suppose (r) is a maximal ideal, and that r is not irreducible. If $r = 0$, then we know $R = R/(r)$ must be a field. Otherwise, we can assume that $r = xy$ for $x, y \in R$ non-units. So $(r) \subseteq (x) \subseteq R$. Since x isn't a unit, $(x) \neq R$. Since (r) is maximal, $(r) = (x)$. So $r = xz$ for some unit $z \in R^\times$. Thus, $r = xz = xy$ so $x(z - y) = 0$ so $z = y$ (since $x \neq 0$ and R is an integral domain). However y isn't a unit whereas z is, which is a contradiction.

($\impliedby$): If $r = 0$ and R is a field, then (r) is maximal. So we instead assume r is irreducible and that (r) is non-maximal. So there is some ideal $(r) \subseteq I \subseteq R$ such that $I \neq (r), R$. Since R is a principal ideal domain, $I = (s)$ for some non-unit $s \in R$. Since $(r) \subseteq (s)$, $r = sz$ for some non-unit $z \in R$, which contradicts irreducibility of r . $\square$

Lemma 1.2.62

Let R be a principal ideal domain. If $r \in R$ is irreducible, it is prime.

Proof. If $r \in R$ is irreducible, then $(r) \subseteq R$ is a maximal ideal, by the previous lemma. Therefore it is also a prime ideal. Since the ideal (r) is prime and $r \neq 0$, a previous lemma implies that r is prime. $\square$

Recall primes are always irreducible in any integral domain, but in the example $\mathbb{Z}[\sqrt{-5}]$, the element 2 was irreducible but not prime. So the lemma shows $\mathbb{Z}[\sqrt{-5}]$ isn't a PID.

Corollary 1.2.63

Let R be a principal ideal domain. Then every non-zero prime ideal is maximal.

Proof. Let $I \subseteq R$ be a non-zero prime ideal. Since R is a PID, $I = (r)$ for some $r \in R$. Since $I \neq 0$ was a prime ideal, r is prime (by a previous lemma). So r is irreducible (recall primes are always irreducible). Therefore by another earlier lemma, (r) is maximal. $\square$

Example 1.2.64

The above isn't true if R is a UFD. For instance, let $R = \mathbb{Z}[X]$ and consider $I = (X)$. R/I is an integral domain, but not a field, and so I is prime, but not maximal.

We already know that maximal ideals are prime. Therefore this corollary says that in a principal ideal domain, prime ideals and maximal ideals are the same (with the exception of the zero ideal, which is always prime if R is a PID but maximal if and only if R is a field).

Proposition 1.2.65

Let R be a principal ideal domain, and $I_1 \subseteq I_2 \subseteq \cdots \subseteq R$ an increasing sequence of ideals in R . Then this sequence is eventually constant, i.e. there is some integer N such that $I_n = I_{n+1}$ for each $n \geq N$.

Proof. Let $I = \cup_{i \geq 1} I_i$. This is an ideal in R . Since R is a PID, $I = (r)$ for some $r \in I = \cup_{i \geq 1} I_i$, so there is some N such that $r \in I_N$. Then,

$$(r) \subseteq I_N \subseteq I_{N+1} \subseteq \cdots \subseteq I = (r).$$

Thus, for $n \geq N$, $I_n = (r)$. □

A bit about Noetherian Rings

Definition 1.2.66. A ring R is called **left-Noetherian** (**right-Noetherian**) if every left (right) ideal is finitely generated. We get that R is **Noetherian** if it is both right and left Noetherian.

Lemma 1.2.67

A ring R is left-Noetherian if and only if it satisfies the ascending chain condition. That is, if

$$I_1 \subset I_2 \subset \cdots I_r \subset I_{r+1} \subset \cdots \subset R,$$

is an infinite chain of left ideals, then there exists N such that for all $r \geq N$, $I_N = I_r$.

Proof. ($\implies$): If $I_1 \subset I_2 \subset \cdots$ is an ascending chain, then setting $\cup_j I_j = I$, this is a left ideal and $I = (a_1, \dots, a_r)$. Thus, there exists an N such that all $a_j \in I_N$ and for all $j \geq N$, $I_N = I_j$.

($\impliedby$): Assuming the ascending chain, let I be a left ideal. Pick $a_1 \in I$. Then, either $I = (a_1) = I_1$ or $\exists a_2 I \setminus I_1$. Then, $I = (a_1, a_2) = I_2$ or $\exists a_3 I \setminus I_2$. Continuing this, either $I = I_n = (a_1, \dots, a_n)$ is finitely generated or we can construct an infinite chain, which contradicts our ascending chain condition. □

With this we can equivalently say that a ring R is left-Noetherian if we have the ascending chain condition in the lemma above.

Example 1.2.68

Let $R = k[x_1, x_2, x_3, \dots] = \cup_k k[x_1, \dots, x_k]$. The ideal $I = (x_1, x_2, x_3, \dots)$ is not finitely generated, so is not Noetherian.

Proposition 1.2.69

If R is a Noetherian ring, then so is R/I for every ideal I in R .

Proof. This follows by the bijection between ideals of R containing I and ideals of R/I . Every ideal in R/I has the form J/I for an ideal J of R such that $I \subset J \subset R$. Since R is Noetherian, J must be finitely generated in R , and that finite generating set for J reduces to a generating set for J/I as an ideal of R/I . □

Theorem 1.2.70 (Hilbert Basis Theorem)

If R is Noetherian, then $R[X]$ is Noetherian.

Proof. We show that every ideal in $R[X]$ is finitely generated. Let $J \subset R[X]$ be an ideal. For all k , let

$$I_k = \{a \mid a = \text{leading coefficient of a polynomial } f \in J \text{ where } \deg(f) = k\}.$$

We claim that I_k is an ideal. Now, let $a_1, a_2 \in I_k$. Then, there exists $f_1, f_2 \in J$ such that

$$\begin{aligned} f_1 &= a_1 x^k + \text{lower order terms} \\ f_2 &= a_2 x^k + \text{lower order terms.} \end{aligned}$$

Then, $f_1 + f_2 = (a_1 + a_2)x^k + \text{lower order terms} \in J$, so $(a_1 + a_2) \in I_k$. $\lambda \in R$, $a \in I_k$ such that there exists $f \in J$, $f = ax^k + \text{lower order terms}$ such that $\lambda f = \lambda ax^k + \text{lower order terms} \in J$, implying $\lambda a \in I_k$. Thus, I_k is an ideal.

There is also the ascending chain condition in R :

$$I_1 \subset I_2 \subset \dots$$

such that for all k , $I_k \subset I_{k+1}$. If $a \in I_k$, there exists an f such that $f = ax^k + \text{lower order terms} \in J$, $x \cdot f = ax^{k+1} + \text{lower order terms} \in J$, implying $a \in I_{k+1}$. R is Noetherian, so we invoke the ascending chain condition for this sequence of ideals: $\exists N$ such that $\forall k \geq N$, $I_k = I_N$. R is Noetherian, so let $I_N = (a_1, \dots, a_m)$. There exists polynomials $f_i = a_i x^N + \text{lower order terms} \in J$. We want to show that these f_i generate J . Choose $f \in J$ so that $f = c_\nu x^\nu + c_{\nu-1} x^{\nu-1} + \dots + c_0$. Suppose $\nu \geq N$. Note that $c_\nu \in I_N = I_N = (a_1, \dots, a_m)$, since the ideals stabilize. Therefore, $c_\nu = \lambda_1 a_1 + \dots + \lambda_m a_m$. Next, consider $\lambda_1 f_1 + \dots + \lambda_m f_m$, where each f_i has leading coefficient a_i and has degree N . This is equal to

$$(\lambda_1 a_1 + \dots + \lambda_m a_m) x^N + \text{lower order terms.}$$

We can then consider $g = f - x^{\nu-N} \lambda_1 f_1 - \dots - x^{\nu-N} \lambda_m f_m$. This is contained in J , since every term is contained in J , and $\deg(g) < \deg(f)$. For $\nu < N$, we can consider $g = f - x^\nu \lambda_1 f_1 - \dots - x^\nu \lambda_m f_m$ to say that $g \in J$ and $\deg(g) < \deg(f)$. Thus, we've shown that the element f_i generate g and we can say the same holds for f , so we're done. $\square$

Remark 1.2.71 — Let us highlight the key mechanism. $\forall k < N$, $I_k = (a_1^k, \dots, a_{m(k)}^k)$. For all k and all $1 \leq j \leq m(k)$, there exists $f_j^k = a_j^k x^k + \text{lower order terms}$. The claim is that these f_j^k , where $1 \leq j \leq \nu$ and $1 \leq j \leq m(k)$, generate J .

Remark 1.2.72 — Note that there exists Noetherian rings that are K -algebras, but not finitely generated.

Example 1.2.73

The formal power series $R[[X]]$ is an example of a Noetherian ring that is a K -algebra, but not finitely generated.

Example 1.2.74

Let K be a field. Then, the ring $K[x, y, z]/(x^2 - yz)$ is Noetherian but not a UFD.

$K[x, y, z]$ is certainly Noetherian by inducting the Hilbert basis theorem on three variables, and we use the result that the quotient of a Noetherian ring is Noetherian (for this, we use the bijection of ideals of R that contain I and sub-ideals of R/I). Then, the ring in concern is Noetherian. However, $x^2 = yz$, so we get non-uniqueness.

Let us get back to our main discussion about $\text{PID} \implies \text{UFD}$.

Theorem 1.2.75

Let R be a principal ideal domain. Then, it is also a unique factorization domain.

Proof. We first show uniqueness. Suppose $p_1 \dots p_n = q_1 \dots q_m$, where each $p_i, q_j \in R$ is irreducible. In particular, $p_1 \mid q_1 \dots q_m$. Since p_1 is irreducible, it's prime, so $p_1 \mid q_j$ for some j . Reordering if necessary, we assume $p_1 \mid q_1$, so $p_1 a = q_1$ for some a . Since q_1 is irreducible, a is a unit, so p_1, q_1 are associates. Since R is an integral domain, we can cancel p_1 to obtain

$$p_2 \dots p_n = (aq_2)q_3 \dots q_m$$

and, replacing q_2 with aq_2 , we get

$$p_2 \dots p_n = q_2 \dots q_m$$

and repeating the process, we find that (possibly after reordering) p_i and q_i are associates for all i . We also find that $n = m$: if not, WLOG $n > m$, and we find that $p_{m+1} \dots p_n = 1$, so p_n is a unit, which is a contradiction.

We next show existence. Suppose $r \in R$ is a non-unit which can't be factored as a product of irreducibles. In particular, r isn't irreducible. So we can write $r = r_1 s_1$, with $r_1, s_1 \in R$ non-units. Since r isn't a product of irreducibles, at least one of r_1, s_1 isn't either, WLOG it's r_1 . So we write $r_1 = r_2 s_2$ with r_2, s_2 non-units. Again, WLOG r_2 isn't a product of irreducibles, so $r_2 = r_3 s_3$ with $r_3, s_3 \in R$ non-units, and we may continue this process indefinitely. By assumption this process doesn't end, so we have an increasing sequence of ideals:

$$(r) \subseteq (r_1) \subseteq (r_2) \subseteq \dots \subseteq R$$

Since R is a PID and hence Noetherian, there is some N such that $(r_N) = (r_{N+1}) = (r_{N+2}) = \dots$. Now since $(r_N) = (r_{N+1})$ and $r_N = r_{N+1} s_{N+1}$, s_{N+1} must be a unit, which is a contradiction (s_{N+1} was a non-unit by construction). So r must be a product of irreducibles. $\square$

Example 1.2.76

$\mathbb{Z}[i]$ is a Euclidean domain, and therefore a PID and also a UFD.

Definition 1.2.77. Let R be a ring. An element $d \in R$ is a **greatest common divisor** (GCD) of a finite sequence $a_1, \dots, a_n \in R$ if $d \mid a_i$ for all i , and if any other $d' \in R$ such that $d' \mid a_i$ for all i , then $d' \mid d$.

Proposition 1.2.78

Let R be a unique factorisation domain. Then for any $a_1, \dots, a_n \in R$ not all zero, they have a greatest common divisor d , and any other greatest common divisor d' is an associate of d .

Proof. Let $p_1, \dots, p_m \in R$ be a collection of irreducibles in R such that any irreducible factor of any a_i is an associate of some p_j , and no p_i, p_j are associates of each other for $i \neq j$. We can write

$$a_i = u_i \prod_{j=1}^m p_j^{k_{ij}},$$

where each $k_{ij} \in \mathbb{Z}_{\geq 0}$ and each u_i is a unit. We let $l_j = \min_i \{k_{ij}\}$ for all j , and define

$$d := \prod_{j=1}^m p_j^{l_j}.$$

Since $l_j \leq k_{ij}$ for all i , it follows that $d \mid a_i$ for all i . Suppose $d' \in R$ also satisfies $d' \mid a_i$ for all i . Then we can write

$$d' := c \prod_{j=1}^m p_j^{c_j},$$

for some unit v , and some $c_j \in \mathbb{Z}_{\geq 0}$. Since $d' \mid a_i$ for all i we must have $c_j \leq l_j$ for all j . So $d' \mid d$, so d is a GCD. Suppose d, d' are both GCDs of $a_1, \dots, a_n$. Then they must divide each other and hence are associates. $\square$

We can summarize some of the main things we've proved so far in the following chain of implications:

$$\mathbb{Z} \implies \text{ED} \implies \text{PID} \implies \text{UFD} \implies \text{ID} \implies \text{Commutative ring} \implies \text{Ring}$$

Neither of these implications goes both ways. Counterexamples to the other direction Lecture 8 of each implication respectively are the following;

1. $\mathbb{Q}, \mathbb{Z}[i]$.
2. $\mathbb{Z} \left[\frac{1+\sqrt{-19}}{2} \right]$. We won't justify that this ring is a PID but not an ED as it would be quite a long digression.
3. $\mathbb{Z}[X]$. We saw already this isn't a PID; we'll see in the next section that it is a UFD.
4. $\mathbb{Z}[\sqrt{-5}]$.
5. $\mathbb{Z}/6$.
6. $M_2(\mathbb{R})$.

We also have that a field is a Euclidean domain, and so it satisfies the chain above (replacing $\mathbb{Z}$ with 'fields'). This is because fields are commutative division rings, and so we don't need to worry about the 'Euclidean map'.

§1.2.iii Factorization in Polynomial Rings

Recall that if R is an integral domain, $R[X]$ also is. This forms an important class of rings. An important special case of this is $F[X]$ when F is a field. $F[X]$ is a Euclidean domain, principal ideal domain and unique factorization domain, which by results of the previous section implies:

- If $I \subseteq F[X]$ is a non-zero ideal, then $I = (f)$ for some non-zero $f \in F[X]$, and I is maximal if and only if I is prime.
- An element $f \in F[X]$ is irreducible if and only if it's prime.

Further, we saw that the following are equivalent for $f \in F[X]$:

- f is irreducible.
- f is prime.
- $F[X]/(f)$ is an integral domain.
- $F[X]/(f)$ is a field.

Constructing The Field Of Fractions

Let R be a commutative ring, not necessarily an integral domain.

Definition 1.2.79. $S \subseteq R$ is a **multiplicative subset** if $1 \in S$ and $S \neq \emptyset$ and $s_1, s_2 \in S \implies s_1 \cdot s_2 \in S$. (we may also call this a multiplicative monoid)

Definition 1.2.80. The **localization** of R in S (for a multiplicative set $S \subseteq R$), denoted $S^{-1}R$, is defined as the set

$$S^{-1}R := \{(a, s) \mid a \in R, s \in S\} / \sim,$$

with the quotient

$$(a_1, s_1) \sim (a_2, s_2) \text{ if } \exists t \in S \text{ such that } t(a_1 s_2 - a_2 s_1) = 0.$$

Remark 1.2.81 — We note that if R is an integral domain and take $S = R \setminus \{0\}$, then $S^{-1}R$ is a field, called the **fraction field** of R .

Remark 1.2.82 — Note that $\sim$ is an equivalence relation. We have the usual addition and multiplication for fractions to hold.

Remark 1.2.83 — Let F be the field of fractions for R . Then, the set

$$\left\{ \frac{r}{1} \mid r \in R \right\} \subseteq F$$

is a subring isomorphic to R .

Example 1.2.84

The field of fractions of $\mathbb{Z}$ is $\mathbb{Q}$.

Example 1.2.85

The field of fractions of $\mathbb{C}[X]$ is

$$\left\{ \frac{p}{q} \mid p, q \in \mathbb{C}[X], q \neq 0 \right\}.$$

Notation 1.2.86. In general, if F is a field, we write $F(X)$ for the field of fractions of the polynomial ring $F[X]$.

Definition 1.2.87. Let R be a UFD, and $f = a_0 + \cdots + a_n X^n \in R[X]$. Then the **content** of f , written $c(f)$, is the greatest common divisor of the coefficients $a_0, \dots, a_n$.

Remark 1.2.88 — This is only well-defined up to associates, though note that the ideal it generates $(c(f))$ is completely well-defined.

Definition 1.2.89. Let R be a UFD and $f \in R[X]$. We say f is **primitive** if $c(f)$ is a unit, i.e. the coefficients of f are all coprime.

Example 1.2.90

Let $f = a_0 \in R[X]$ be a constant polynomial. Then $c(f) = a_0$ (up to associates).

Example 1.2.91

Let $f = 2 + 3X + 4X^2 \in \mathbb{Z}[X]$. Then $c(f)$ is a unit so f is primitive.

Example 1.2.92

Let $g = 2 + 4X + 6X^2 \in \mathbb{Z}[X]$. Then $c(f) = 2$ (up to associates) so g isn't primitive. However, 2 is a unit in $\mathbb{Q}$ so g is primitive in $\mathbb{Q}[X]$.

Lemma 1.2.93

Let R be a UFD, and $f \in R[X]$. Then

$$f = c(f) \cdot f',$$

for some primitive $f' \in R[X]$.

Proof. By the definition of GCD, $c(f) \mid a_i$ for all i , so $a_i = c(f)b_i$ for some $b_i \in R$, such that the GCD of $b_1, \dots, b_N$ is 1. Let $f' = b_0 + \cdots + b_n X^n$. Then f' is primitive and $f = c(f) \cdot f'$. $\square$

Lemma 1.2.94

Let R be a UFD. If $f, g \in R[X]$ are primitive, so is $f \cdot g$.

Proof. Write $f = a_0 + \cdots + a_n X^n$ and $g = b_0 + \cdots + b_m X^m$, with $a_n, b_m \neq 0$ and $f, g \in R[X]$ primitive. Suppose fg isn't primitive. So $c(fg)$ isn't a unit. Since R is a UFD, there's some irreducible $p \in R$ with $p \mid c(fg)$. By assumption, $c(f)$ and $c(g)$ are units, so p doesn't divide $c(f)$ or $c(g)$. This implies p doesn't divide all a_i or all b_j . So there are some $k, l \geq 0$ such that $p \mid a_0, p \mid a_1, \dots, p \mid a_{k-1}$ but $p \nmid a_k$. Similarly $p \mid b_0, p \mid b_1, \dots, p \nmid b_{l-1}$ but $p \nmid b_l$.

Consider the X^{k+l} th coefficient in $f \cdot g$. This is

$$\sum_{i+j=k+l} a_i b_j = (a_{k+l} b_0 + \cdots + a_{k+1} b_{l-1}) + a_k b_l + (a_{k-1} b_{l+1} + \cdots + a_0 b_{l+k}).$$

Note that in the first set of brackets in the RHS, p divides each b_j , and in the final set of brackets, p divides each a_i , by assumption. Furthermore p divides this whole coefficient by assumption. So p must divide the only other term, which is $a_k b_l$. Since p is irreducible and hence prime, $p \mid a_k$ or $p \mid b_l$, either of which provides a contradiction. Thus, $c(f \cdot g)$ is a unit so $f \cdot g$ is primitive. $\square$

Corollary 1.2.95

Let R be a UFD. Then for $f, g \in R[X]$, $c(f \cdot g)$ is an associate of $c(f) \cdot c(g)$.

Proof. Write $f = c(f) \cdot f'$ and $g = c(g) \cdot g'$ where $f', g' \in R[X]$ are primitive, which we can do by an earlier lemma. Then $f \cdot g = c(f)c(g)(f' \cdot g')$ and f', g' are primitive so by the previous lemma, $f' \cdot g'$ is primitive, so $c(f)c(g)$ is a GCD of the coefficients of $f \cdot g$. $\square$

Remark 1.2.96 — Note that this would imply the previous lemma, but we will use the previous lemma to prove this corollary.

Also note that we can't really say $c(f \cdot g) = c(f) \cdot c(g)$, since both are only well-defined up to associates.

Lemma 1.2.97 (Gauss's Lemma)

Let R be a UFD, and $f \in R[X]$ a primitive polynomial. Let F be the field of fractions of R . Then f is irreducible in $R[X]$ if and only if it's irreducible in $F[X]$.

Proof. ($\implies$): Suppose f is reducible in $R[X]$. Then, $f = gh$, where $g, h \in R[X]$ are both non-units. Note that f is primitive, so both g and h are primitive as well. If g has degree 0, it would be a unit (since it's primitive), which it's not. Thus, $\deg(g) > 0$, and for the same reason, $\deg(h) > 0$. Since $\deg(g), \deg(h) > 0$, g and h aren't units in $F[X]$. Thus, writing $f = gh$ and viewing f, g, h as lying in $F[X]$, we see f is reducible in $F[X]$.

($\impliedby$): Suppose f is now reducible in $F[X]$. Then, $f = g \cdot h$, where $g, h \in F[X]$ are non-units. This implies that $\deg(g), \deg(h) > 0$. We now clear denominators. That is,

we choose $a, b \in R \setminus \{0\}$ such that ag and bh lie in $R[X]$. Note that we can do this, for example by taking a to be the product of all denominators of coefficients of g and similarly for b . Now let $g' = ag$ and $h' = bh$. These both lie in $R[X]$. Since h may not be in $R[X]$, so writing $h' = bh$ isn't a factorization on $R[X]$, and similarly for $g' = ag$. Plugging in, we see

$$abf = g'h' \in R[X],$$

and all factors in this equation lie in $R[X]$. Thus, we can write $g' = c(g') \cdot g''$ and $h' = c(h') \cdot h''$, where $g'', h'' \in R[X]$ are both primitive. Since f is primitive, $c(abf) = ab$. However, by the above equation, this is equal to $c(g'h') = c(g')c(h')$ (up to associates). Therefore, ab is an associate of $c(g')c(h')$, i.e. $uab = c(g')c(h')$, where $u \in R^\times$ is some unit. Therefore,

$$abf = g'h' = c(g')c(h')g''h'' = uabg''h''.$$

Since $R[X]$ is an integral domain and $ab \neq 0$, we can cancel ab to get

$$f = u \cdot g'' \cdot h''.$$

Here, $u \in R^\times$ is a unit and $g'', h'' \in R[X]$ both lie in $R[X]$, and have positive degree and hence are non-units. Thus, f is reducible in $R[X]$. $\square$

This is useful because checking irreducibility in $F[X]$ is generally harder than checking it in $R[X]$. Here is an example to illustrate this.

Example 1.2.98

Let $f = 1 + X + X^3 \in \mathbb{Z}[X]$. Note that $c(f) = 1$, so f is primitive. Now suppose f is reducible in $\mathbb{Q}[X]$. Then by Gauss's lemma, f is reducible in $\mathbb{Z}[X]$. Thus, $f = g \cdot h$, where $g, h \in \mathbb{Z}[X]$ are non-units. Since g and h are non-units, they can't be constants since all coefficients of f are 0 and 1, so they would be either 0 or ± 1 .

$\deg(g) + \deg(h) = \deg(f) = 3$, so WLOG $\deg(g) = 1$ and $\deg(h) = 2$. Thus, we can write $g = b_0 + b_1X$ and $h = c_0 + c_1X + c_2X^2$, where $b_i, c_j \in \mathbb{Z}$. Since $f = g \cdot h$, the X^0 coefficient of f is

$$1 = b_0c_0$$

and the X^3 coefficient is

$$1 = b_1c_2.$$

Thus, since all b_i, c_j are integers, all of b_0, c_0, b_1, c_2 must be ± 1 . Particularly, $g = \pm 1 \pm X$. Since g is a factor of f , this implies ± 1 is a root of f . However, we can compute $f(1)$ and $f(-1)$, and they're both non-zero, which is a contradiction. Thus, f is irreducible in $\mathbb{Q}[X]$. Particularly, $\mathbb{Q}[X]/(f)$ is a field.

Note that when we concluded b_0 and b_1 are ± 1 , we really needed to be working in $\mathbb{Z}$. If we were in $\mathbb{Q}$ this would tell us a lot less. It was Gauss' lemma that allowed us to work over $\mathbb{Z}$ rather than $\mathbb{Q}$.

Theorem 1.2.99

Let R be a UFD. Then, $R[X]$ is a UFD.

Proof. We first prove existence of factorizations. Let $f \in R[X]$ be a nonzero and not a unit. We first write $f = c(f) \cdot f_1$, with $f_1 \in R[X]$ primitive. Since R is a UFD, we can factor the content as

$$c(f) = p_1 \cdots p_n,$$

where each $p_i \in R$ is irreducible in R (and so irreducible in $R[X]$). Next, we deal with f_1 . Suppose f_1 isn’t a product of irreducibles. Particularly, f_1 itself isn’t irreducible, so we can write $f_1 = f_2 g_2$, with $f_2, g_2 \in R[X]$ non-units. Also, since f_1 is primitive, f_2 and g_2 must also be primitive, and also non-constant. Thus, we must have $\deg(f_2), \deg(g_2) > 0$. Then, by $\deg(f_1) = \deg(f_2) + \deg(g_2)$, we must also have $\deg(f_2), \deg(g_2) < \deg(f_1)$. Now, if both f_2, g_2 were products of irreducibles, then f_1 would be too. Thus, we can assume WLOG that f_2 isn’t a product of irreducibles. We apply the same argument to f_2 to write $f_2 = f_3 \cdot g_3$ as a product of non-units. If we continue like this, we get a sequence $f_1, f_2, f_3, \dots \in R[X]$ of nonzero elements, with $\deg(f_1) > \deg(f_2) > \deg(f_3) > \dots > 0$. However, there can’t be an infinite sequence of positive integers that strictly decrease, so we have a contradiction. Thus, we can write $f_1 = q_1 \cdots q_m$, with each $q_i \in R[X]$ irreducible. So, $f = p_1 \cdots p_n \cdot q_1 \cdots q_m$, where all p_i are irreducible constant polynomials and all the q_j are irreducible nonconstant polynomials. Thus, factorization into irreducibles exist.

Next, we prove uniqueness of factorizations of a non-zero non-unit $f \in R[X]$. First, we deal with the content. We write $c(f) = p_1 \cdots p_n$ as before, a factorization of $c(f)$ into irreducibles in R (noting this is unique up to reordering and associates since R is a UFD). If we cancel the content out, we only need to show that primitive polynomials can be factored uniquely. Suppose $f' \in R[X]$ is primitive and that we are given two factorizations

$$f' = q_1 \cdots q_m = r_1 \cdots r_l,$$

where each $q_i, r_j \in R[X]$ is irreducible. Note that f' is primitive, so since each $c(q_i)$ and $c(r_j)$ is a factor of $c(f')$, which is a unit, each q_i and r_j must be primitive. Let F be the field of fractions of R , and consider $q_i, r_j \in F[X]$. By Gauss’s lemma since the q_i and r_j are irreducible in $R[X]$, they’re also irreducible in $F[X]$. Since $F[X]$ is a Euclidean domain, it’s a PID and hence a UFD, by uniqueness of factorization in $F[X]$, we find that $m = l$, and (after some reordering), r_i and q_i are associates in $F[X]$ for all i , i.e. $r_i = u_i q_i$ for some units $u_i \in F[X]$. Note that we want to prove the same thing but with each $u_i \in R[X]$ instead. Further note that each u_i must be a constant polynomial since it’s a unit. So, $u_i = \frac{a_i}{b_i}$, for some elements $a_i, b_i \in R$, which must both be non-zero. Now, $a_i r_i = b_i q_i$, and all factors in this equation are in $R[X]$. r_i and q_i are primitive, so taking contents of both sides, we find that a_i and b_i are associates. Therefore, we can write $b_i = v_i a_i$, for some $v_i \in R^\times$. Thus,

$$a_i r_i = v_i a_i q_i,$$

and cancelling a_i , since it’s nonzero, we find that $r_i = v_i q_i$. v_i is a unit in $R^\times$, so it’s also a unit in $R[X]^\times$. Thus, r_i and q_i are associates in $R[X]$ for each i , implying factorization into irreducibles are unique in the appropriate sense. $\square$

Remark 1.2.100 — Showing existence of factorizations was similar to showing (PID $\implies$ UFD): first we found factorizations $f_1 = f_2 g_2, f_2 = f_3 g_3, \dots$ into non-units, and then we showed this process must terminate; here we did this second step by showing that at each stage we decrease degrees, whereas earlier we did this using the Noetherian property.

To show uniqueness, we first took out out factors of the contents to turn everything into primitives, used Gauss' lemma to argue reducibility in $R[X]$ and $F[X]$ are the same (noting we're working with primitive polynomials at this stage), and used the fact that $F[X]$ is a UFD.

Corollary 1.2.101

Let R be a UFD. Then, $R[X_1, \dots, X_n]$ is a UFD, for $n \geq 1$.

Proof. We iterate the above proof. □

Remark 1.2.102 — In fact, the converse is true: $R[X]$ is UFD implies R is a UFD. However, the reason for this is not because R is a subring of $R[X]$. In fact, in general, a subring of a UFD need not to be a UFD. To show that $R[X]$ being a UFD implies that R is a UFD, we must consider 0-th degree polynomials and see that irreducibles in R are equivalent to irreducibles in $R[X]$ and prove uniqueness via UFD property of $R[X]$.

Proposition 1.2.103 (Generalized Eisenstein's Criterion)

Let R be a commutative ring,

$$f = a_0X^n + a_1X^{n-1} + \dots + a_n \in R[X],$$

a polynomial, and $P \subseteq R$ a prime ideal, such that $a_0 \notin P$, $a_i \in P$ for $0 < i \leq n$, and $a_n \notin P^2$. Then, if $(a_0, \dots, a_n) = 1$, f is irreducible.

Proof. Consider $f \bmod P$; that is,

$$\bar{f} = (a_0 + P)X^n + (a_1 + P)X^{n-1} + \dots + (a_n + P) \in (R/P)[X].$$

Then, P is prime so R/P is an integral domain. Now, for the sake of contradiction, suppose $f = gh$, for $g, h \in R[X]$ and write $\bar{g}, \bar{h}$ for the reduction mod P , respectively. Note that we have

$$\bar{f} = (a_0 + P)X^n,$$

so we must have that

$$\bar{g} = cX^m, \quad \bar{h} = dX^{n-m},$$

for some $c, d \in R$ (as R/P is an integral domain). Now, if $m, n-m \neq 0$, we have that the constant coefficients of $\bar{g}, \bar{h}$ must both be in P , implying that the constant coefficient of f must be in P^2 , which is a contradiction. Therefore, WLOG, assume that $m = 0$. Then, if $\deg(g)$ is nonzero, we get that the leading coefficient of g must be in P . Thus, the leading coefficient of f must also lie in P , which is a contradiction. Now, g is a constant polynomial, and as f is primitive, it follows that g must be a unit. □

Proposition 1.2.104 (Eisenstein's Criterion)

Let R be a UFD, and let $f = a_0 + a_1X + \cdots + a_nX^n \in R[X]$ be a primitive polynomial with $a_n \neq 0$. Let $p \in R$ be irreducible (and hence prime) such that

1. $p \nmid a_n$
2. $p \mid a_i$, for all $0 \leq i < n$
3. $p^2 \nmid a_0$.

Then, f is irreducible in $R[X]$ (And hence also in $F[X]$, by Gauss's lemma).

Proof. Follows by the above generalization to a UFD. □

Remark 1.2.105 — Eisenstein leads to the following criterion as well. Let p be prime, and given $f \in R[X]$, let $\bar{f} \in \mathbb{F}_p[X]$ be the reduction of f modulo p . Then, if $\bar{f}$ is irreducible then f is irreducible.

We show an example that illustrates the above.

Example 1.2.106

Show that $f(X) = X^6 + X^3 + 1 \in \mathbb{Q}[X]$ is irreducible.

Solution. Note that by Gauss's lemma, it suffices to show that $f(X)$ is irreducible in $\mathbb{Z}[X]$. Next, the reduction of $f(X)$ mod 2 provides the same function. However, we can check cases. $X^2 + X + 1$ is not a factor. Next, $(X^3 + X + 1)(X^3 + X^2 + 1) \neq X^6 + X^3 + 1$. Thus, this polynomial is irreducible. □

Example 1.2.107

Let $p \in \mathbb{N}$ is prime. Take $f(X) = X^p - T \in K[X]$, where K is the fraction field of $\mathbb{F}_p[T]$. This satisfies Eisenstein's criterion for $p = T$.

Example 1.2.108

Consider $f = X^n - p \in \mathbb{Z}[X]$, where $p \in \mathbb{N}$ is prime. Note all the conditions for Eisenstein's criterion hold. So f is irreducible in $\mathbb{Z}[X]$ and hence by Gauss' lemma also in $\mathbb{Q}[X]$. In particular, $X^n - p$ has no rational zeroes, i.e. $\sqrt[n]{p}$ is irrational for $n > 1$.

Example 1.2.109

Let $f = X^{p-1} + X^{p-2} + \cdots + 1 \in \mathbb{Z}[X]$, for $p \in \mathbb{N}$ prime. Note that

$$f = \frac{X^p - 1}{X - 1}.$$

Since all the coefficients are 1, Eisenstein's criterion can't be applied with any prime. Let $Y = X - 1$. Then $f(X) = \hat{f}(Y)$ where $\hat{f} \in \mathbb{Z}[Y]$ is given by

$$\hat{f}(Y) = \frac{(Y+1)^p - 1}{Y} = Y^{p-1} + \binom{p}{1}Y^{p-2} + \binom{p}{2}Y^{p-3} + \cdots + \binom{p}{p-1}.$$

Now Eisenstein's criterion can be applied with the prime p (note $\hat{f}$ is primitive since the top coefficient is 1, and $p \mid \binom{p}{i}$ for all $1 \leq i \leq p-1$, but $p \nmid \binom{p}{p-1}$). Thus $\hat{f} \in \mathbb{Z}[Y]$ is irreducible. If we had a factorization $f = gh$ in $\mathbb{Z}[X]$, then $\hat{f}(Y) = g(Y+1)h(Y+1)$. So $f \in \mathbb{Z}[X]$ is irreducible. In particular, none of the roots of f are rational. Note we already knew this: the roots are $e^{\frac{2\pi ik}{p}}$ for $1 \leq k \leq p-1$, which aren't even real for $p > 2$.

§1.2.iv Algebraic Integers

Definition 1.2.110. A complex number $\alpha \in \mathbb{C}$ is called an **algebraic integer** if it's the root of some monic polynomial $f \in \mathbb{Z}[X]$, i.e. $f(\alpha) = 0$.

Remark 1.2.111 — Note not all complex numbers can be algebraic integers: there are only countably many polynomials with integer coefficients, each of which only has finitely many roots, so the set of algebraic integers is countable. But $\mathbb{C}$ is uncountable.

Definition 1.2.112. For α an algebraic integer, we write $\mathbb{Z}[\alpha] \leq \mathbb{C}$ for the smallest subring containing α . More explicitly,

$$\mathbb{Z}[\alpha] := \bigcap_{S \leq \mathbb{C}, \alpha \in S} S,$$

where we take the intersection over all subrings $S \leq \mathbb{C}$ containing α . Equivalently, we can let $\phi_\alpha: \mathbb{Z}[X] \rightarrow \mathbb{C}$ be the homomorphism sending f to $f(\alpha)$, and let $\mathbb{Z}[\alpha] = \text{Im}(\phi_\alpha) \leq \mathbb{C}$.

Remark 1.2.113 — Note that by the first isomorphism theorem, we have

$$\mathbb{Z}[\alpha] \cong \mathbb{Z}[X] / \ker(\phi_\alpha)$$

and since α is an algebraic integer, $\ker(\phi_\alpha)$ is nonempty.

Proposition 1.2.114

Let $\alpha \in \mathbb{C}$ be an algebraic integer, and $\phi_\alpha: \mathbb{Z}[X] \rightarrow \mathbb{C}$ the homomorphism sending f to $f(\alpha)$. let $I = \ker(\phi_\alpha) \subseteq \mathbb{Z}[X]$. Then, the ideal I is principal, and $I = (f_\alpha)$, for some $f_\alpha \in \mathbb{Z}[X]$ which is irreducible and monic.

Proof. We know there is some monic $f \in \mathbb{Z}[X]$ such that $f(\alpha) = 0$, so $f \in I$ (so $I \neq 0$). Let $f_\alpha \in I$ be a non-zero polynomial of minimal degree in I . If f_α isn't primitive, we can write $f_\alpha = c(f_\alpha) \cdot f'_\alpha$ for some primitive $f'_\alpha \in I$, so we can in fact assume that we have chosen f_α to be primitive. We now claim that f_α generates I , f_α is irreducible, and f_α is monic (up to associates).

Let $h \in I$. By the Euclidean algorithm in $\mathbb{Q}[X]$, we can write

$$h = f_\alpha \cdot q + r,$$

for some $q, r \in \mathbb{Q}[X]$, with $r = 0$ or $\deg(r) < \deg(f_\alpha)$. We then clear denominators: that is, there is some $0 \neq a \in \mathbb{Z}$ such that $aq, ar \in \mathbb{Z}[X]$. Thus,

$$ah = f_\alpha \cdot (aq) + (ar).$$

Evaluating this at α shows that $ar(\alpha) = 0$, so $ar \in I$. Since $f_\alpha \in I$ has minimal degree among non-zero elements, we must have that $ar = 0$, so $r = 0$.

Thus, $ah = (aq) \cdot f_\alpha$ is a factorization in $\mathbb{Z}[X]$. But we wanted to factorize h , not ah . We take the contents:

$$ac(h) = c(ah) = c(f_\alpha) \cdot c(aq) = c(aq).$$

Note that the last equality is up to associates and is due to the fact that f_α is primitive. Thus, $a \mid c(aq)$, and so $aq = a\bar{q}$, for some $\bar{q} \in \mathbb{Z}[X]$. This gives $h = f_\alpha \cdot q \in I$, implying $I = (f_\alpha)$.

As $\mathbb{Z}[\alpha] \leq \mathbb{C}$, and $\mathbb{Z}[X]/I$ is an integral domain, it follows that I is a prime ideal, so f_α is prime and therefore irreducible. Since I contained some monic polynomial f and $f_\alpha \mid f$, the top coefficient of f_α must be a unit in $\mathbb{Z}$, i.e. ± 1 . If it's -1 , we take $-f_\alpha$. Either way, we may assume that f_α is monic. $\square$

As we require that f_α is monic, these conditions determine f_α uniquely.

Definition 1.2.115. We call the polynomial f_α as above the **minimal polynomial** of α .

Example 1.2.116

i is an algebraic integer with minimal polynomial $X^2 + 1$.

Example 1.2.117

$\sqrt{2}$ is an algebraic integer with minimal polynomial $X^2 - 2$.

Example 1.2.118

$\frac{1}{2}(1 + \sqrt{-3})$ is an algebraic integer with minimal polynomial $X^2 - X - 1$.

Lemma 1.2.119

Let $\alpha \in \mathbb{Q}$ be an algebraic integer. Then, $\alpha \in \mathbb{Z}$.

Proof. Let $f_\alpha \in \mathbb{Z}[X]$ be the minimal polynomial of α . Then, f_α is irreducible and primitive, so by Gauss's lemma, f_α is irreducible in $\mathbb{Q}[X]$. Since $f_\alpha(\alpha) = 0$ and $\alpha \in \mathbb{Q}$, $x - \alpha \mid f_\alpha$ in $\mathbb{Q}[X]$. Since f_α is irreducible and monic, $f_\alpha = X - \alpha$. However, since $f_\alpha \in \mathbb{Z}[X]$, we must have that $\alpha \in \mathbb{Z}$. $\square$

Remark 1.2.120 — It turns out that the set of algebraic integers is a subring of $\mathbb{C}$.

Proposition 1.2.121

Let $p \in \mathbb{N}$ be a prime number. Then, p is prime in $\mathbb{Z}[i]$ if and only if p can't be written as $a^2 + b^2$, for $0 \neq a, b \in \mathbb{Z}$.

Proof. ($\implies$): If $p = a^2 + b^2$, then $p = (a + ib)(a - ib)$ isn't irreducible, so it isn't prime.

($\impliedby$): Suppose p isn't irreducible in $\mathbb{Z}[i]$. Then $p = uv$ with $u, v \in \mathbb{Z}[i]$ non-unites. Taking norm squares, we find that $p^2 = |u|^2|v|^2$. As u, v aren't unites, we must have that their norm-squares are not equal to 1. Thus, $|u|^2 = p$. If $u = a + ib$, this says that $p = a^2 + b^2$, $\square$

Lemma 1.2.122

Let $p \in \mathbb{N}$ be a prime number, and let $\mathbb{F}_p \cong \mathbb{Z}/p$ be the field with p elements. Let $\mathbb{F}_p^\times = \mathbb{F}_p \setminus \{0\}$ be its group of units. Then, $\mathbb{F}_p^\times \cong C_{p-1}$ is cyclic of order $p - 1$.

2 Rings of Invariants

§2.1 Rings of Invariants

Remark 2.1.1 — All rings will be commutative and $k = \mathbb{C}$ (but any other k with characteristic 0 and k has sufficiently many roots of 1).

We consider $\mathbb{C}[x_1, \dots, x_n]$, the ring of polynomial functions on $\mathbb{C}^n$. $\mathbb{C}^n$ is a vector space, so GL_n acts on $\mathbb{C}^n$ and hence $\mathbb{C}[x_1, \dots, x_n]$. With this, we can consider a group G acting on a ring R : $G \times R \rightarrow R$ defined by $\gamma, f \mapsto \gamma f$, and $\gamma f: X \rightarrow \mathbb{C}$ such that for all $a \in X$, we have $\gamma f(a) = f(\gamma^{-1}a)$. Why do we make f act on the inverse of gamma? The reason is as follows: $(\gamma_1\gamma_2)f$ has the following mapping:

$$a \mapsto f((\gamma_1\gamma_2)^{-1}a) = f(\gamma_2^{-1}(\gamma_1^{-1}a)) = \gamma_2 f(\gamma_1^{-1}a) = \gamma_1(\gamma_2 f)(a).$$

Example 2.1.2

Consider $GL_2(\mathbb{C})$ acting on $\mathbb{C}[x, y]$. An element $\Gamma \in GL_2(\mathbb{C})$ is a matrix

$$\Gamma = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}.$$

It acts on (x, y) by Γ^{-1} . We have

$$\begin{aligned} \Gamma^{-1} \begin{pmatrix} a \\ b \end{pmatrix} &= \begin{pmatrix} 1 & \\ \alpha\delta - \beta\gamma & \end{pmatrix} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} \\ &= \begin{pmatrix} 1 & \\ \alpha\delta - \beta\gamma & \end{pmatrix} \begin{pmatrix} \delta a - \beta b \\ -\gamma a + \alpha b \end{pmatrix}. \end{aligned}$$

Therefore,

$$\begin{aligned} x &\mapsto \frac{1}{\alpha\delta - \beta\gamma}(\delta x - \beta y) \\ y &\mapsto \frac{1}{\alpha\delta - \beta\gamma}(-\gamma x + \alpha y) \end{aligned}$$

We now consider finite subgroups $G \subset GL_n(\mathbb{C})$ and the resulting action G on the ring $\mathbb{C}[x_1, \dots, x_n]$.

Remark 2.1.3 — We may denote that a group G acts on a ring R by a circle with an arrow inside G and R .

Definition 2.1.4. The **ring of invariants** is the following ring

$$\mathbb{C}[x_1, \dots, x_n]^G = \{f(x_1, \dots, x_n) \in \mathbb{C}[x_1, \dots, x_n] \mid \forall \gamma \in G, \gamma f = f\}.$$

Remark 2.1.5 — Note that $\mathbb{C}[x_1, \dots, x_n]^G$ is equal to the set of polynomials f_n on the quotient space $\mathbb{C}^n/G$.

Here is now an example of finite groups acting on $\mathbb{C}[x_1, \dots, x_n]$.

Example 2.1.6 (S_n acting on $\mathbb{C}[x_1, \dots, x_n]$)

If σ is a permutation on n letters, then

$$\sigma: f(x_1, \dots, x_n) \rightarrow f^\sigma = f(x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(n)}).$$

For instance σ_2 , with $f(x, y) \mapsto f(y, x)$, where $x \mapsto y$ and $y \mapsto x$.

As another example, we also have σ_3 that sends $x \mapsto y, y \mapsto x, z \mapsto z$ or $x \mapsto y, y \mapsto z, z \mapsto x$.

Here are now some examples of invariants.

Example 2.1.7 (Elementary Symmetric Functions)

Here are some examples of invariants:

$$\begin{aligned} x_1 + x_2 + \dots + x_n &= \sigma_1 \\ \sum_{i < j} x_i x_j &= \sigma_2 \\ &\vdots \\ \sum_{i_1 < i_2 < \dots < i_k} x_{i_1} x_{i_2} \dots x_{i_k} &= \sigma_k \\ &\vdots \\ x_1 x_2 \dots x_n &= \sigma_n \end{aligned}$$

Remark 2.1.8 — Note that these are the elementary ones because we can build more:

$$x_1^2 + x_2^2 + \dots + x_n^2 = \sigma_1^2 - 2\sigma_2$$

We ask now whether there are more invariants.

Theorem 2.1.9

$\mathbb{C}[x_1, \dots, x_n]^{\sigma_n}$ is generated by $1, \sigma_1, \sigma_2, \dots, \sigma_n$. In fact, there are no algebraic relations between the σ_i s. That is, there is no ring homomorphism

$$\begin{aligned} \Phi: \mathbb{C}[y_1, \dots, y_n] &\rightarrow \mathbb{C}[x_1, \dots, x_n]^{\sigma_n} \\ \Phi: y_i &\mapsto \sigma_i \end{aligned}$$

Proof.

□

Example 2.1.10 (Cyclic Group Actions)

Let $\mu_n = \{\zeta \mid \zeta^n = 1\} \subset \mathbb{C}^\times$ cyclic generated by $\zeta = e^{\frac{2\pi i}{n}}$. We consider the action of μ_n on $\mathbb{C}[x, y]$, where $\zeta: x \mapsto \zeta x$ and $y \mapsto \zeta^{-1}y$. We now want to know invariants of this action.

Some invariants are $u = xy$, $v = x^n$, and $w = y^n$. It turns out that $\Phi: \mathbb{C}[u, v, w] \rightarrow \mathbb{C}[x, y]^{\mu_n}$ with

$$u \mapsto x^n$$

$$v \mapsto y^n$$

$$w \mapsto xy$$

is surjective. This is a finitely generated $\mathbb{C}$ -algebra. It turns out that $\ker(\Phi) = (uv - w^n)$.

Example 2.1.11

Again, consider the action μ_n on $\mathbb{C}[x, y]$ such that ζ sends

$$x \mapsto \zeta x$$

$$y \mapsto \zeta y.$$

It turns out that $\mathbb{C}[x, y]^{\mu_n}$ is generated by the homogeneous polynomials of degree n : $x^n, x^{n-1}y, \dots, y^n$ (with lots of relations). If we call $u_0 = x^n, u_1 = x^{n-1}y, \dots, u_n = y^n$, then write the following matrix.

$$\begin{pmatrix} u_0 & u_1 & \dots & u_{n-1} \\ u_1 & u_2 & \dots & u_n \end{pmatrix}.$$

It turns out that the ideal of relations is generated by the 2×2 minors of the matrix above.

Example 2.1.12

We can also consider the action μ_n on $\mathbb{C}[x, y]$ such that ζ sends

$$x \mapsto \zeta x$$

$$y \mapsto \zeta^q y,$$

where $q \leq n$.

Example 2.1.13

Let $G = BD_{4n} \subset SL_2(\mathbb{C})$, the binary dihedral group. This group is

$$\left\langle \mu_{2n} = \left\{ \begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix} \mid \zeta^{2n} = 1 \right\}, B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right\rangle.$$

We consider the action of G on $\mathbb{C}[x, y]$. For $\zeta \in \mu_{2n}$, we have $x \mapsto \zeta x, y \mapsto \zeta y$, and

$$B: x \mapsto y, \quad y \mapsto -x.$$

We look for invariants of $R = \mathbb{C}[x, y]^G$.

The ring of μ_{2n} -invariants is generated by $u = x^{2n}, v = y^{2n}, w = xy$. Then, BD_{4n} -invariants are to all be found $\mathbb{C}[u, v, w]$. Then,

$$B: u \mapsto v$$

$$B: v \mapsto u$$

$$B: w \mapsto -w.$$

$w^2 =: Z, u + v =: Y, w(u - v) =: X$ are invariants. It turns out that X, Y, Z generate the ring of invariants. We now search for a relation for X, Y, Z . We have $X^2 = Z(Y^2 - 4Z^n)$, noting that $uv = w^{2n}$. It turns out that this relation generates the ideal of relations:

$$\mathbb{C}[X, Y, Z] / \underbrace{X^2 + Y^2 Z + Z^{n+1}}_{=D_{n+2} \text{ singularity}}$$

Remark 2.1.14 — Note that the two claims in the last paragraph is something quite big that might not be covered.

Remark 2.1.15 — There is the classification of finite subgroups $G \leq SL_2(\mathbb{C})$. Given the following, we have the binary tetrahedral, binary octahedral, binary icosahedral groups.

1. $\mu_n \subset SL_2(\mathbb{C})$ as $\begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix}$.
2. BD_{4n} , as in the example above.

§2.2 Emmy Noether's Theorem

Definition 2.2.1. A ring R is **graded** if

$$R = \bigoplus_{n \geq 0} R_n,$$

such that $R_n R_m \subseteq R_{n+m}$ as additive groups. Here, R_n consists of elements that are **homogeneous** of degree n .

Example 2.2.2

$R = \mathbb{C}[x_1, \dots, x_n]$ is a graded ring. This is obvious.

Definition 2.2.3. Let k be a field. R is a **finitely generated k -algebra** if

1. $k \subset R$
2. $\exists a_1, \dots, a_m \in R$ such that the ring homomorphism $\phi: k[y_1, \dots, y_n] \rightarrow R$ characterized by $\phi(y_i) = a_i$, for all i , is surjective. Equivalently, $R \cong k[y_1, \dots, y_n]/I$, for any ideal $I \triangleleft k[y_1, \dots, y_n]$.

Theorem 2.2.4 (Emmy Noether's Theorem)

Consider the action $G \leq GL_n(\mathbb{C})$ on $R = \mathbb{C}[x_1, \dots, x_n]$. The ring of invariants R^G is a finitely generated $\mathbb{C}$ -algebra.

Proof. First note that the ring of polynomials is a graded ring, and G preserves the grading. $\forall n \geq 0, \forall \gamma \in G, \forall f \in R_n$, we have $\gamma f \in R_n$. Therefore, R^G is graded too ($= \bigoplus_n R_n$).

Secondly, note that there is an operation, which we call the **averaging over G** , $\rho: R \rightarrow R^G$, where

$$\rho(f) = \frac{1}{|G|} \sum_{\gamma \in G} \gamma f.$$

First, we have that $\rho(f) = f \iff f \in R^G$. Then, we can get

- ρ is additive: $\rho(f + g) = \rho(f) + \rho(g)$.
- ρ is R^G -linear: $f \in R^G$ and $g \in R$, then $\rho(fg) = f\rho(g)$.
- If $f \in R_n$, then $\rho(f) \in R_n$.

We're now ready for the proof. Let I be the ideal of R generated by R_+^G . Here,

$$R_+ = \bigoplus_{d>0} R_d.$$

Then, by the Hilbert basis theorem, there exists $a_1, \dots, a_m \in R$ such that the ideal $I = (a_1, \dots, a_m)$. These a_i are not necessarily invariant. We now want a set of generators that are invariant. For all i , there is a linear combination

$$a_i = \lambda_{i_1} r_{i_1} + \lambda_{i_2} r_{i_2} + \dots + \lambda_{i_{k(i)}} r_{i_{k(i)}},$$

where $r_{i_j} \in R^G$. The ideal I is generated by the (finitely many) r_{i_j} . By splitting r_{i_j} into necessarily invariant homogeneous components, we can say that there exists homogeneous invariants $r_1, r_2, \dots, r_N$ and $I = (r_1, r_2, \dots, r_N)$. We now go on to claim that $r_1, \dots, r_N$ generate R^G . Let S be the ring generated by the r_j s, so $S \subset R^G$. We will now show, via induction, that $R_n^G \subset S$. For $n = 0$, since the constants are in S by definition of S . Thus, assume $R_0^G \oplus R_1^G \oplus \dots \oplus R_{n-1}^G \subset S$. Let $f \in R_n^G$ (and so invariant of degree n). In particular, $f \in I$, so $\exists a_i \in R$ that are homogeneous so that

$$f = a_1 r_1 + a_2 r_2 + \dots + a_N r_N.$$

If the a_i s are invariant, we are finished, then we’re done since r_j s are invariant (and since $\deg a_i < n$), since we can then apply the inductive hypothesis. For this, we take the average:

$$f = \rho(f) = \rho(a_1)r_1 + \rho(a_2)r_2 + \cdots + \rho(a_N)r_N.$$

Then, for all i , $b_i = \rho(a_i) \in R_N^G$, so we’re done. $\square$

Remark 2.2.5 — The averaging function ρ , may also be called the **Reynold’s operator**.

Remark 2.2.6 — Not all subrings $S \subset \mathbb{C}[x_1, \dots, x_n]$ are finitely generated.

Example 2.2.7

Let $R = \mathbb{C}[x_1, x_2]$. Consider the monoid generated by $x_1^{n_1}x_2^{n_2}$, where $a = 0$. In other words, this can be the set $P = \{(n_1, n_2) \in \mathbb{N}^2 \mid n_2 > 0\} \cup \{(0, 0)\}$ is a monoid.

We now give an alternative proof of Emmy Noether’s theorem, which is Emmy Noether’s original proof of the theorem:

Alternative Proof of Emmy Noether’s Theorem. We work in $R[T]$, where T is an additional variable. For all $f \in R$, define $P^f(T) = \prod_{\gamma \in G} (T - \gamma f)$. This is a polynomial of degree $N = |G|$ and coefficients that are in R^G . We apply this for $f = x_i$:

$$P^{x_i}(T) = \prod_{\gamma \in G} (T - \gamma x_i) =: T^N + a_{N-1}^i T^{N-1} + \cdots + a_0^i,$$

where $a_j^i \in R^G$. We denote by

$$S = \mathbb{C} [a_j^i \mid i = 1, \dots, n, 1 \leq j \leq |G|] \subset R^G.$$

This is a subring. We note that $\deg(a_j^i) \leq N$.

Now, for a monomial $x^r = x_1^{r_1}x_2^{r_2} \cdots x_n^{r_n}$, define $|r| = \max(r_1, r_2, \dots, r_n)$. Then, we claim that

$$\hat{S} = \left\{ \sum f_r x^r \mid f_r \in S, |r| < N \right\} = R.$$

Note that the x^r , with $|r| < N$ generate R as a module over S . We now prove the claim. The key point is that for all r , $x^r \in \hat{S}$. If $|r| < N$, then $x^r \in \hat{S}$ by definition. Thus, we consider when $|r| \geq N$. For this, note that x_i is a root of the polynomial $P^{x_i}(T)$, since γ can equal to $I \in G$. Thus,

$$0 = P^{x_i}(x_i) = x_i^N + a_{N-1}^i x_i^{N-1} + \cdots + a_0^i = 0.$$

We can use this expression to reduce the degree so that $|r|$ eventually becomes less than N .

Now, we claim that the ring of invariants R^G is generated as a ring by S . Hence, ultimately by the a_j^i and

$$\{\rho(x^r) \mid |r| < N\}.$$

Note that the maximum degree of the above is $\deg x_1^{N-1} \dots x_n^{N-1} = n(|G| - 1)$. Let $f \in R^G$. By the previous claim, we can write

$$f = \sum_{\substack{|r| < N \\ s_r \in S}} s_r x^r.$$

We now apply the averaging operator ρ :

$$f = \rho(f) = \sum s_r \rho(x^r),$$

so we've proven the claim. $\square$

Remark 2.2.8 — We can see that there is a bound on the maximum degree of a set of invariant generators, from Emmy Noether's original proof. The bound is the one we see at the end: $\sim n(|G| - 1)$.

Exercise 2.2.9. Use the above theorem to prove the statements about μ_n -invariants, BD_{4n} -invariants.

Remark 2.2.10 — Symmetric polynomials will be dealt with differently.

Theorem 2.2.11

Let S_n act on $\mathbb{C}[x_1, \dots, x_n]$. Then,

1. $R^G = \mathbb{C}[e_1, \dots, e_n]$, where for all j , $e_j(x_1, \dots, x_n)$ is the j th elementary symmetric polynomial

$$\sum_{0 \leq k_1 < k_2 < \dots < k_j \leq n} x_{k_1} x_{k_2} \dots x_{k_j}.$$

2. The homomorphism $\phi: \mathbb{C}[y_1, \dots, y_n] \rightarrow R^G$ defined by $y_j \mapsto e_j$, for all j , is an isomorphism.

Proof. We first define order on monomials lexicographically. Suppose $x^r = x_1^{r_1} \dots x_n^{r_n} > x^s = x_1^{s_1} \dots x_n^{s_n}$. This happens if $r_1 > s_1$ or $r_1 = s_1$. In the latter, we must have $r_2 > s_2$ or $r_1 = s_1, r_2 = s_2$. We can then continue by considering the equality case and repeating. For $f \in R$, we define $\text{hm}(f)$ to be the highest monomial that appears in f . Using the example below, we have for $f \in R^G$, $\exists k_1, \dots, k_n$ such that

$$\text{hm}(f) = \text{hm}(e_1^{k_1} \dots e_n^{k_n}).$$

$\square$

Remark 2.2.12 — Note that we can also write the elementary symmetric polynomials as

$$\prod_{i=1}^n (T - x_i) = T^n - e_1 T^{n-1} + e_2 T^{n-2} - \dots \pm e_n$$

Example 2.2.13

We consider $\text{hm}(e_1^{k_1} e_2^{k_2} \dots e_n^{k_n})$. We have

$$e_1^{k_1} e_2^{k_2} \dots e_n^{k_n} = (x_1 + x_2 + \dots)^{k_1} (x_1 x_2 + x_1 x_3 + \dots)^{k_2} \dots (x_1 \dots x_n)^{k_n}.$$

Therefore,

$$\text{hm}(e_1^{k_1} e_2^{k_2} \dots e_n^{k_n}) = x_1^{k_1 + \dots + k_n} x_2^{k_2 + \dots + k_n} \dots x_n^{k_n}.$$

Thus, if $f \in R^G$, then

$$\text{hm}(f) = x_1^{m_1} x_2^{m_2} \dots x_n^{m_n}$$

where $m_1 \geq m_2 \geq \dots \geq m_n$.

3 Modules

Remark 3.0.1 — In this section, we will let R be a ring that is not necessarily commutative.

Definition 3.0.2. An R -**module** is a set M with functions $+: M \times M \rightarrow M$ and $\cdot: R \times M \rightarrow M$, along with an element $0 = 0_M \in M$, such that $(M, +)$ is an abelian group with $0 = 0_M$, and the following hold:

- $(r + r') \cdot m = r \cdot m + r' \cdot m$
- $r \cdot (m + m') = r \cdot m + r \cdot m'$
- $r \cdot (r' \cdot m) = (r \cdot r') \cdot m$
- $1_R \cdot m = m$

Note that $r, r' \in R$ and $m, m' \in M$.

Remark 3.0.3 — We always multiply elements of M by elements of R on the left (like ideals or cosets). Because of this, what we call a module is sometimes called a *left* module. We can consider the notion of *right* modules, but they're essentially the same.

Definition 3.0.4. An R -module is an abelian group M equipped with a ring homomorphism $\phi: R \rightarrow \text{End}_{\mathbb{Z}}(M)$.

Abuse of Notation 3.0.5. Given such a ϕ , and $r \in R$ and $m \in M$, we write $r \cdot m$ for $(\phi(r))(m)$.

Proposition 3.0.6

These two definitions of an R -module are equivalent.

Proof Sketch. We think of an R -modules as ' R acting on an abelian group', just like how groups can act on sets. $\square$

Example 3.0.7

An abelian group is the same thing as a $\mathbb{Z}$ -module. If $m \in \mathbb{Z}$ and $a \in A$, then

$$m \cdot a = \underbrace{(a + \cdots + a)}_{m \text{ times}},$$

for $m \geq 0$. If $m < 0$, then we have

$$m \cdot a = -\underbrace{(a + \cdots + a)}_{m \text{ times}}.$$

Example 3.0.8

$R^n = \{(r_1, \dots, r_n) \mid r_i \in R, \forall i\}$ is an R -module.

Example 3.0.9

A left ideal $I \subset R$ is an R -module. Further, $R/I = \{r + I \mid r \in R\}$ is an R -module.

Example 3.0.10

Given a field F , an F -vector space is the same thing as an F -module (the axioms for both are exactly the same).

§3.1 Constructions

Definition 3.1.1. Let $M_1, \dots, M_k$ be R -modules. Their **direct sum**, written $M_1 \oplus \dots \oplus M_k$ is the abelian group $M_1 \times \dots \times M_k$, with

$$r(m_1, \dots, m_k) := (r \cdot m_1, \dots, r \cdot m_k).$$

Definition 3.1.2. A subset $N \subseteq M$ is an (R) -**submodule** if it's a subgroup of M and if $r \in R$ and $n \in N$, then $r \cdot n \in N$. In this case, we write $N \leq M$.

Remark 3.1.3 — A subset $I \subseteq R$ is a submodule if and only if I is an ideal.

Example 3.1.4

$\{0\}, M \leq M$ are always submodules.

Example 3.1.5

If F is a field, submodules and sub-vector spaces are the same, since the axioms are the same.

Example 3.1.6

Let k be a field. Then, k^n is an $M_n(k)$ -module. More generally, $M_{n \times m}(k)$ is a left $M_n(k)$ -module and $M_{m \times n}(k)$ is a right $M_n(k)$ -module

Definition 3.1.7. Let $N \leq M$ be a submodule. The **quotient module** M/N is the abelian group M/N , with R action given by

$$r \cdot (m + N) := (rm) + N.$$

Remark 3.1.8 — Groups have both subgroups and normal subgroups. Rings have subrings and ideals. However, modules just have submodules.

Definition 3.1.9. A function $\phi: M \rightarrow N$ between R -modules is an **R -module homomorphism** if it's a homomorphism of abelian groups, and $\phi(r \cdot m) = r \cdot \phi(m)$ for all $r \in R$ and $m \in M$. An isomorphism is once again a bijective homomorphism.

Definition 3.1.10. Let $\phi: M \rightarrow N$ be an R -module homomorphism. Then,

- The **image** $\phi(M) = \text{Im}(\phi) \subseteq N$ is a submodule.
- The **kernel** $\ker(\phi) = \{m \in M \mid \phi(m) = 0\} \subset M$ is a submodule.
- The **cokernel** $N/\text{Im}(\phi) := \text{coker}(\phi)$.

Remark 3.1.11 — The *world* of modules, with homomorphisms $\text{Im}, \ker, \text{coker}, \oplus$, is an abelian **category**. The reason why we say this is because modules provide for more *flexibility* than ideals and rings.

Remark 3.1.12 — If M is a module and $N \leq M$ is a submodule and if we have the quotient M/N , we may think of this as an exact sequence:

$$(0) \longrightarrow N \longrightarrow M \longrightarrow M/N \longrightarrow (0)$$

Example 3.1.13

The ideals $(2), (3) \subseteq \mathbb{Z}$ are different, but they're isomorphic as R -modules, via the isomorphism $\phi: (2) \rightarrow (3)$ sending $2k$ to $3k$.

In fact, if R is an integral domain and $0 \neq r \in R$, then R and (r) are always isomorphic as R -modules.

Definition 3.1.14. Let R, R' be rings, M an R -module, and M' an R' -module. Then $M \times M'$ is an $R \times R'$ -module, with R -action

$$(r, r') \cdot (m, m') = (r \cdot m, r' \cdot m').$$

§3.1.i The Universal Property of Direct Sums

We have the following theorem for direct sums, which we won't prove here.

Theorem 3.1.15 (The Universal Property of Direct Sums)

Let R be a ring, N be an R -module, and $\{M_i\}_{i \in I}$ be a family of R -modules. Further, let $M = \bigoplus_{i \in I} M_i$ be the direct sum, and $\psi_i: M_i \rightarrow N$ for each i . Then, there exists a unique morphism $\Phi: M \rightarrow N$ such that for all i , $\psi_i = \Phi \circ \phi_i$, where $\phi_i: M_i \rightarrow M$ is the i -th projection.

$$\begin{array}{ccc} M = \bigoplus_i M_i & & \\ \phi_i \uparrow & \searrow \Phi & \\ M_i & \xrightarrow{\psi_i} & N \end{array}$$

§3.2 Some Theorems

Note that some of the first results have proofs analogous to that of ring theory.

Theorem 3.2.1 (First Isomorphism Theorem)

Let $\phi: M \rightarrow N$ be an R -module homomorphism. Then:

1. $\ker(\phi) \leq M$ is an R -submodule.
2. $\text{Im}(\phi) \leq N$ is an R -submodule.
3. There is an isomorphism of R -modules $M/\ker(\phi) \cong \text{Im}(\phi)$.

Theorem 3.2.2 (Second Isomorphism Theorem)

Let $A, B \leq M$ be submodules.

1. $A + B := \{a + b \mid a \in A, b \in B\} \leq M$ is a sub-module.
2. $A \cap B \leq M$ is a sub-module.
3. There is an isomorphism of R -modules $(A + B)/A \cong B/(A \cap B)$.

Remark 3.2.3 — In general, if $\{A_i\}_{i \in S}$ is a collection of submodules, then

$$\sum_{i \in S} A_i \leq M$$

is a submodule, where the LHS is defined to be the set of finite sums of elements of the A_i . Similarly,

$$\bigcap_{i \in S} A_i \leq M$$

is a submodule.

Theorem 3.2.4 (Third Isomorphism Theorem)

Let $N \leq L \leq M$ be submodules. Then

1. $L/N \leq M/N$ is a submodule.
2. There is an isomorphism of R -modules $M/L \cong (M/N)/(L/N)$.

With this, we have the following proposition:

Proposition 3.2.5

Let $N \leq M$ be a submodule. There is a natural bijection

$$\{\text{submodules of } M/N\} \leftrightarrow \{\text{submodules of } M \text{ containing } N\}$$

Proposition 3.2.6

The composition of two R -modules homomorphisms is an R -module homomorphism.

Definition 3.2.7. Let $m \in M$. The submodule **generated** by m , $Rm \leq M$, is $\{r \cdot m \mid r \in R\}$. Alternatively, there is a homomorphism $\phi_m: R \rightarrow M$ sending r to $r \cdot m$, and $R \cdot m := \text{Im}(\phi_m)$.

Also note that $\ker(\phi_m) \leq R$ is a submodule. We define the **annihilator** of m , written $\text{Ann}(m)$, to be

$$\text{Ann}(m) := \{r \in R \mid r \cdot m = 0\} = \ker(\phi_m).$$

Remark 3.2.8 — Note that by the first isomorphism theorem, $R \cdot m \cong R/\text{Ann}(m)$.

Definition 3.2.9. An R -module M is **finitely generated** if there are elements $m_1, \dots, m_n \in M$, such that

$$M = Rm_1 + \dots + Rm_n = \{r_1m_1 + \dots + r_nm_n \mid r_i \in R\}$$

Remark 3.2.10 — In the course, we only do one *serious* thing with modules (with commutative rings): The classification of finitely generated modules over commutative PIDs.

Example 3.2.11

If F is a field, and M an F -module, then M is finitely generated (as an F -module) if and only if it's finite-dimensional (as an F -vector space).

Lemma 3.2.12

An R -module M is finitely generated if and only if there's a surjective homomorphism $R^n \rightarrow M$.

Proof. ($\implies$): If $M = Rm_1 + \dots + Rm_n$, let $\phi: R^n \rightarrow M$ be the homomorphism sending $(r_1, \dots, r_n)$ to $r_1m_1 + \dots + r_nm_n$. It's surjective by assumption.

($\impliedby$): Suppose $\phi: R^n \rightarrow M$ is a surjective homomorphism. Take $m_1, \dots, m_n$ to be the images under ϕ of the standard basis vectors. Then $M = Rm_1 + \dots + Rm_n$. $\square$

Corollary 3.2.13

If $N \leq M$ is a submodule and M is finitely generated, then M/N is finitely generated too.

Proof. Let $\phi: R^n \rightarrow M$ be a surjective homomorphism. Note the quotient $M \rightarrow M/N$ is also surjective. So the composition $R^n \rightarrow M \rightarrow M/N$ is also surjective. $\square$

Remark 3.2.14 — Note that if $N \leq M$ is a submodule and M is finitely generated, then N isn’t necessarily finitely generated, as can be seen in the following example.

Example 3.2.15

Let $R = \mathbb{Z}[X_1, X_2, \dots]$, and let I be the ideal $I = (X_1, X_2, \dots) \subseteq R$. Then R is a finitely generated R -module but $I \leq R$ is a submodule which isn’t finitely generated.

Proof. Suppose $I = (f_1, \dots, f_k)$ is finitely generated. Let $p \in \mathbb{N}$ be the largest number such that X_p appears in any of the f_i . Then, $X_{p+1} \in I$ but $X_{p+1} \notin (f_1, \dots, f_k)$, a contradiction. $\square$

Example 3.2.16

Here is an example of a finitely generated module with a finitely generated submodule. Let $R = K[X, Y]$ and the ideal $I = (X, Y)$. Then, R is finitely generated (by 1) and I is finitely generated.

§3.3 Free Modules

Definition 3.3.1. Let S be a set. The **free module** over S , written $R^{(S)}$, is

$$R^{(S)} = \oplus_{i \in S} R := \{(x_i)_{i \in S} \mid x_i \in R, x_i = 0 \text{ for all but finitely many } i\},$$

with coordinate-wise addition and R -action.

Proposition 3.3.2

Let R be a ring, and S a set. Then $R^{(S)}$ is finitely generated if and only if S is finite.

Proof. ($\Leftarrow$): If $|S| = n$, then $R^{(S)} \cong R^n$, so there is a surjective homomorphism $R^n \rightarrow R^{(S)}$.

($\Rightarrow$): Suppose S is infinite and $R^{(S)} = Rm_1 + \dots + Rm_n$. Write each $m_k = (x_i^{(k)})_{i \in S} \in R^{(S)}$. Define the subset $T \subseteq S$ by

$$T := \{i \in S \mid x_i^{(k)} \neq 0 \text{ for some } 1 \leq k \leq n\}.$$

Note T is a finite set. However S is infinite, so we can find some $s \in S \setminus T$. Let $a = (a_i)_{i \in S} \in R^{(S)}$ be the element with

$$a_i := \begin{cases} 1 & \text{if } i = s \\ 0 & \text{if } i \neq s \end{cases}.$$

Then $a \notin Rm_1 + \dots + Rm_n$, a contradiction. $\square$

Theorem 3.3.3

Let F be a field. Any F -module is free.

Remark 3.3.4 — This is equivalent to the axiom of choice. We can give an alternative characterization of freeness.

Definition 3.3.5. A subset $S \subseteq M$ generates M **freely** if the following hold:

1. S generates M as an R -module, i.e. $R \cdot S = M$ (where $R \cdot S$ is defined to be the set of finite sums of elements of the form rs , with $r \in R$ and $s \in S$).
2. For any other R -module N , any function $\psi: S \rightarrow N$ extends to an R -module homomorphism $\phi: M \rightarrow N$.

Remark 3.3.6 — Suppose $S \subseteq M$ generates M freely, and suppose $\phi, \phi': M \rightarrow N$ are two such extensions. Then $\phi - \phi': M \rightarrow N$ is a homomorphism, sending all of S to 0. Then, $S \subseteq \ker(\phi - \phi') \subseteq M$. But since S generates M , this implies $M \subseteq \ker(\phi - \phi')$. This implies that $\ker(\phi - \phi') = M$. This implies that $\phi = \phi' = 0$, so the extension ϕ is unique.

Definition 3.3.7. An R -module M is **free** if it's freely generated by some subset $S \subseteq M$. Such a subset $S \subseteq M$ is called a **basis** for M .

Proposition 3.3.8

These two definitions of free module are equivalent.

Proof. First suppose $M \cong R^{(S)}$. We view S as a subset in M by identifying $s \in S$ with the element $(x_i^s)_{i \in S}$ defined by

$$x_i^s = \begin{cases} 1 & \text{if } i = s \\ 0 & \text{otherwise.} \end{cases}$$

This subset generates M as an R -module. Let $\psi: S \rightarrow N$ be any function, for N some other R -module. Define a homomorphism $\phi: M \rightarrow N$ to send $(x_i)_{i \in S}$ to

$$\phi((x_i)_{i \in S}) := \sum_{i \in S} x_i \cdot \psi(i) \in N.$$

This is a homomorphism extending ψ , so S generates M freely.

Conversely, now suppose some subset $S \subseteq M$ freely generates M . We claim $M \cong R^{(S)}$. Let $\psi: S \rightarrow R^{(S)}$ send $s \in S$ to the element $(x_i)_{i \in S}$ defined above. This extends to a homomorphism $\phi: M \rightarrow R^{(S)}$. This is an isomorphism, with inverse sending $(x_i)_{i \in S}$ to $\sum_{i \in S} x_i \cdot i$. $\square$

Definition 3.3.9. A set of elements $m_1, \dots, m_n \in M$ are **linearly independent** if whenever $r_1 \cdot m_1 + \dots + r_n \cdot m_n = 0$ for $r_i \in R$, we have $r_1 = \dots = r_n = 0$.

Abuse of Notation 3.3.10. We may just write LI to say linearly independent.

Proposition 3.3.11

For a subset $S = \{m_1, \dots, m_n\} \subseteq M$, the following are equivalent

1. S generates M freely (equivalently, M is free).
2. S generates M and S is linearly independent.
3. Every element of M is uniquely expressible as $r_1 \cdot m_1 + \dots + r_n \cdot m_n$, for $r_i \in R$, $m_i \in M$

Proof. (2 $\iff$ 3): This follows from an analogous argument to linear algebra.

(1 $\iff$ 2): Let S generate M freely. If S is not independent, then we can write $r_1 m_1 + \dots + r_k m_k = 0$, with $r_i \in R$ and, say, $r_1 \neq 0$. We define the set function $\psi: S \rightarrow R$ by sending $m_1 \mapsto 1_R$ and $m_i \mapsto 0$ for all $i \neq 1$. As S generates M freely, this extends to an R -module homomorphism $\theta: M \rightarrow R$. But then, we get that

$$0 = \theta(r_1 m_1 + \dots + r_k m_k) = r_1,$$

so we reach a contradiction. For the converse, suppose every element can be uniquely written as $r_1 m_1 + \dots + r_k m_k$. Given any set function $\psi: S \rightarrow N$, we define $\theta: M \rightarrow N$ by

$$\theta(r_1 m_1 + \dots + r_k m_k) = r_1 \psi(m_1) + \dots + r_k \psi(m_k).$$

This is well-defined by uniqueness, and is clearly a homomorphism. So it follows that S generates M freely. $\square$

Example 3.3.12

Consider the $\mathbb{Z}$ -module $\mathbb{Z}/2\mathbb{Z}$. This isn’t free: if it were freely generated by $S \subseteq \mathbb{Z}/2\mathbb{Z}$, we would have $S = \{1\}$. Then, this implies there is a homomorphism $\theta: \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}$ sending 1 to 1. But it does not send $0 = 1 + 1$ to $1 + 1$, since homomorphisms send 0 to 0. So $\mathbb{Z}/2\mathbb{Z}$ is not freely generated.

Example 3.3.13

The subset $\{2, 3\} \subseteq \mathbb{Z}$ generates $\mathbb{Z}$, but not freely, as it’s not linearly independent: $3 \cdot 2 + (-2) \cdot 3 = 0$.

Recall that if S spans an F -vector space V and S isn’t linearly independent, we can *throw away* some of S and find some basis $T \subseteq S$. However no subset of $\{2, 3\}$ freely generates $\mathbb{Z}$.

Definition 3.3.14. Let M be a finitely generated R -module. There is some surjective homomorphism $\phi: R^n \rightarrow M$. We call the submodule $\ker(\phi) \leq R^n$ the **relation module** for these generators.

Definition 3.3.15. A finitely generated R -module M is **finitely presented** if there’s a surjective homomorphism $R^n \rightarrow M$, with finitely generated kernel. In other words, this means M is isomorphic to the cokernel of some homomorphism $\phi: R^n \rightarrow R^m$ for some n, m .

Remark 3.3.16 — Finitely presented modules are finitely generated, by definition. However, the converse isn't always true.

Proposition 3.3.17

Let $\phi: R^n \rightarrow R^m$ be an R -module homomorphism. Let $u_1, \dots, u_m \in R^m$ and $v_1, \dots, v_n \in R^n$ be the standard bases. Then we can write $\phi(u_j) = \sum_{i=1}^n A_{ij}v_i$, for some matrix $A \in M_{m \times n}(R)$, i.e. $\phi(r) = r \cdot A$, so ϕ is given by multiplying by some matrix on the right. If R is commutative, we could similarly write $\phi(r) = B \cdot r$, i.e. we could have represented ϕ by multiplying by some matrix on the left. This provides a bijection

$$\begin{aligned} M_{m \times n}(R) &\leftrightarrow \{R\text{-module homomorphisms } R^m \rightarrow R^n\} \\ A &\mapsto (r \mapsto r \cdot A) \end{aligned}$$

Recall from linear algebra, that if G is a field and $F^n \cong F^m$, then $n = m$. The same is true for any commutative ring:

Theorem 3.3.18

Let R be a commutative ring. Then, $R^n \cong R^m$ as R -modules if and only if $n = m$.

Proof Sketch. To prove this, we use the idea that any commutative ring contains a maximal ideal (theorem 3.3.20) to reduce the theorem to the case of a field.

Let $I \subseteq R$ be a maximal ideal, and let $F = R/I$ be a field. Then, we're done. For M an R -module, we define IM to be the submodule of M generated by elements $i \cdot m$, for $i \in I$ and $m \in M$:

$$IM: \{i_1 \cdot m_1 + \dots + i_k \cdot m_k \mid i_i \in I, m_i \in M\} \leq M.$$

Then M/IM is an R -module. In fact, M/IM is an R/I , which is equivalent to an F -module: for $r + I \in F$ and $m + IM \in M/IM$, we define

$$(r + I) \cdot (m + IM) := (r \cdot m) + IM.$$

If $R^n \cong R^m$ as R -modules, this then implies that

$$R^n/IR^n \cong R^m/IR^m$$

as F -modules. We claim that $R^k/IR^k \cong F^k$ are isomorphic F -modules. Indeed there is an isomorphism $\psi: R^k/IR^k \rightarrow F^k = (R/I)^k$ sending $(r_1, \dots, r_k) + IR^k$ to $(r_1 + I, \dots, r_k + I)$. From this, it follows that $F^n \cong F^m$ as F -modules. Since F is a field, this implies that $n = m$. $\square$

Remark 3.3.19 — Note that for the theorem above, it's false for non-commutative rings. There exist rings R such that, for example, $R^2 \cong R^1$ are isomorphic R -modules.

Theorem 3.3.20

Any commutative ring contains a maximal ideal.

Proof Sketch. Select an ideal I . If it's maximal, then we're done. If it's not maximal, then there is a bigger one. we can continue this to get an infinite sequence of ideals. If we're working over Noetherian rings, this is enough. If not, then the cardinality is too large and so we need the axiom of choice. $\square$

§3.4 Modules over PIDs

Recall that a commutative integral domain R is a PID if every ideal of R is principal.

Example 3.4.1

Euclidean domains are PIDs. For instance, there are $\mathbb{Z}, k[X], \mathbb{Z}[i], \mathbb{Z}[\sqrt{\pm 2}]$.

Our next goal is to prove the following theorem.

Theorem 3.4.2 (Classification of finitely generated modules over a PID)

Let R be a PID (in particular, R is commutative), and M a finitely generated R -module. Then,

$$M \cong R^n \oplus (R/d_1) \oplus (R/d_2) \oplus \cdots \oplus (R/d_r)$$

are isomorphic R -modules, for some $d_1, \dots, d_r \in R$, with $d_1 \mid \cdots \mid d_r$ and $n = \text{rank}(R)$. Furthermore, n and the d_i are unique up to associates.

Remark 3.4.3 — The uniqueness up to associates part will not be proven.

Remark 3.4.4 — Note that this theorem generalizes the classification theorem for finitely generated abelian groups, since they're just $\mathbb{Z}$ -modules.

Remark 3.4.5 — We will see that the proof is slightly easier for Euclidean domain.

Here is the outline of the proof:

1. First, we show M is in fact finitely presented, so $M \cong \text{coker}(\phi_A: R^n \rightarrow R^m)$ for some matrix $A \in M_{n \times m}(R)$, where $\phi_A(x) := A \cdot x$.
2. We then show that if $B = PAQ$, for invertible matrices P, Q , then ϕ_A and ϕ_B have isomorphic cokernels.
3. We show that for any matrix A over a PID R , there are invertible matrices P, Q such that PAQ is a rectangular diagonal matrix. (This is the hardest step).
4. Then we put all this together to prove the theorem.

Lemma 3.4.6

Let R be a PID, and $N \leq R^n$ a submodule. Then $N \cong R^k$ for some $k \leq n$, so N is finitely generated and free.

Proof. We induct on n . If $n = 0$, $R^0 = 0$, so $n = 0 \cong R^0$. If $n = 1$, then $N \subseteq R$ is an ideal, so $N = (\alpha)$ for some $\alpha \in R$. Since PIDs are integral domains, $N \cong R^1$ if $\alpha \neq 0$ and $N \cong R^0$ if $\alpha = 0$. Now, let $\pi_n: R^n \rightarrow R$ be the homomorphism sending $(r_1, \dots, r_n)$ to r_n . Then, $\ker(\pi_n) \cong R^{n-1}$. $\pi_n(N) \leq R$ is a submodule, so it's (α) for some $\alpha \in R$ since R is a PID.

If $\alpha = 0$, this means $N \leq \ker(\pi_n) \cong R^{n-1}$, so the inductive hypothesis finishes the proof. If $\alpha \neq 0$, we can select some $\beta \in N$ such that $\pi_n(\beta) = \alpha$. By the inductive hypothesis, we can choose some $x_1, \dots, x_k \in N \cap \ker(\pi_n)$ which freely generate $N \cap \ker(\pi_n)$, with $k \leq n - 1$. We let $S = \{x_1, \dots, x_k, \beta\} \subseteq N$ and claim that S freely generates N , which will complete the proof.

We first show S generates N . Let $y \in N$. Then, $\pi_n(y) = r\alpha$, for some $r \in R$. Thus, $y - r\beta \in N \cap \ker(\pi_n)$ is spanned by the $x_1, \dots, x_k$. Thus, y is generated by S . Now suppose $0 = r_1 \cdot x_1 + \dots + r_k x_k + r' \cdot \beta$, for $r_i, r' \in R$. Then, $0 = \pi_n(0) = r'\alpha$, so $r' = 0$. Since $x_1, \dots, x_k$ are linearly independent, we must have that $r_1 = \dots = r_k = 0$, so S is linearly independent. $\square$

Corollary 3.4.7

Let M be a finitely generated module over a PID R . Then M is finitely presented.

Proof. There's a surjective homomorphism $\phi: R^n \rightarrow M$. $\ker(\phi) \leq R^n$ is a submodule, so by the lemma, $\ker(\phi) \cong R^k$, for some k . Let $\psi: R^k \rightarrow R^n$ have image $\ker(\phi)$. Then, $M \cong \text{coker}(\psi)$. Thus, M is finitely presented. $\square$

This is the first part of the proof for the theorem.

Definition 3.4.8. Two matrices $A, B \in M_{m \times n}(R)$ are **equivalent** if there are invertible matrices $P \in GL_n(R), Q \in GL_m(R)$ such that $B = PAQ$.

By considering the homomorphisms $\phi_A, \phi_B: R^n \rightarrow R^m$, this is equivalent to there being isomorphisms $f = \phi_P: R^n \rightarrow R^n$ and $g = \phi_Q: R^m \rightarrow R^m$ such that $f \circ \phi_B = \phi_A \circ g$. In other words, A and B are the same after changing bases.

Remark 3.4.9 — This is an equivalence relation.

Proposition 3.4.10

Let $A, B \in M_{m \times n}(R)$ be matrices. If A and B are equivalent, then $\text{coker}(\phi_A) \cong \text{coker}(\phi_B)$ are isomorphic R -modules.

Proof. Let $f: R^n \rightarrow R^n$ and $g: R^m \rightarrow R^m$ be isomorphisms such that $f \circ \phi_B = \phi_A \circ g$. Note that $\text{Im}(\phi_A) = \text{Im}(\phi_A \circ g) = \text{Im}(f \circ \phi_B) = f(\text{Im}(\phi_B))$. Let $\psi: \text{coker}(\phi_B) = R^n / \text{Im}(\phi_B) \rightarrow \text{coker}(\phi_A) = R^n / \text{Im}(\phi_A)$ send $x + \text{Im}(\phi_B)$ to $f(x) + \text{Im}(\phi_A)$. This is a well-defined homomorphism, and is an isomorphism, since f is. $\square$

Remark 3.4.11 — Suppose that $R = F$ is a field. Then, equivalent matrices have the same rank, so $\text{coker}(\phi_A)$ and $\text{coker}(\phi_B)$ have the same dimension, so they're isomorphic. Thus, we already knew the proposition in this case.

This completes the second step of the proof of the theorem. We now consider special types of equivalences.

Definition 3.4.12. An **elementary matrix** is an $n \times n$ matrix A over R of one of the following forms:

1. A has 1s along the diagonal, $A_{ij} = c$ for some fixed $c \in R$ and $1 \leq i, j \leq n$, and all other entries are 0.
2. Let $1 \leq i, j \leq n$ be distinct and fixed. Then, A is I except in 4 entries: $A_{ii} = A_{jj} = 0$ and $A_{ij} = A_{ji} = 1$.
3. Let $1 \leq i$ and $c \in R^\times$ be a fixed unit. Then, A is I except in 1 entry: $A_{ii} = c$.

Definition 3.4.13. Now let $A \in M_{m \times n}$ be any matrix. An **elementary row/column operation** on A is given by replacing A with PA or AP respectively, for P an elementary matrix.

1. Elementary matrices of type 1. correspond to adding $c \in R$ times the i th row/column to the j th row/column.
2. Elementary matrices of type 2. correspond to swapping the i th and j th rows/columns.
3. Elementary matrices of type 3. correspond to multiplying the i th row/column by $c \in R^\times$.

Remark 3.4.14 — All elementary matrices are invertible, which gives the following proposition.

Proposition 3.4.15

Let $A \in M_{m \times n}(R)$ be a matrix. If B is attained from A by row and column operations, then A and B are equivalent matrices. In particular, if this holds, we have

$$\text{coker}(\phi_A) \cong \text{coker}(\phi_B).$$

Definition 3.4.16. A matrix $A \in M_{m \times n}(R)$ is in **Smith normal form** if A is a rectangular diagonal matrix, i.e

$$A = \begin{pmatrix} d_1 & & & & 0 \\ & \ddots & & & \\ & & d_r & & \\ & & & 0 & \\ 0 & & & & \ddots \\ & & & & & 0 \end{pmatrix}$$

and all $d_i \neq 0$ and $d_1 \mid \cdots \mid d_r$.

Example 3.4.17

Here is a matrix that is in Smith normal form:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 6 & 0 \end{pmatrix}$$

Theorem 3.4.18

Let $A \in M_{m \times n}(R)$ be a matrix over a PID R . Then, A is equivalent to a matrix in Smith normal form.

Proof. Note that R is a PID and so a UFD, so GCDs exist. These are only defined up to units, so each time we take GCDs, we'll just fix some choice, for the sake of our sanity.

Let $A = (A_{ij}) \in M_{m \times n}(R)$. If $A = 0$, we're done, so assume that $A \neq 0$.

Claim 3.4.19 (Claim 1). Given two entries A_{ij} and A_{kl} in the same row (so $i = k$), we can modify A so that the greatest common divisor of A_{ij}, A_{kl} appears in A .

Proof of Claim 1 (3.4.19). First assume that they're in the same column. Since invertible 2×2 matrices can be extended to invertible $m \times m$ matrices, it suffices to prove the claim when $m = 2$. That is, for any vector $\begin{pmatrix} a \\ b \end{pmatrix} \in R^2$, we can left-multiply by some matrix $P \in M_2(R)$ to attain a vector with $\text{GCD}(a, b)$ in. Now that R is a PID, so $(a, b) = (d) \subseteq R$ are equal ideals, for some $d \in R$. Thus, $d = \text{GCD}(a, b)$. $d \in (a, b)$, so there are some $x, y \in R$ such that $xa + yb = d$. Since $\text{GCD}(a, b) = d$, we must have that $\text{GCD}(x, y) = 1$. Similar to the above, there exists $u, v \in R$ such that $xu + yv = 1$. Now, let

$$P := \begin{pmatrix} x & y \\ -v & u \end{pmatrix}.$$

Note that $\det(P) = xu + yv = 1$, so P is invertible. Now,

$$P \cdot \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} xa + yb \\ -va + ub \end{pmatrix} = \begin{pmatrix} d \\ 0 \end{pmatrix},$$

so we're done. If, instead A_{ij} and A_{kl} are in the same column, we can do the same, but with the row matrix $\begin{pmatrix} a & b \end{pmatrix}$ and right multiplication by P^T . $\square$

Claim 3.4.20 (Claim 2). We can modify A so that A_{11} divides the rest of the first row and column, i.e. $A_{11} \mid A_{i1}, A_{1j}$, for all i, j .

Proof of claim 2 (3.4.20). For $r \in R \setminus \{0\}$, let $\delta(r) \in \mathbb{Z}_{\geq 0}$ be the number of (possibly repeated) irreducible factors of r . So for example if $r = p_1 \dots p_k$, with all p_i irreducible, then $\delta(r) = k$; and r is a unit if and only if $\delta(r) = 0$. Note that R is a UFD so this is well-defined.

Suppose that A isn't of the required form. First, A must contain some non-zero entry A_{ij} . Using row and column operations, we can move this to the top left. Let $\alpha_1 = A_{11} \neq 0$. If A still isn't of the required form, there are some $1 \leq i, j \leq n$, not

both equal to 1, such that $A_{11} \nmid A_{ij}$. By claim 1 (3.4.19), we can modify A so that $\alpha_2 := \text{GCD}(A_{11}, A_{ij})$ appears in our matrix. Using row and column operations, we can assume this entry is in the top left.

Note that $\alpha_2 \mid \alpha_1$, but α_2 and α_1 aren't associates since $\alpha_2 \mid A_{ij}$, but $\alpha_1 \nmid A_{ij}$. Thus, $\delta(\alpha_2) < \delta(\alpha_1)$. Therefore, if A wasn't of the required form, we could modify it so that the top left entry had strictly lower δ . If we do this process repeatedly, it must terminate, and we must find that $A_{11} \mid A_{ij}$ whenever $i = 1, j = 1$, as required. $\square$

Now we modify A to be of the form in claim 2 (3.4.20). Since $A_{11} \mid A_{1j}$, for all $j > 1$, subtracting multiples of the first column/ row from the other columns/ row, we can modify A so that $A_{1j} = 0$ for all $j > 1$ and $A_{i1} = 0$ for all $i > 1$. We thus have

$$A = \begin{pmatrix} d_1 & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & A' & \\ 0 & & & \end{pmatrix},$$

for some smaller matrix $A' \in M_{(m-1) \times (n-1)}$. We apply this process to A' to modify A (without changing its equivalence class) so that it's in rectangular diagonal form:

$$A = \begin{pmatrix} d_1 & & & & 0 \\ & \ddots & & & \\ & & d_r & & \\ & & & 0 & \\ 0 & & & & \ddots \\ & & & & & 0 \end{pmatrix}$$

However, we're not done yet, We still need to modify A so that $d_1 \mid \dots \mid d_r$. First, we do the case $r = 2$:

$$\begin{aligned} \begin{pmatrix} d_1 & 0 \\ 0 & d_2 \end{pmatrix} &\sim \begin{pmatrix} d_1 & \text{GCD}(d_1, d_2) \\ 0 & d_2 \end{pmatrix} \\ &\sim \begin{pmatrix} d_1 & \text{GCD}(d_1, d_2) \\ -\lambda d_1 & 0 \end{pmatrix} \\ &\sim \begin{pmatrix} \text{GCD}(d_1, d_2) & 0 \\ 0 & \lambda d_1 \end{pmatrix} \end{aligned}$$

Here, $d_2 = \lambda \cdot \text{GCD}(d_1, d_2)$ for some $\lambda \in R$. We now proceed as in the proof of claim 2(3.4.20). Let $\alpha_1 = d_1$. If $d_1 \nmid d_i$, for some $i \geq 2$, let $\alpha_2 = \text{GCD}(d_1, d_i)$, and use row and column operations to move it to the place of d_1 . $\alpha_2 \mid \alpha_1$, but $\alpha_1 \nmid \alpha_2$, so we have that $\delta(\alpha_2) < \delta(\alpha_1)$. Thus, once again, this process terminates, and we find that $d_1 \mid d_i$, for $i \geq 2$. If we repeat this, we can modify A so that $d_2 \mid d_i$ for all $i \geq 3$. This process preserves $d_1 \mid d_i$, for $i \geq 2$, and iterating gives the required result. $\square$

Remark 3.4.21 — When we modified A to an equivalent matrix, we mostly did this by applying row and column operations, but not always. in proving claim 1(3.4.19), we used an equivalence of matrices that wasn't necessarily of this form.

In fact, if R is a Euclidean domain, any $A \in GL_n(R)$ is a product of elementary

matrices. Therefore, if R is a Euclidean domain, and $A \in M_{m \times n}(R)$ can be put into Smith normal form, just use elementary matrices.

Proposition 3.4.22

Let $A \in M_{m \times n}(R)$ be in Smith normal form:

$$A = \begin{pmatrix} d_1 & & & & 0 \\ & \ddots & & & \\ & & d_r & & \\ & & & 0 & \\ 0 & & & & \ddots \\ & & & & & 0 \end{pmatrix},$$

with $d_1, \dots, d_r \in R$ non-zero, and $d_1 \mid \dots \mid d_r$. Then, there is an isomorphism of R -modules

$$\text{coker}(\phi) \cong R/(d_1) \oplus \dots \oplus R/(d_r) \oplus R^{m-r},$$

where ϕ_A is given by left multiplication by A .

Proof. Let $\psi: R^m \rightarrow R/(d_1) \oplus \dots \oplus R/(d_r) \oplus R^{m-r}$ be the homomorphism sending $(x_1, \dots, x_m)$ to

$$(x_1 + (d_1), \dots, x_r + (d_r), x_{r+1}, \dots, x_m).$$

Then, $\text{Im}(\phi_A) = \ker(\psi) = (d_1) \oplus \dots \oplus (d_r) \oplus 0 \oplus \dots \oplus 0$ (where there are $m - r$ 0s at the end) Therefore, the first isomorphism theorem proves the result. $\square$

Theorem 3.4.23 (Classification of Finitely Generated Modules over a PID)

Let R be a PID (in particular, R is commutative), and M a finitely generated R -module. Then,

$$M \cong R^n \oplus R/(d_1) \oplus \dots \oplus R/(d_r),$$

for some $d_1, \dots, d_r \in R$ non-zero, with $d_1 \mid \dots \mid d_r$.

Proof. 1. Since R is a PID, M is finitely presented. Thus, $M \cong \text{coker}(\phi_A)$ for some $A \in M_{m \times n}(R)$.

2. If $A, B \in M_{m \times n}(R)$ are equivalent, then $\text{coker}(\phi_A) \cong \text{coker}(\phi_B)$.

3. A is equivalent to a matrix in Smith normal form, i.e. we can assume that

$$A = \begin{pmatrix} d_1 & & & & \\ & \ddots & & & \\ & & d_r & & \\ & & & 0 & \\ & & & & \ddots \\ & & & & & 0 \end{pmatrix}.$$

4. Therefore,

$$M \cong R^n \oplus R/(d_1) \oplus \cdots \oplus R/(d_r),$$

and $d_1 \mid \cdots \mid d_r$ are all nonzero. $\square$

Remark 3.4.24 — There's another type of decomposition that we could consider, called *prime decomposition*:

$$M \cong R^n \oplus R/(p_1^{n_1}) \oplus \cdots \oplus R/(p_k^{n_k}),$$

for $n_i \in \mathbb{N}$ and $p_i \in R$ irreducible.

Note that this follows from the classification result that we proved, along with the fact that if $d = p_1^{n_1} \cdots p_k^{n_k}$, then

$$R/(d) \cong R/(p_1^{n_1}) \oplus \cdots \oplus R/(p_k^{n_k})$$

Example 3.4.25

Let $R = \mathbb{Z}$; then, R -modules are abelian groups. Let A be the abelian group generated by the elements a, b, c with relations

$$\begin{aligned} 2a + 3b + c &= 0 \\ a + 2b &= 0 \\ 5a + 6b + 7c &= 0. \end{aligned}$$

This means that

$$A = \mathbb{Z}^3 / \left\langle \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 5 \\ 6 \\ 7 \end{pmatrix} \right\rangle = \text{coker}(\phi_X),$$

where X is the matrix

$$X = \begin{pmatrix} 2 & 1 & 5 \\ 3 & 2 & 6 \\ 1 & 0 & 7 \end{pmatrix}.$$

We then put X into Smith normal form:

$$\begin{pmatrix} 2 & 1 & 5 \\ 3 & 2 & 6 \\ 1 & 0 & 7 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Thus, we have

$$A \cong \mathbb{Z}/\mathbb{Z} \oplus \mathbb{Z}/\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} \cong C_3.$$

§3.5 Jordan Normal Form

Here, we study modules over polynomial rings of fields using the theorem from the previous section, and amongst other things deduce Jordan normal form. Let F be a

field, V an F -vector space, and $\alpha: V \rightarrow V$ a linear map. Then, we can make V into an $F[X]$ -module: If $f \in F[X]$ and $v \in V$, then we set

$$f \cdot v := (f(\alpha))(v).$$

We write V_α for V , when we want to view it as an $F[X]$ -module (so V and V_α are in particular the same set). We use the structure of V_α as an $F[X]$ -module to study the linear map α .

Lemma 3.5.1

If V is a finite dimensional F -vector space, then V_α is a finitely generated $F[X]$ -module.

Proof. If $v_1, \dots, v_n \in V$ is a basis for V as a vector space, then they generate V_α as an $F[X]$ -module. $\square$

Example 3.5.2

Suppose $V_\alpha \cong F[X]/(X^r)$ as $F[X]$ -modules. Particularly, they're isomorphic as F -vector spaces.

$1, X, \dots, X^{r-1} \in F[X]/(X^r)$ are an F -basis, and the action of multiplication by X has the matrix

$$\begin{pmatrix} 0 & & 0 \\ 1 & 0 & \\ & 1 & \ddots \\ 0 & & 1 & 0 \end{pmatrix}$$

with respect to this basis. Thus, in V_α , α is represented by the same matrix with respect to the corresponding basis.

Example 3.5.3

Suppose that $V_\alpha \cong F[X]/((X - \lambda)^r)$, as $F[X]$ -modules, for some $\lambda \in F$. Let $\beta = \alpha - \lambda I$. This is a linear map $\beta: V \rightarrow V$. Then, $V_\beta \cong F[Y]/(Y^r)$, for $Y = X - \lambda$. Therefore, by the previous example, there is a basis for V such that β is given by the matrix

$$\begin{pmatrix} 0 & & 0 \\ 1 & 0 & \\ & 1 & \ddots \\ 0 & & 1 & 0 \end{pmatrix}$$

Thus, $\alpha = \beta + \lambda I$ has the matrix

$$\begin{pmatrix} \lambda & & 0 \\ 1 & \lambda & \\ & 1 & \ddots \\ 0 & & 1 & \lambda \end{pmatrix}$$

a Jordan block.

Example 3.5.4

Suppose $V_\alpha \cong F[X]/(f)$, for $f = a_0 + \cdots + a_{r-1}X^{r-1} + X^r \in F[X]$ some monic polynomial. V_α has an F -basis given by $1, X, \dots, X^{r-1}$, in which α is given by

$$C(f) := \begin{pmatrix} 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & 0 & 1 & -a_{r-1} \end{pmatrix}$$

This matrix $C(f)$ is called the **companion matrix** of f .

Theorem 3.5.5 (Rational Canonical Form)

Let F be any field, V a finite dimensional F -vector space, and $\alpha: V \rightarrow V$ some linear map. Then

$$V_\alpha \cong F[X]/(f_1) \oplus \cdots \oplus F[X]/(f_r),$$

with $f_1 \mid \cdots \mid f_r$ all non-zero polynomials over F . Therefore there is a basis for V in which α is given by the block diagonal matrix

$$\begin{pmatrix} C(f_1) & & 0 \\ & \ddots & \\ 0 & & C(f_r) \end{pmatrix}.$$

Proof. $F[X]$ is a PID so by the classification theorem for f.g. modules over a PID, we must have

$$V_\alpha \cong F[X]/(f_1) \oplus \cdots \oplus F[X]/(f_r) \oplus F[X]^n,$$

with $f_1 \mid \cdots \mid f_r$. As $F[X]$ isn't a finite-dimensional F -vector space, but V_α is, we must have that $n = 0$. Let b_i for each $\deg(f_i)$. With respect to the basis

$$X_1^0, X_1, \dots, X_1^{b_1-1}, \dots, X_r^0, \dots, X_r^{b_r-1},$$

α is then in the required form. Note that X_j is the X in the j th term in the direct sum $F[X]/(f_j)$. $\square$

Remark 3.5.6 — If we required all the f_i to be monic, this is completely canonical and there is no ambiguity.

Example 3.5.7

We have

$$\mathbb{Q}[X]/(X^2 - 1) \cong \mathbb{Q}[X]/(X - 1) \oplus \mathbb{Q}[X]/(X + 1),$$

but only the LHS is in rational canonical form (since neither of $X - 1$ and $X + 1$ divide each other).

Let us now focus on when $F = \mathbb{C}$.

Lemma 3.5.8

The primes in $\mathbb{C}[X]$ are, up to associates, $X - \lambda$, for $\lambda \in \mathbb{C}$.

Proof. We first note that any linear polynomial $X - \lambda \in \mathbb{C}[X]$ is prime. Now let $f \in \mathbb{C}[X]$ be prime. f is non-zero and not a unit so it's non-constant. So by the fundamental theorem of algebra, f has some root, $\lambda \in \mathbb{C}$. So $(X - \lambda) \mid f$. Since f is irreducible, we must have that $X - \lambda$ and f are associates. $\square$

Theorem 3.5.9 (Jordan Normal Form)

Let $\alpha: V \rightarrow V$ be a linear map, where V is a finite dimensional complex vector space. Then there is an isomorphism of $\mathbb{C}[X]$ -modules:

$$V_\alpha \cong \mathbb{C}[X]/((X - \lambda_1)^{n_1}) \oplus \cdots \oplus \mathbb{C}[X]/((X - \lambda_r)^{n_r})$$

for some $n_i \in \mathbb{N}, \lambda_i \in \mathbb{C}$. Furthermore, there's a $\mathbb{C}$ -basis for V in which α has the form of the following block diagonal matrix:

$$\begin{pmatrix} J_{n_1}(\lambda_1) & & 0 \\ & \ddots & \\ 0 & & J_{n_r}(\lambda_r) \end{pmatrix}$$

where $J_n(\lambda)$ is the $n \times n$ Jordan block

$$J_n(\lambda) = \begin{pmatrix} \lambda & & 0 \\ 1 & \ddots & \\ \vdots & & \vdots \\ 0 & & 1 & \lambda \end{pmatrix}.$$

Proof. For the first part, we apply the prime decomposition theorem to V_α , and use the fact that all primes in $\mathbb{C}[X]$ are of the form $X - \lambda$. Then take a basis as in the proof of rational canonical form, and use the fact that

$$C((X - \lambda)^n) = J_n(\lambda)$$

$\square$

Lemma 3.5.10

Let V be a finite-dimensional F -vector space, and $\alpha, \beta: V \rightarrow V$ two linear maps. Then $V_\alpha \cong V_\beta$ are isomorphic as $F[X]$ -modules if and only if α and β are conjugate, i.e. there is some isomorphism $\gamma: V \rightarrow V$ such that $\gamma \circ \alpha = \beta \circ \gamma$.

Proof. ($\implies$): Let $\gamma: V_\alpha \rightarrow V_\beta$ be an $F[X]$ -module isomorphism; so it's in particular an isomorphism of F -vector spaces. Then for $v \in V$, $\beta(v) = X \cdot_{V_\beta} v$ (here $\cdot_{V_\beta}$ denotes the $F[X]$ -module action on V_β), and similarly $\alpha(v) = \cdot_{V_\alpha} v$. Then,

$$\beta \circ \gamma(v) = X \cdot_{V_\beta} \gamma(v) = \gamma(X \cdot_{V_\alpha} v) = \gamma \circ \alpha(v).$$

Thus, $\gamma \circ \alpha = \beta \circ \gamma$, so α and β are conjugate.

($\Leftarrow$): Suppose α and β are conjugate, so there is an isomorphism $\gamma: V \rightarrow V$ such that $\gamma \circ \alpha = \beta \circ \gamma$. We claim that $\gamma: V_\alpha \rightarrow V_\beta$ is an $F[X]$ -module isomorphism. Indeed for $f \in F[X]$ and $v \in V$:

$$\gamma(f \cdot_{V_\alpha} (v)) = \gamma(f(\alpha)(v)) = f(\beta)(\gamma(v)),$$

where we note that $\gamma \circ \alpha^i = \alpha^i \circ \beta$ for all $i \geq 0$. Thus, γ is an $F[X]$ -module homomorphism; it's a bijection since it's an F -linear isomorphism. $\square$

Reinterpreting this plus rational canonical form, we have a classification result for square matrices over a field; it's weaker than Jordan normal form but works over any field.

Corollary 3.5.11

Let F be a field. There's a bijection between conjugacy classes of $n \times n$ matrices over F and sequences of monic polynomials $f_1, \dots, f_r \in F[X]$, such that $f_1 \mid \dots \mid f_r$, and $\sum_i \deg(f_i) = n$.

Proof. Let A be an $n \times n$ matrix over F , let $V = F^n$ and $\alpha: V \rightarrow V$ send v to $A \cdot v$. Then by rational canonical form, we have

$$V_\alpha \cong F[X]/(f_1) \oplus \dots \oplus F[X]/(f_r)$$

for some unique sequence $f_i \in F[X]$, with $f_1 \mid \dots \mid f_r$ all non-zero monic polynomials in $F[X]$. By counting dimensions of both sides, $\sum_i \deg(f_i) = n$. Furthermore, if A and B are conjugate matrices, then V_α and V_β are isomorphic $F[X]$ -modules (by the lemma), so they define the same sequence of polynomials. If $f_1 \mid \dots \mid f_r$ is any sequence of monic polynomials in $F[X]$ with $\sum_i \deg(f_i) = n$, then

$$V := F[X]/(f_1) \oplus \dots \oplus F[X]/(f_r)$$

is a finite dimensional F -vector space of dimension n , with a linear map $\alpha: V \rightarrow V$ sending v to $X \cdot v$, which determines a matrix up to conjugation, by picking a basis. $\square$

Example 3.5.12

Let's look at conjugacy classes in $M_{2 \times 2}(F)$. This corresponds to classifying $F[X]$ -modules of the form

$$F[X]/(f_1) \oplus \cdots \oplus F[X]/(f_r),$$

with $f_1 \mid \cdots \mid f_r$ a sequence of monic polynomials, with $\sum_i \deg(f_i) = 2$. Therefore, either $r = 1$ and $\deg(f_1) = 2$, or $r = 2$ and $\deg(f_1) = \deg(f_2) = 1$. In the latter case of $r = 2$, $f_1 \mid f_2$ and both are monic, so $f_1 = f_2 = X - \lambda$, for some $\lambda \in F$. So since the companion matrix is the 1×1 matrix (λ) , the corresponding matrix in $M_{2 \times 2}(F)$ is

$$\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}.$$

In the case $r = 1$, we write $f_1 = X^2 + aX + b$, for some $a, b \in F$. Then the matrix we get is the companion matrix of f_1 , which is

$$C(f_1) = \begin{pmatrix} 0 & -b \\ 1 & -a \end{pmatrix}.$$

If f_1 is irreducible, we can't simplify this any further for an arbitrary field F . However if for example $F = \mathbb{Z}/3$, we could find all irreducible degree 2 polynomials, and get a better classification. If f_1 is reducible, then $f_1 = (X - \lambda)(X - \mu)$, for some $\lambda, \mu \in F$. If $\lambda = \mu$, then $f_1 = (X - \lambda)^2$ so we saw in an earlier example that the matrix is conjugate to the Jordan block:

$$\begin{pmatrix} \lambda & 0 \\ 1 & \lambda \end{pmatrix}$$

If $\lambda \neq \mu$, the matrix

$$\begin{pmatrix} 0 & -\lambda\mu \\ 1 & \lambda + \mu \end{pmatrix}$$

has distinct eigenvalues so it's conjugate to

$$\begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}.$$

4 Matrix Lie Groups

§4.1 Matrix Groups and Topology Recap

Let R be a commutative ring. Recall $M_n(R)$ is the ring of $n \times n$ matrices over R , which is non-commutative for $n \geq 2$.

Definition 4.1.1. The **general linear group** $GL_n(R)$ over R is defined to be the unit group of $M_n(R)$, $M_n(R)^\times$.

Remark 4.1.2 — One can define the determinant function $\det: M_n(R) \rightarrow R$ with the same formula as we saw in linear algebra, and the same proof shows that M is invertible if and only if $\det(M) \in R^\times$.

Proposition 4.1.3

The center of $GL_n(R)$ is the set of matrices of the form λI , for $\lambda \in R^\times$.

Proof. Let S lie in the center $Z(GL_n(R))$. For $1 \leq u, v \leq n$ distinct, let $E^{uv} \in GL_n(R)$ be the matrix with 1s on the diagonal, $(E^{uv})_{uv} = 1$, and all other entries 0. Explicitly,

$$(E^{uv})_{ij} = \delta_{ij} + \delta_{ui}\delta_{vj}.$$

We must have that $AE^{uv} = E^{uv}A$, and so for all $1 \leq i, j \leq n$,

$$0 = (E^{uv}A - AE^{uv})_{ij} = E_{ik}^{uv}A_{kj} - A_{ik}E_{kj}^{uv} = \delta_{ui}A_{vj} - \delta_{vj}A_{ui}.$$

For $u = i \neq v = j$, this tells us $A_{ii} = A_{jj}$, so all the diagonal entries are the same. If $i = j = u \neq v$, this tells us $A_{vj} = 0$, so all other entries are 0. So A must be of the form λI . For A to be invertible, λ must be invertible. $\square$

§4.2 Topology Recap

Now, recall some topological facts. Particularly, recall the following:

- The composition of two continuous functions is continuous.
- Multiplication and addition of two continuous functions are continuous.
- A function $f: \mathbb{F}^m \rightarrow \mathbb{F}^n$ is continuous if and only if each of its components $f_i: \mathbb{F}^m \rightarrow \mathbb{F}$ is continuous.
- Projection maps are also continuous, so combining these, any polynomial (in multiple variables) $f: \mathbb{F}^n \rightarrow \mathbb{F}$ is continuous.

With these, we get that the multiplication map

$$\mu: M_n(\mathbb{F}) \times M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$$

sending sending (A, B) to $A \cdot B$ is continuous, since each entry of $A \cdot B$ is a polynomial in the entries of A and B .

There is also the notion of density, path connectedness, and compactness which we will refer to:

Definition 4.2.1. If $Y \subseteq \mathbb{F}^n$ has the Euclidean topology and $X \subseteq Y$ is a subset, X is **dense** in Y if $\overline{X}$, the closure of X in Y , is the whole of Y . Equivalently, this means that for all $y \in Y$ and for all $\epsilon > 0$, there is some $x \in X$ with $|x - y| < \epsilon$.

Example 4.2.2

$\mathbb{Q} \subseteq \mathbb{R}$ is dense.

Definition 4.2.3. A topological space X is **path-connected** if for all $x, y \in X$, there is a path from x to y , i.e. there is a continuous function $f: [0, 1] \rightarrow X$ sending 0 to x and 1 to y .

Example 4.2.4

We'll show $U(1)$ is compact:

$$\begin{aligned} U(1) &= \{(z) \in M_1(\mathbb{C}) \mid (z)^\dagger(z) = 1\} \\ &= \{z \in \mathbb{C} \mid z^*z = 1\} \\ &= \{z \in \mathbb{C} \mid |z| = 1\} \end{aligned}$$

This is bounded as $|z| \leq 1$. Further, it's a closed subset of $\mathbb{C}$ since $|\cdot|$ is a continuous function.

Example 4.2.5

$GL_n(\mathbb{F}) \subseteq \mathbb{F}^{n \times n}$ isn't bounded, and hence isn't compact. To see this, consider the matrix

$$\begin{pmatrix} 1 & 0 & \dots & k \\ 0 & 1 & & \\ \vdots & & \ddots & \\ 0 & & & 1 \end{pmatrix} \in GL_n(\mathbb{F}),$$

for $k \in \mathbb{N}$. As $k \rightarrow \infty$, the norm of this matrix goes to ∞ too, so $GL_n(\mathbb{F})$ isn't a bounded subset of $M_n(\mathbb{F})$.

Example 4.2.6

$GL_1(\mathbb{R}) = \mathbb{R}^\times$ isn't path-connected, since there's no path between 1 and -1 . In fact, for any n , $GL_n(\mathbb{R})$ isn't path-connected.

Proof. We'll show there's no path in $GL_n(\mathbb{R})$ between

$$A := \begin{pmatrix} -1 & \dots & 0 \\ \vdots & 1 & \\ & & \ddots \\ 0 & & & 1 \end{pmatrix}$$

and I . Suppose $\gamma: [0, 1] \rightarrow GL_n(\mathbb{R})$ is a path from I to A . Then $\det \circ \gamma: [0, 1] \rightarrow \mathbb{R}$ is continuous, and sends 0 to 1 and 1 to -1 . So by the intermediate value theorem there would be some t such that $\det(\gamma(t)) = 0$. But $\gamma(t)$ was an invertible matrix, so this is a contradiction. $\square$

Remark 4.2.7 — We can show that $GL_n^+(\mathbb{R})$ the space of positive-determinant real matrices is path-connected. Then, we can use this fact the fact that the Gram-Schmidt process is continuous, to show that groups such as $SO(n)$ or $SU(n)$ are path-connected.

Example 4.2.8

$GL_n(\mathbb{C})$ is always path-connected.

Proof Sketch. Let $A \in GL_n(\mathbb{C})$. We can write $A = PBP^{-1}$, where B is in Jordan normal form. We assume for simplicity that B is a single Jordan block:

$$B = \begin{pmatrix} \lambda & 1 & 0 & \dots & 0 \\ 0 & \lambda & 1 & & \\ \vdots & & \ddots & & \\ 0 & & & & \lambda \end{pmatrix}$$

$\lambda \neq 0$, so $\lambda = e^z$, for some $z \in \mathbb{C}$. We can now write down the path we want:

$$\gamma(t) := \begin{pmatrix} e^{tz} & t & 0 & \dots & 0 \\ 0 & e^{tz} & t & & \\ \vdots & & \ddots & & \\ 0 & & & & e^{tz} \end{pmatrix}$$

This is invertible for all i , and $\gamma(0) = A$ and $\gamma(1) = I$. The case where B isn't just a single Jordan block is similar. $\square$

There is also the notion of differentiability and continuous differentiability. Recall that by the implicit function theorem, we can also locally invert matrices.

Example 4.2.9

Consider $\exp: \mathbb{R} \rightarrow \mathbb{R}$, sending $x \mapsto e^x$. This isn't a homeomorphism as it's not surjective. But the derivative at 0 is $1 \in M_1(\mathbb{R})$, which is invertible. We can find the open neighborhood in the theorem explicitly

$$\exp|_{(-1,1)}: (-1,1) \mapsto (e^{-1}, e^1)$$

is a homeomorphism, it's continuously differentiable, and its inverse is $\log$ which is continuously differentiable.

Example 4.2.10

Consider $\exp: \mathbb{C} \rightarrow \mathbb{C}$, sending $z \mapsto e^z$. The derivative at 0, $d\exp_0: \mathbb{C} = \mathbb{R}^2 \rightarrow \mathbb{C} = \mathbb{R}^2$, is the identity $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, which is invertible. Now it's less obvious what the preimage U and the image V should be, though we can still do it; it turns out we can take U to be the disc of radius 1 around 0.

§4.3 Matrix Lie groups

Remark 4.3.1 — For the rest of this part, we will work with matrices over $\mathbb{R}$ or $\mathbb{C}$. We write $\mathbb{F}$ to mean either $\mathbb{R}$ or $\mathbb{C}$.

Remark 4.3.2 — We will work with closed subgroups $G \leq GL_n(\mathbb{F})$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. The point of this is that $GL_n(\mathbb{F})$ is a *topological group*.

We equip $GL_n(\mathbb{F}) \subseteq M_n(\mathbb{F})$ with the subspace topology. Since $GL_n(\mathbb{F}) = \{M \in M_n(\mathbb{F}) \mid \det(M) \neq 0\}$, $GL_n(\mathbb{F}) \subseteq M_n(\mathbb{F})$ is an open subset.

Definition 4.3.3. A **topological group** is a group G that is also a topological space (in the case of point-set topology) such that the multiplication map, $m: G \times G \rightarrow G$ such that $g_1, g_2 \mapsto g_1 g_2$, and the inverse map, $i: G \rightarrow G$ such that $g \mapsto g^{-1}$, are both continuous, and G is Hausdorff.

Remark 4.3.4 — A homomorphism of topological groups is a homomorphism of groups which is continuous. An isomorphism is an isomorphism of groups which is a homeomorphism.

Lemma 4.3.5

$GL_n(\mathbb{F})$ is a topological group in a natural way.

Proof. We give it a topology:

$$G = \{A \in M_n(\mathbb{F}) \mid \det(A) \neq 0\} \subset M_n(\mathbb{F}) \stackrel{(*)}{\cong} \mathbb{F}^{n^2} = \begin{cases} \mathbb{R}^{n^2} \\ \mathbb{C}^{n^2} \end{cases},$$

where on the very right, these have a natural *Euclidean* topology. In $(*)$, this is isomorphic as a vector space/ set. Given this, we view it with the induced topology: $U \subset G$ is open $\iff U \subset F^{n^2}$ is open. The maps $m: G \times G \rightarrow G$ is continuous and m is a polynomial, and $i: G \rightarrow G$ is a rational function, so this is true. $\square$

Lemma 4.3.6

Let G be a topological group. Any subgroup $H \leq G$, when equipped with the subspace topology, is a topological group.

Proof. The product $H \times H \rightarrow H$ is the restriction of the product $G \times G \rightarrow G$, so is continuous in the subspace topology; similar for the inverse. $\square$

Definition 4.3.7. A **matrix Lie group** is a topological group G , which is isomorphic (as a topological group) to a closed subgroup $H \leq GL_n(\mathbb{F})$ for some n .

Remark 4.3.8 — Note that since $GL_n(\mathbb{F})$ is Hausdorff and in fact metrizable, so is any subspace. So the same applies to any matrix Lie group.

We will have examples of matrix Lie groups underneath.

Example 4.3.9

There is an isomorphism of topological groups $\phi: \mathbb{R}/\mathbb{Z} \rightarrow U(1)$, defined by $\phi(t) := e^{2i\pi t}$.

Lemma 4.3.10

Let G be a topological group, and let $H \leq G$ be any subgroup. Then, the closure $\overline{H}$ of H in G is a closed subgroup.

Proof. $\overline{H} \subseteq G$ is certainly closed, so we check it's a subgroup. Let $g, h \in \overline{H}$. Then, there are sequences $\{g_i\}_i$ and $\{h_i\}_i$ in H , such that $g_i \rightarrow g$ and $h_i \rightarrow h$. Then, since multiplication is continuous, $g_i \cdot h_i \rightarrow g \cdot h$. Since each $g_i \cdot h_i \in H$, the limit of this sequence, $g \cdot h$, lies in $\overline{H}$. Similarly, each $g_i^{-1} \in H$, so $g^{-1} \in \overline{H}$ too. $\square$

Therefore, if we take any subgroup $G \leq GL_n(\mathbb{F})$, its closure $\overline{G}$ is a matrix Lie group.

Example 4.3.11

The closure of $\mathbb{Q}^\times$ in $\mathbb{R}^\times$ is the entirety of $\mathbb{R}^\times$.

Proposition 4.3.12

$GL_n(\mathbb{R})$ is a closed subgroup in $GL_n(\mathbb{C})$.

Proof. Consider the continuous map $f: GL_n(\mathbb{C}) \rightarrow M_n(\mathbb{R})$ sending A to $\text{Im } A$. $GL_n(\mathbb{R}) = f^{-1}(0)$, so it's a closed subset. $\square$

Proposition 4.3.13

$GL_n(\mathbb{C})$ is isomorphic to a closed subgroup in $GL_{2n}(\mathbb{R})$.

Proof. We want to produce an injective group homomorphism $\phi: GL_n(\mathbb{C}) \hookrightarrow GL_{2n}(\mathbb{R})$ which is a homeomorphism onto its image. Then ϕ is an isomorphism of topological groups between $GL_n(\mathbb{C})$ and $\text{Im}(\phi)$. For $M \in GL_n(\mathbb{C})$, we define $\phi(M)$ to be the block matrix

$$\begin{pmatrix} \text{Re } M & \text{Im } M \\ -\text{Im } M & \text{Re } M \end{pmatrix} \in GL_{2n}(\mathbb{R}).$$

It is immediate that ϕ is continuous, injective, and $\phi(I) = I$. Let's check that it's a homomorphism.

Let $M, N \in GL_n(\mathbb{C})$, and write $M = A + iB$ and $N = C + iD$, where A, B, C, D are in $M_n(\mathbb{R})$ (Ex: why can't we assume they're in $GL_n(\mathbb{R})$?). Then,

$$\phi(M)\phi(N) = \begin{pmatrix} A & B \\ -B & A \end{pmatrix} \begin{pmatrix} C & D \\ -D & C \end{pmatrix}$$

and

$$\phi(MN) = \phi((AC - BD) + i(BC + AD)).$$

Expanding, we see that both sides are the same. Thus, it remains to check that $(\phi) \subseteq GL_{2n}(\mathbb{R})$ is closed.

Let $J \in GL_{2n}(\mathbb{R})$ be the matrix

$$\begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}.$$

We will show that $\text{Im}(\phi) = \{M \in GL_{2n}(\mathbb{R}) \mid MJ = JM\}$, which is a closed subset. Writing $M = A + iB \in GL_{2n}(\mathbb{C})$ with $A, B \in M_n(\mathbb{R})$, we can compute

$$\phi(M)J - J\phi(M) = \begin{pmatrix} A & B \\ -B & A \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} - \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} A & B \\ -B & A \end{pmatrix},$$

which we can expand to get 0. Now suppose $P \in GL_{2n}(\mathbb{R})$ satisfies $PJ = JP$. Then, P is of the form

$$\begin{pmatrix} A & B \\ -B & A \end{pmatrix}$$

for some matrices $A, B \in M_n(\mathbb{R})$. Let $M = A + iB \in M_n(\mathbb{C})$. If M is invertible, then $\phi(M) = P$, so $P \in \text{Im}(\phi)$. Multiplying the equation $PJ = JP$ on the left and right by P^{-1} , we see that $P^{-1}J = JP^{-1}$. Therefore, P^{-1} must also be of the form

$$\begin{pmatrix} C & D \\ -D & C \end{pmatrix}$$

for some matrices $C, D \in M_n(\mathbb{R})$. Let $N = C + iD \in M_n(\mathbb{C})$. Then, we can check that N is an inverse to M . $\square$

Remark 4.3.14 — This tells us that a topological group G is a matrix Lie group with respect to $\mathbb{F} = \mathbb{R}$ if and only if it's one with respect to $\mathbb{F} = \mathbb{C}$.

Lemma 4.3.15

Let $G \leq GL_n(\mathbb{F})$ be a closed subgroup, and $H \leq G$ a closed subgroup. Then, $H \leq GL_n(\mathbb{F})$ is a closed subgroup.

Proof. Subgroups of subgroups are subgroups, so $H \leq GL_n(\mathbb{F})$ is a subgroup. Similarly closed subsets of closed subsets are closed subsets. So $H \leq GL_n(\mathbb{F})$ is closed; together, these imply it's a closed subgroup. $\square$

Corollary 4.3.16

We have:

1. Any closed subgroup $G \leq GL_n(\mathbb{R})$ is a closed subgroup on $GL_n(\mathbb{C})$.
2. Any closed subgroup $H \leq GL_m(\mathbb{C})$ is isomorphic to a closed subgroup in $GL_{2m}(\mathbb{R})$.

Proof. 1. Let $G \leq GL_n(\mathbb{R})$ be a closed subgroup; since $GL_n(\mathbb{R}) \leq GL_n(\mathbb{C})$ is a closed subgroup, by the lemma, $G \leq GL_n(\mathbb{C})$ is a closed subgroup.

2. Let $H \leq GL_m(\mathbb{C})$ be a closed subgroup, and $\phi: GL_m(\mathbb{C}) \rightarrow K$ an isomorphism, where $K \leq GL_{2m}(\mathbb{R})$ is some closed subgroup. Then $H \cong \phi(H)$, and $\phi(H) \leq K$ is a closed subgroup. So by the lemma, $H \cong \phi(H) \leq GL_{2m}(\mathbb{R})$, where $\phi(H) \leq GL_{2m}(\mathbb{R})$ is a closed subgroup. $\square$

§4.3.i Examples of Topological Groups

Example 4.3.17

Any group G , when equipped with the discrete topology, becomes a topological group.

Example 4.3.18

$(\mathbb{R}, +)$ is a topological group.

Example 4.3.19

$(\mathbb{Q}, +)$ is a topological group, and the inclusion $\mathbb{Q} \hookrightarrow \mathbb{R}$ is a homomorphism of topological groups.

Example 4.3.20

$(\mathbb{R}^\times, \cdot)$ and $(\mathbb{C}^\times, \cdot)$ are topological groups. There is a continuous homomorphism $(\mathbb{R}, +) \rightarrow (\mathbb{C}^\times, \cdot)$ sending t to $e^{2\pi it}$.

Example 4.3.21

We have $SL_n(\mathbb{F})$ as an example.

Example 4.3.22

$SL_2(\mathbb{C})$ acts on the Riemann sphere $\mathbb{C} \cup \{\infty\}$ by $z \mapsto \frac{az+b}{cz+d}$

Example 4.3.23

$SL_2(\mathbb{R})$ acts on the upper half plane $h^+ = \{z \mid \text{Im}(z) > 0\}$ by $z \mapsto \frac{az+b}{cz+d}$

Example 4.3.24

$U_n = \{g \in GL_n(\mathbb{C}) \mid \bar{A}^t \cdot A = I_n\}$ acts on the sphere S^{2n-1} . These preserve the Hermitian product.

There's also $SU_n = \{g \in GL_n(\mathbb{C}) \mid \det(A) = 1\}$ acts on S^{2n-1} .

Example 4.3.25

$\text{Aff}_n(\mathbb{F}) = \mathbb{F}^n \rtimes GL_n(\mathbb{F})$.

Example 4.3.26

$O_n(\mathbb{R}) = \{g \in GL_n(\mathbb{R}) \mid g^t g = I\}$, which is equivalent by: $\forall x, y \in \mathbb{R}^n, gx \cdot gy = x \cdot y$. These preserve standard inner products.

We also have $SO_n = \{A \in O_n \mid \det A = \pm 1\}$.

Remark 4.3.27 — We have that $\text{Aff}_n(\mathbb{F}) = \mathbb{F}^n \rtimes GL_n(\mathbb{F}) \leq GL_{n+1}\mathbb{F}$. This is the subgroup

$$\{g \in GL_{n+1}(\mathbb{F}) \mid g\Pi \subset \Pi \text{ where } \Pi = \{x_{n+1} = 1\} \subset \mathbb{F}^{n+1}\}.$$

Example 4.3.28

$\text{Eucl}_n(\mathbb{R})$, which is equal to the set of rigid motions of $\mathbb{R}^n$ with its *natural* metric.

There's also $\text{Eucl}_n^+(\mathbb{R}) = \mathbb{R}^n \rtimes SO_n$.

Remark 4.3.29 — Consider $\mathbb{R}^n \rightarrow \text{Eucl}_n(\mathbb{R}) \rightarrow O_n(\mathbb{R})$ that maps $x \mapsto x + v$. $\mathbb{R}^n \trianglelefteq \text{Eucl}_n$, so we may take the quotient $\text{Eucl}_n/\mathbb{R}^n$. Choosing our origin $O \in \mathbb{R}^n$, $\text{Eucl}_m = \mathbb{R}^n \rtimes O_n(\mathbb{R})$.

Example 4.3.30

$$PGL_n(\mathbb{F}) = [PSL_n(\mathbb{F})] = GL_n(\mathbb{F}) / \{\lambda I_n \mid \lambda \in \mathbb{F}^\times\}.$$

Remark 4.3.31 — Why is this a subgroup of $GL_m(\mathbb{F})$, for some m ?

Example 4.3.32

$O_{p,q} = \{A \in GL_n(\mathbb{R}) \mid A^t B_{p,q} A = B_{p,q}\}$, where $B_{p,q}$ is the indefinite inner product on $\mathbb{R}^n$:

$$B_{p,q} = \begin{pmatrix} I_p & \mathbf{0} \\ \mathbf{0} & -I_q \end{pmatrix}, \quad p + q = n.$$

Here, for $v, w \in \mathbb{R}^n$, we have $B(v, w) = v^t B_{p,q} w$.

Any finite subgroup of any of the above is a very special case.

Example 4.3.33

Let $SL_2(\mathbb{C})$ act on $\mathbb{R}^4$ by the following way. Let $\mathbf{x} \in \mathbb{R}^4$, so that

$$\mathbf{x} = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

To this, we associate a 2×2 complex matrix

$$x = \begin{pmatrix} x_0 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix}$$

Then, we have $\bar{x}^t = x$. We can then write

$$\mathbb{R}^4 = \{x \in M_2(\mathbb{C}) \mid \bar{x}^t = x\}.$$

This is, in fact, an $\mathbb{R}$ -vector space isomorphism. Let $g: x \mapsto gx\bar{g}^t$. This defines an action $SL_2(\mathbb{C})$ on $\mathbb{R}^4$. Note that this action preserves $\det x = x_0^2 - x_1^2 - x_2^2 - x_3^2$. This defines a group homomorphism from $SL_2(\mathbb{C}) \rightarrow O_{1,3}$. Note that $SO_3 \subset O_{1,3}$, which is the stabilizer of

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \in \mathbb{R}^4.$$

Also note that we have

$$SU_2 \subset SL_2(\mathbb{C}),$$

where $SU_2 = \{A \in SL \mid \bar{A}^t = I_2\} = \left\{ \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} \mid |a|^2 + |b|^2 = 1 \right\}$. Thus, we have another map $SU_2 \rightarrow SO_3$. Note that by the second representation of SU_2 , SU_2 is topologically a 3d sphere; thus, we see that $SU_2 \cong S^3$. As $S^3 \cong Q$, where Q denotes the quaternionic group, we also have that $Q \cong SU_2$ as well. Note that the map from SU_2 to SO_3 is a 2-to-1 map (surjective with kernel $\pm I$). Further, note that SO_3 is, topologically, $\mathbb{P}^3(\mathbb{R})$.

§4.3.ii Examples of Matrix Lie Groups**Example 4.3.34**

$SL_n(\mathbb{F}) \leq GL_n(\mathbb{F})$. It's a closed subset since $SL_n(\mathbb{F}) = \{M \in GL_n(\mathbb{F}) \mid \det(M) = 1\}$, and $\det$ is a continuous function.

Example 4.3.35

The set of diagonal matrices in $GL_n(\mathbb{F})$.

Example 4.3.36

$O(n) \leq GL_n(\mathbb{R})$, the orthogonal group of orthogonal matrices, meaning $A^T A = I$.

Example 4.3.37

$SO(n) \leq GL_n(\mathbb{R})$, the special orthogonal group of orthogonal matrices with determinant 1.

Example 4.3.38

$U(n) \leq GL_n(\mathbb{C})$, the unitary group of unitary matrices, meaning $A^\dagger A = \bar{A}^T A = I$.

Example 4.3.39

$SU(n) \leq GL_n(\mathbb{C})$, the special group of unitary matrices with determinant 1.

Example 4.3.40

$\mathbb{Z}$ or any finite group, equipped with the discrete topology.

Example 4.3.41

We know that the center of $GL_n(\mathbb{F})$ is $Z(GL_n(\mathbb{F})) = \{\lambda I \mid \lambda \in F^\times\}$. We define the projective general linear group $PGL_n(\mathbb{F})$ to be the quotient $GL_n(\mathbb{F})/Z(GL_n(\mathbb{F}))$ (equipped with the quotient topology).

Example 4.3.42

The Heisenberg group $H_n(\mathbb{F}) \subseteq GL_n(\mathbb{F})$ is defined to be the set of upper triangular matrices with 1s on the diagonal, i.e. matrices of the following form:

$$\begin{pmatrix} 1 & ? & \dots & ? \\ 0 & 1 & & \\ \vdots & & \ddots & \\ 0 & & & 1 \end{pmatrix}$$

§4.4 Matrix Exponentiation

Definition 4.4.1. For $A \in M_n(\mathbb{F})$, its exponential, written e^A or $\exp(A)$, is defined to be the infinite sum

$$e^A := \sum_{k=0}^{\infty} \frac{1}{k!} A^k.$$

Remark 4.4.2 — For $n = 1$, this is the usual exponential.

Theorem 4.4.3

The power series for $\exp$ converges for all A . Moreover,

1. The power series for $\exp: M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$ converges absolutely, and uniformly on compact subsets of $M_n(\mathbb{F})$.
2. The power series for its partial derivatives (in the components of A) of $\exp$ all converge absolutely, and uniformly on compact subsets.

In particular, this implies that $\exp: M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$ is continuously differentiable.

Proof. The case for $n = 1$ is seen in analysis. The general case is longer but involves similar analysis. $\square$

Example 4.4.4

If $A = 0$, $e^A = I + 0 + 0 + \dots = I$. If $A = I$,

$$e^A = I + I + \frac{1}{2}I + \frac{1}{6}I + \dots = eI.$$

Example 4.4.5

Let

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

We compute e^A . We can see inductively that

$$A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}.$$

Then,

$$e^A = \sum_{i=0}^{\infty} \begin{pmatrix} \frac{1}{i!} & \frac{i}{i!} \\ 0 & \frac{1}{i!} \end{pmatrix} = \begin{pmatrix} e & e \\ 0 & e \end{pmatrix}.$$

Definition 4.4.6. For $A \in M_n(\mathbb{F})$, its logarithm, $\log(A)$ is defined to be the infinite sum

$$\log(A) := \sum_{k=1}^{\infty} (-1)^{k+1} \frac{(A-I)^k}{k} = (A-I) - \frac{(A-I)^2}{2} + \frac{(A-I)^3}{3} - \dots$$

Remark 4.4.7 — Unfortunately, this does not converge for all A ; however when A is close enough to I , all the same properties as we had for the exponential hold.

Theorem 4.4.8

There is an open neighborhood U of I in $M_n(\mathbb{F})$ such that the power series for $\log$ converges for all $A \in U$. Moreover,

1. The power series for $\log$ converges absolutely on U , and uniformly on compact subsets of U .
2. The power series for its partial derivatives (in the components of A) of $\log$ all converge absolutely, and uniformly on compact subsets of U .

In particular, this implies that $\log: U \rightarrow M_n(\mathbb{F})$ is continuously differentiable.

Example 4.4.9

Consider the case $n = 1$. Then this is the usual power series for $\log$; this converges for z if $|z - 1| < 1$.

Example 4.4.10

Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

We compute $\log(A)$. Note that $(A - I)^k = 0$ for $k \geq 2$, so only the first term of the sum is non-zero, so the series converges, and we find that $\log(A) = B$, where

$$B := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

In fact, we can compute that $e^B = A$; so in this case $e^{\log(A)} = A$.

Note that when both sums converge, $\exp$ and $\log$ are inverse to each other:

Proposition 4.4.11

Suppose the power series for $\log(A)$ is absolutely convergent. $\exp(\log(A)) = A$. Similarly, if the power series for $\log(\exp(B))$ is convergent, $\log(\exp(B)) = B$.

Proof. This is by calculation of writing a double sum. □

Proposition 4.4.12

Let $A, B \in M_n(\mathbb{F})$. If $AB = BA$, then $e^{A+B} = e^A e^B$.

Proof. We have the following calculation:

$$\begin{aligned}
 e^{A+B} &= \sum_{k=0}^{\infty} (A+B)^k \\
 &= \sum_{k=0}^{\infty} \frac{1}{k!} \left(A^k + \binom{k}{1} A^{k-1} B + \cdots + B^k \right) \\
 &= \sum_{k=0}^{\infty} \sum_{l=0}^k \frac{1}{k!} \binom{k}{l} A^{k-l} B^l \\
 &= \left(\sum_{i=0}^{\infty} \frac{1}{i!} B^i \right) \left(\sum_{j=0}^{\infty} \frac{1}{j!} A^j \right) \\
 &= e^A e^B.
 \end{aligned}$$

Here we use the fact that $AB = BA$ to go from the first to the second line. Also note that when we rearrange these infinite series, we needed to use the fact that these series absolutely converge. $\square$

Remark 4.4.13 — The condition that $AB = BA$ is necessary.

Corollary 4.4.14

For any $A \in M_n(\mathbb{F})$, $e^A e^{-A} = e^0 = I$. In particular, e^A is always invertible. So $\exp$ is really a map $\exp: M_n(\mathbb{F}) \rightarrow GL_n(\mathbb{F})$.

Proof. A and $-A$ always commute, so $e^A e^{-A} = e^{A-A} = e^0 = I$. $\square$

Remark 4.4.15 — The last part just says that after applying $\exp$, we can get an invertible matrix.

Example 4.4.16

For $n = 1$, this just says that $e^\lambda e^{-\lambda} = 1$ for all $\lambda \in \mathbb{F}$.

Example 4.4.17

A and A^k always commute, so A and e^A also commute. This implies the identity:

$$e^{Ae^A} = e^A \cdot e^{e^A}.$$

Lemma 4.4.18

Let $A, B \in M_n(\mathbb{F})$, and suppose that $A \in GL_n(\mathbb{F})$ is invertible. Then

$$e^{ABA^{-1}} = A e^B A^{-1}.$$

Proof. First note that $(ABA^{-1})^k = AB^kA^{-1}$. Thus,

$$\begin{aligned} e^{ABA^{-1}} &= \sum_{k=0}^{\infty} \frac{1}{k!} (ABA^{-1})^k \\ &= \sum_{k=0}^{\infty} \frac{1}{k!} AB^kA^{-1} \\ &= Ae^BA^{-1}. \end{aligned}$$

□

Lemma 4.4.19

For any $A \in M^n$, $(e^A)^T = e^{A^T}$.

Proof. Note that $(A^T)^k = (A^k)^T$, for all k . Thus,

$$e^{A^T} = \left[\sum_{k=0}^{\infty} \frac{1}{k!} (A^k)^T \right]^T = (e^A)^T$$

□

Example 4.4.20

Suppose A is an upper triangular matrix:

$$A = \begin{pmatrix} \lambda_1 & & ? \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

Then, A^k is of a similar form:

$$A^k = \begin{pmatrix} \lambda_1^k & & ? \\ & \ddots & \\ 0 & & \lambda_n^k \end{pmatrix}.$$

Therefore,

$$e^A = \begin{pmatrix} \sum_{k=0}^{\infty} \frac{1}{k!} \lambda_1^k & & ? \\ & \ddots & \\ 0 & & \sum_{k=0}^{\infty} \frac{1}{k!} \lambda_n^k \end{pmatrix},$$

which is also upper triangular. From this description though we can immediately see the trace, determinant and eigenvalues, and in fact

$$\det(e^A) = e^{\lambda_1} \dots e^{\lambda_n} = e^{\text{Tr}(A)}$$

Lemma 4.4.21

For any $A \in M_n(\mathbb{F})$, we have the identity

$$\det(e^A) = e^{\text{Tr}(A)}$$

Proof. Write $A = PBP^{-1}$, where $P \in GL_n(\mathbb{C})$ (even if $\mathbb{F} = \mathbb{R}$ we can do this) and B is in Jordan normal form (upper triangular will suffice).

$$\det(e^A) = \det(e^{PBP^{-1}}) = \det(Pe^BP^{-1}) = \det(e^B),$$

and

$$e^{\text{Tr}(A)} = e^{\text{Tr}(PBP^{-1})} = e^{\text{Tr}(B)}.$$

Therefore, from the previous example, $\det(e^B) = e^{\text{Tr}(B)}$, so it follows that $\det(e^A) = e^{\text{Tr}(A)}$. $\square$

Now recall that $\exp: M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$ is differentiable. For any $A \in M_n(\mathbb{F})$, by differentiability of $\exp$, the function $\gamma_A(t) := e^{tA}$ is a continuously differentiable function $\mathbb{R} \rightarrow GL_n(\mathbb{F})$.

Notation 4.4.22. We will continue with the notation $\gamma_A(t) = e^{tA}$ from here (for $A \in M_n(\mathbb{F})$).

Lemma 4.4.23

For any $A \in M_n(\mathbb{F})$, the derivative of γ_A at 0 is $\dot{\gamma}_A(0) = A$.

Proof. We have

$$\begin{aligned} \dot{\gamma}_A(0) &= \lim_{t \rightarrow 0} \frac{e^{tA} - e^0}{t} \\ &= \lim_{t \rightarrow 0} \frac{\sum_{k=0}^{\infty} \frac{1}{k!} t^k A^k - I}{t} \\ &= \lim_{t \rightarrow 0} \sum_{k=1}^{\infty} \frac{1}{k!} t^{k-1} A^k \\ &= A \end{aligned}$$

$\square$

Lemma 4.4.24

For any $A \in M_n(\mathbb{F})$, $\gamma_A(t) = e^{tA}$ is a continuous (group) homomorphism $\gamma_A: \mathbb{R} \rightarrow GL_n(\mathbb{F})$.

Proof. Since $\exp$ is continuous, γ_A is too, so we just need to check γ_A is a homomorphism.

$$\gamma_A(s+t) = e^{(s+t)A} = e^{sA} \cdot e^{tA} = \gamma_A(s) \cdot \gamma_A(t),$$

where we use the fact that sA and tA commute. So γ_A is a homomorphism. $\square$

Corollary 4.4.25

For any $t_0 \in \mathbb{R}$, the derivative of γ_A at t_0 is $\dot{\gamma}_A(t_0) = A \cdot e^{t_0 A} = A \cdot \gamma_A(t_0)$.

Proof. Since γ_A is a homomorphism,

$$\gamma_A(t) = (t - t_0)\gamma_A(t_0).$$

Differentiating both sides at $t = t_0$ and using the $t_0 = 0$ case, we find that

$$\dot{\gamma}_A(t_0) = \dot{\gamma}_A(0) \cdot \gamma_A(t_0) = A \cdot e^{t_0 A}.$$

□

Remark 4.4.26 — Note that A and $e^{t_0 A}$ commute:

$$A \cdot e^{t_0 A} = e^{t_0 A} A.$$

In particular, this tells us the derivative of $\exp$ at 0. Recall that this is a linear map $d\exp_0: M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$.

Corollary 4.4.27

The derivative of $\exp$ at 0, $d\exp_0: M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$, is the identity map.

Proof. By the chain rule, the derivative of $\gamma_A(t) = e^{tA}$ at $t = 0$ is $d\exp_0(A)$. So $A = d\exp_0(A)$ for all $A \in M_n(\mathbb{F})$, i.e. $d\exp_0$ is the identity map. □

Proposition 4.4.28

$\exp$ restricts to a continuously differentiable homeomorphism from an open neighborhood of 0 in $M_n(\mathbb{F})$ to an open neighborhood of I in $GL_n(\mathbb{F})$, with continuously differentiable inverse.

Proof. $\exp$ sends 0 to I , and is differentiable at 0, and its derivative at 0, $d\exp_0$, is an invertible linear map. The result then follows from the inverse function theorem. □

Remark 4.4.29 — In the proof above, we could alternatively use $\log$ to construct an inverse.

Example 4.4.30

For $n = 1$, we can take $U = (-1, 1)$:

$$\exp|_{(-1,1)}: (-1, 1) \mapsto (e^{-1}, e^1) =: V$$

is a homeomorphism, with inverse $\log$, which is continuously differentiable.

Remark 4.4.31 — Recall that we showed that any continuous homomorphism $\phi: (\mathbb{R}, +) \rightarrow (\mathbb{R}, +)$ is of the form $\phi(t) = \lambda t$ for some $\lambda \in \mathbb{R}$.

Claim 4.4.32. Any continuous homomorphism $\psi: (\mathbb{R}, +) \rightarrow (\mathbb{R}^\times, \cdot)$ is of the form $\psi(t) = e^{\lambda t}$ for some $\lambda \in \mathbb{R}$.

Proof of Claim. Let $\psi: (\mathbb{R}, +) \rightarrow (\mathbb{R}^\times, \cdot)$ be a continuous homomorphism. $\psi(0) = 1 > 0$. Thus, by the intermediate value theorem, $\psi(t) > 0$ for all $t \in \mathbb{R}$, as otherwise there would be some $s \in \mathbb{R}$ such that $\psi(s) = 0$, a contradiction.

Let $\phi: \mathbb{R} \rightarrow \mathbb{R}$ be defined to send $t \in \mathbb{R}$ to $\log(\psi(t))$. This is defined since $\psi(t)$ is always positive. Then ϕ is continuous, and we'll show it's a homomorphism: let $s, t \in \mathbb{R}$. Then

$$\phi(s + t) = \log(\psi(s + t)) = \log(\psi(s) \cdot \psi(t)) = \phi(s) + \phi(t).$$

So ϕ is a continuous homomorphism $(\mathbb{R}, +) \rightarrow (\mathbb{R}, +)$, so $\phi(t) = \lambda t$ for all t , for some λ . Thus, $\psi(t) = e^{\phi(t)} = e^{\lambda t}$. $\square$

Here, we used the fact that $\log$ was defined on all of $\mathbb{R} > 0$, which isn't true for $n > 0$; similarly if we wanted to study continuous homomorphisms $\mathbb{R} \rightarrow GL_1(\mathbb{C}) = \mathbb{C}^\times$, we'd need to e.g. carefully choose a branch of logarithm.

Theorem 4.4.33

Let $\gamma: \mathbb{R} \rightarrow GL_n(\mathbb{F})$ be a continuous homomorphism of topological groups. Then $\gamma(t) = e^{tA}$, for some $A \in M_n(\mathbb{F})$.

Proof. First, choose open neighborhoods U of 0 in $M_n(\mathbb{F})$ and V of I in $GL_n(\mathbb{F})$ such that $\exp(U) \subset V$ and $\exp$ restricts to a homeomorphism

$$\exp|_U: U \rightarrow V.$$

(the existence of these neighborhoods was shown in class based on the inverse function theorem and the computation of the differential of $\exp$ at 0.)

By continuity of γ , there exists $\delta > 0$ such that $\gamma([- \delta, \delta]) \subset V$. Let $\beta: [- \delta, \delta] \rightarrow U$ be defined as:

$$\beta(t) := \exp|_U^{-1}(\gamma(t)),$$

i.e. β is defined so that $\gamma(t) = e^{\beta(t)}$, whenever $|t| \leq \delta$. Further, we can shrink δ and assume that for all $t_1, t_2 \in [- \delta, \delta]$, $\beta(t_1) + \beta(t_2) \in U$ (using the continuity of $+$).

Claim 4.4.34. For all $r \in [0, 1]$ and all $t \in [- \frac{\delta}{2}, \frac{\delta}{2}]$, $\beta(rt) = r\beta(t)$.

Proof of Claim 4.4.34. Step 1: We first show that for all $t \in [- \delta, \delta]$, we have $\beta(t) = 2\beta(\frac{t}{2})$. Note that both $\beta(t)$ and $2\beta(\frac{t}{2})$ are in U . To show that they are equal, it is enough to show that their exponentials are equal, and this in turn follows from the calculations:

$$\begin{aligned} \exp(\beta(t)) &= \gamma(t) \\ \exp\left(2\beta\left(\frac{t}{2}\right)\right) &= \left(\exp\left(\beta\left(\frac{t}{2}\right)\right)\right)^2 = \gamma\left(\frac{t}{2}\right)^2 = \gamma(t). \end{aligned}$$

Thus, $\beta(t) = 2\beta(\frac{t}{2})$.

Step 2: We show that for all $t \in [-\frac{\delta}{2}, \frac{\delta}{2}]$, and for all integers k, p with $0 \leq p \leq 2^k$, $\beta\left(\frac{p}{2^k}t\right) = \frac{p}{2^k}\beta(t)$.

We prove this by induction on k . The base case is proven in the prior step. Thus, assume the claim is true for all the way up to $k-1$. If p is even, we're done by the inductive hypothesis. Next, suppose that p is odd. We have

$$\frac{p}{2^k} = \frac{1}{2} \left(\frac{p-1}{2^k} + \frac{p+1}{2^k} \right).$$

Note that our inductive hypothesis tells us the following:

$$\beta\left(\frac{p-1}{2^k}t\right) = \frac{p-1}{2^k}\beta(t), \quad \beta\left(\frac{p+1}{2^k}t\right) = \frac{p+1}{2^k}\beta(t)$$

By the first step, it's enough to show that

$$\beta\left(\frac{p}{2^{k-1}}t\right) = \beta\left(\frac{p-1}{2^k}t\right) + \beta\left(\frac{p+1}{2^k}t\right).$$

Note that we are also using $t \in [-\frac{\delta}{2}, \frac{\delta}{2}]$, because we multiplied by 2. Both the LHS and the RHS are in U , so it's enough to show their exponentials are equal. First we examine the LHS:

$$\exp\left(\beta\left(\frac{p}{2^{k-1}}t\right)\right) = \exp\left(p\beta\left(\frac{t}{2^{k-1}}\right)\right) = \left(\exp\left(\beta\left(\frac{t}{2^{k-1}}\right)\right)\right)^p = \left(\gamma\left(\frac{t}{2^{k-1}}\right)\right)^p.$$

Where we used induction on k and Step 1. Next we examine the RHS. Let $p = 2q + 1$. Then, we have the following:

$$\begin{aligned} \exp\left(\beta\left(\frac{p-1}{2^k}t\right) + \beta\left(\frac{p+1}{2^k}t\right)\right) &\stackrel{(*)}{=} \exp\left(q\beta\left(\frac{t}{2^{k-1}}\right) + (q+1)\beta\left(\frac{t}{2^{k-1}}\right)\right) \\ &\stackrel{(\dagger)}{=} \exp\left(q\beta\left(\frac{t}{2^{k-1}}\right)\right) \exp\left((q+1)\beta\left(\frac{t}{2^{k-1}}\right)\right) \\ &= \gamma\left(\frac{t}{2^{k-1}}\right)^q \gamma\left(\frac{t}{2^{k-1}}\right)^{q+1} \\ &= \left(\gamma\left(\frac{t}{2^{k-1}}\right)\right)^{2q+1} \end{aligned}$$

Finally, as $p = 2q + 1$, we're done. Note that the equality $(*)$ is justified by some calculations using the inductive hypothesis and the first step. The equality $(\dagger)$ is justified by the commutativity of the two matrices that appear.

Step 3: We have shown that Claim 4.4.34 holds for all r rational of the form $\frac{p}{2^k}$. The set of these rational numbers is dense in $[0, 1]$, and hence the result follows by continuity of β . □

Claim 4.4.35. For all $t \in \mathbb{R}$, we have that

$$\gamma(t) = e^{t \cdot \frac{1}{\delta} \beta(\delta)}.$$

Proof. Let $t \in \mathbb{R}$. Then there is some large integer $N > 0$ such that $|\frac{t}{N}| \leq \delta$. Equivalently, we have $\frac{t}{N} = r\delta$ for some $r \in [-1, 1]$. So by the previous claim,

$$\gamma\left(\frac{t}{N}\right) = \gamma(r\delta) = e^{\beta(r\delta)} = e^{r\beta(\delta)}.$$

Thus,

$$\gamma(t) = \gamma\left(\frac{t}{N}\right)^N = e^{rN\beta(\delta)} = e^{t\frac{1}{\delta}\beta(\delta)},$$

since $\frac{t}{N} = r\delta$. $\square$

This second claim (4.4.35) then implies that the theorem holds, for $A = \frac{1}{\delta}\beta(\delta)$. $\square$

Remark 4.4.36 — In particular, this implies that γ is continuously differentiable, which is not at all obvious. Further, this means that we can use exponential functions to describe all continuous group homomorphisms from $\mathbb{R} \rightarrow GL_n(\mathbb{F})$.

Corollary 4.4.37

Let $G \leq GL_n(\mathbb{F})$ be a closed subgroup. Then any continuous homomorphism $\gamma: \mathbb{R} \rightarrow G$ is of the form $\gamma = \gamma_A$ for some $A \in GL_n(\mathbb{F})$.

Proof. Since $G \leq GL_n(\mathbb{F})$, any continuous homomorphism $\gamma: \mathbb{R} \rightarrow G$ is in particular a continuous homomorphism $\mathbb{R} \rightarrow GL_n(\mathbb{F})$, so the result follows from the theorem. $\square$

Example 4.4.38

We find all the continuous homomorphisms $\gamma: \mathbb{R} \rightarrow SL_n(\mathbb{R})$.

Solution.

Claim 4.4.39. For $B \in M_n(\mathbb{R})$, we have that $e^B \in SL_n(\mathbb{R})$ if and only if $\text{Tr}(B) = 0$.

Proof. $\det(e^B) = e^{\text{Tr}(B)}$ by an earlier lemma, so this is 1 if and only if $\text{Tr}(B) = 0$ (since $\text{Tr}(B)$ is a real number). $\square$

Thus $\gamma_A(t) = e^{tA} \in SL_n(\mathbb{R})$ if and only if $\text{Tr}(tA) = 0$, so γ_A is a continuous homomorphism $\mathbb{R} \rightarrow SL_n(\mathbb{R})$ if and only if $\text{Tr}(A) = 0$. $\square$

§4.5 The Lie Algebra of a Matrix Lie Group

Definition 4.5.1. Let $G \leq GL_n(\mathbb{F})$ be a closed subgroup, and $A \in G$. Then, **the tangent space** $T_A G$ of G at A is the subset of $\dot{\gamma}(0) \in M_n(\mathbb{F})$, for $\gamma: \mathbb{R} \rightarrow G$ a continuously differentiable (C^1) path sending 0 to A . (note we don't require γ to be a homomorphism)

Remark 4.5.2 — Note that this subset was assigned to matrix Lie groups inside GL_n , not an abstract matrix Lie group.

Proposition 4.5.3

For any $A \in G$,

$$T_A G = A \cdot T_I G = T_I G \cdot A.$$

Proof. Let M lie in $T_I G$. Then $M = \dot{\gamma}(0)$ for some continuously differentiable function $\gamma: \mathbb{R} \rightarrow G$ sending 0 to I . Consider $A \cdot \gamma: \mathbb{R} \rightarrow G$, sending t to $A \cdot \gamma(t)$. This is continuously differentiable and sends 0 to A . Therefore, $\frac{d}{dt} \big|_{t=0} A \gamma(t) = A \dot{\gamma}(0)$ lies in $T_A G$. This shows $A \cdot T_I G \subseteq T_A G$. The same argument with A^{-1} shows that $A^{-1} \cdot T_A G \subseteq T_I G$, so $T_A G = A \cdot T_I G$, and a similar argument shows that $T_A G = (T_I G) \cdot A$. $\square$

Proposition 4.5.4

For any A , $T_A G$ is a real vector space.

Proof. We prove this first when $A = I$, the general case then follows from the previous proposition. Let $M, N \in T_I G$, and $\lambda \in \mathbb{R}$. We need to show that $M + N \in T_I G$ and $\lambda M \in T_I G$. Pick continuously differentiable paths $\alpha, \beta: \mathbb{R} \rightarrow G$ sending 0 to I , with $\dot{\alpha}(0) = M$ and $\dot{\beta}(0) = N$. Let $\gamma(t) = \alpha(\lambda t)$. Then $\gamma: \mathbb{R} \rightarrow G$ is a continuously differentiable path with $\gamma(0) = I$ and $\dot{\gamma}(0) = \lambda \dot{\alpha}(0) = \lambda M$, so $\lambda M \in T_I G$. Now let $\delta(t) = \alpha(t)\beta(t)$. Then $\delta: \mathbb{R} \rightarrow G$ is a continuously differentiable path with $\delta(0) = I$ and, using the product rule,

$$\dot{\delta}(0) = \dot{\alpha}(0)\beta(0) + \alpha(0)\dot{\beta}(0) = MI + IN = M + N \in T_I G.$$

$\square$

Remark 4.5.5 — In the above, even if $\mathbb{F} = \mathbb{C}$, $T_I G$ might not be a complex vector space.

Definition 4.5.6. For $G \leq GL_n(\mathbb{F})$ a closed subgroup, its **dimension** is the dimension of $T_A G$, as a real vector space.

Recall the commutator of two matrices A and B is $[A, B] := AB - BA$.

Proposition 4.5.7

Let $G \leq GL_n(\mathbb{F})$ be a closed subgroup, and let $M, N \in T_I G$. Then $[M, N] \in T_I G$.

Proof. Pick continuously differentiable paths $\alpha, \beta: \mathbb{R} \rightarrow G$ sending 0 to I , with $\dot{\alpha}(0) = M$ and $\dot{\beta}(0) = N$. For $s, t \in \mathbb{R}$, let $\delta_s(t) = \alpha(s)\beta(t)\alpha(s)^{-1}$. This defines a continuously differentiable function $\mathbb{R}^2 \rightarrow G$. For fixed s , δ_s is a continuously differentiable path $\mathbb{R} \rightarrow G$ so $\dot{\delta}_s(0) \in T_I G$. We can compute this derivative with the product rule: $\dot{\delta}_s(0) = \alpha(s)\dot{\beta}(0)\alpha(s)^{-1} = \alpha(s) \cdot N \cdot \alpha(s)^{-1}$. Allowing s to vary, $\dot{\delta}_s(0)$ defines a continuously differentiable path $\mathbb{R} \rightarrow T_I G$. So its derivative at 0 must also lie in $T_I G$, i.e.

$$\frac{d}{ds} \bigg|_{s=0} \dot{\delta}_s(0) \in T_I G.$$

Note that by the chain rule, $\frac{d}{ds} \big|_{s=0} \alpha(s)^{-1} = -\dot{\alpha}(0)$. Then, by the product rule, we have

$$\begin{aligned} \frac{d}{ds} \bigg|_{s=0} \dot{\delta}_s(0) &= \left(\frac{d}{ds} \bigg|_{s=0} \dot{\alpha}(s) \right) \cdot N \cdot \alpha(0) + \alpha(0) \cdot N \cdot \left(\frac{d}{ds} \bigg|_{s=0} \dot{\alpha}(-s) \right) \\ &= \dot{\alpha}(0) \cdot N \cdot \alpha(0) - \alpha(0) \cdot N \cdot \dot{\alpha}(0) \\ &= M \cdot N \cdot I - I \cdot N \cdot M \\ &= [M, N], \end{aligned}$$

which therefore lives in $T_I G$. □

Proposition 4.5.8

For matrices A, B, C in $M_n(\mathbb{F})$, the **Jacobi identity** holds:

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0.$$

Proof. □

Definition 4.5.9. A (real) **Lie algebra** is a pair $(V, [\cdot, \cdot])$, where V is a real vector space, and $[\cdot, \cdot]: V \times V \rightarrow \mathbb{R}$ is a bilinear map, such that the following hold:

- $[\cdot, \cdot]: V \times V \rightarrow \mathbb{R}$ is antisymmetric: $[A, B] = -[B, A]$, for all $A, B \in V$.
- The Jacobi identity holds:

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$$

Abuse of Notation 4.5.10. We will frequently drop $[\cdot, \cdot]$ from the notation and refer to the Lie algebra just as V .

Definition 4.5.11. A **homomorphism of Lie algebras** from V to W consists of a linear map $f: V \rightarrow W$, which respects the brackets, meaning that $f([A, B]_V) = [f(A), f(B)]_W$ for all $A, B \in V$.

Definition 4.5.12. For a closed subgroup $G \leq GL_n(\mathbb{F})$, its **Lie algebra** is defined to be the vector space $T_I G$, with $[\cdot, \cdot]$ given by the commutator of matrices.

Notation 4.5.13. We've been using capital letters to denote matrix Lie groups (both abstract ones, like G , and the standard ones, like $GL_n(\mathbb{R})$ and $SU(2)$). A common convention is to use lowercase Fraktur font to denote Lie algebras. So the Lie algebra of $GL_n(\mathbb{R})$ is written $\mathfrak{gl}_n(\mathbb{R})$, the Lie algebra of $SU(2)$ is written $\mathfrak{su}(2)$, and often the Lie algebra of G or H is written $\mathfrak{g}$ or $\mathfrak{h}$ respectively.

Remark 4.5.14 — In other words, a Lie algebra of a set G is

$$\mathfrak{g} = \text{Lie}(G) = \{X \in M_n(\mathbb{F}) \mid \exists \gamma: (-\epsilon, \epsilon) \rightarrow G \text{ such that } \gamma(0) = I, \dot{\gamma}(0) = X\},$$

where γ is a C^1 -path.

Example 4.5.15

Consider $U_n \leq GL_n(\mathbb{C}) = \{A \in M_n(\mathbb{C}) \mid \bar{A}^T A = I\}$. Then, let $\gamma_A(t)$ be a path such that $\gamma_A(0) = I$ and $\dot{\gamma}_A(0) = A$. Then, we have that

$$0 = \frac{d}{dt} I \big|_{t=0} = \frac{d}{dt} \gamma_{\bar{A}^T} \gamma_A \big|_{t=0} = \dot{\gamma}_{\bar{A}^T} \gamma_A + \gamma_{\bar{A}^T} \dot{\gamma}_A \big|_{t=0} = \bar{A}^T \cdot I + A \cdot I.$$

Therefore,

$$\mathfrak{u}_n = \{A \in M_n(\mathbb{C}) \mid \bar{A}^T + A = 0\}$$

Note that this implies, for all i , $a_{ii} \in \mathbb{R}$. We also note that $\mathfrak{u}_n$ is a real vector space and U_n is a *real* Lie group.

Example 4.5.16

Consider SL_n . We have

$$1 = \det(\gamma(t)) = \det(I + tX + o(t)) = 1 + t \cdot \text{Tr}(X) + o(t).$$

This implies that $\text{Tr}(X) = 0$. (in fact, if the trace is 0, the converse is also true)

Example 4.5.17

Let $G = O_n(\mathbb{R}) \subset GL_n(\mathbb{R})$. We have $\text{Lie}(G) = \mathfrak{so}_n(\mathbb{R}) = \{X \in M_n(\mathbb{R}) \mid X^T + X = 0\}$, the set of antisymmetric matrices. Thus, $X \in \text{Lie}(G)$ implies that X is antisymmetric.

For the converse, suppose $X^T + X = 0$. Then, X^T and X commute, and so we consider $\gamma(t) = e^{tX}$. We claim that $\gamma(t) \in O_n(\mathbb{R})$. Indeed,

$$\gamma(t)^T \cdot \gamma(t) = (e^{tX})^T e^{tX} = e^{tX^T} e^{tX} = e^{t(X^T + X)} = I.$$

Since $\gamma(0) = I$, $\dot{\gamma}(0) = X$, $X \in \mathfrak{so}_n(\mathbb{R})$.

The next goal we'll work towards is to characterize $T_I G$ in an alternative way.

Proposition 4.5.18

Let $\gamma: \mathbb{R} \rightarrow GL_n(\mathbb{F})$ be a continuously differentiable function sending 0 to I , and let $t \in \mathbb{R}$. Then

$$e^{t\dot{\gamma}(0)} = \lim_{n \rightarrow \infty} \gamma\left(\frac{t}{n}\right)^n.$$

In particular, the limit on the right hand side exists.

Proof. We can write $\gamma(t) = e^{\beta(t)}$, for $|t| \leq \delta$, for some $\delta > 0$ and some continuously differentiable function $\beta: [-\delta, \delta] \rightarrow M_N(\mathbb{F})$. Since the derivative of $\exp$ at 0, $d\exp_0$ is the identity map $M_N(\mathbb{F}) \rightarrow M_N(\mathbb{F})$, by the chain rule, $\dot{\beta}(0) = \dot{\gamma}(0)$. Let $M = \dot{\beta}(0) = \dot{\gamma}(0)$. Then,

$$\begin{aligned} \lim_{n \rightarrow \infty} \gamma\left(\frac{t}{n}\right)^n &= \lim_{n \rightarrow \infty} \left(e^{\beta(\frac{t}{n})}\right)^n \\ &= \lim_{n \rightarrow \infty} \left(e^{\frac{t}{n}\dot{\gamma}(0) + o(\frac{t}{n})}\right)^n \\ &= \lim_{n \rightarrow \infty} e^{t\dot{\gamma}(0) + o(\frac{t}{n})} \\ &= e^{t\dot{\gamma}(0)}. \end{aligned}$$

Particularly, this limit exists. □

Example 4.5.19

Suppose $n = 1$, and $\gamma(t) = 1 + t$. Then, the proposition says that

$$e^t = \lim_{n \rightarrow \infty} \left(1 + \frac{t}{n}\right)^n,$$

which is already known.

Proposition 4.5.20

Let $G \leq GL_n(\mathbb{F})$ be a closed subgroup, and let $M \in \mathfrak{g} = T_I G$. Then $e^M \in G$.

Proof. Let $\gamma: \mathbb{R} \rightarrow G$ be a continuously differentiable curve sending 0 to I , with $\dot{\gamma}(0) = M$. Then,

$$e^{tM} = \lim_{n \rightarrow \infty} \gamma\left(\frac{t}{n}\right)^n,$$

and the limit on the RHS exists. Each $\gamma\left(\frac{t}{n}\right)^n$ lives in G , and since $G \subseteq GL_n(\mathbb{F})$ is a closed subset, the limit lives in G as well. $\square$

Corollary 4.5.21

The continuous homomorphisms $\gamma: \mathbb{R} \rightarrow G$ are exactly those given by $\gamma_A(t) = e^{tA}$, for some A in $\mathfrak{g} = T_I G$.

Proof. We first prove any such continuous homomorphism is of this form. Any continuous homomorphism $\gamma: \mathbb{R} \rightarrow G$ is in particular a continuous homomorphism to $GL_n(\mathbb{F})$ (since $G \leq GL_n(\mathbb{F})$), so by the earlier theorem, $\gamma(t) = \gamma_A(t) = e^{tA}$ for some $A \in M_n(\mathbb{F})$. But γ_A is always continuously differentiable, so $\dot{\gamma}_A(0) \in \mathfrak{g}$. But this derivative is A , so $A \in \mathfrak{g}$. Conversely, if $A \in \mathfrak{g}$, then $tA \in \mathfrak{g}$ for all $t \in \mathbb{R}$, so by the previous proposition, $\gamma_A(t) = e^{tA} \in G$ for all t . We also showed earlier that γ_A defines a continuous homomorphism $\mathbb{R} \rightarrow GL_n(\mathbb{F})$. Since it lands in G when $A \in \mathfrak{g}$, this implies that γ_A is a continuous homomorphism $\mathbb{R} \rightarrow G$. $\square$

This corollary says there is a bijection

$$\mathfrak{g} \longleftrightarrow \{\text{continuous homomorphisms } \mathbb{R} \rightarrow G\}$$

sending $A \in \mathfrak{g}$ to γ_A , and in the other direction sending γ to $\dot{\gamma}(0)$.

Remark 4.5.22 — We could have defined $\mathfrak{g}$ to be the set of continuous homomorphisms $\mathbb{R} \rightarrow G$. However with this definition it would be less clear how to define the bracket $[\cdot, \cdot]$, nor even why $\mathfrak{g}$ is a vector space.

Remark 4.5.23 — Let $\gamma, \delta: \mathbb{R} \rightarrow G$ two continuous homomorphisms, where $G \leq GL_n(\mathbb{F})$ is a closed subgroup. Then they're continuously differentiable, and $\dot{\gamma}(0) + \dot{\delta}(0) \in \mathfrak{g} = T_s IG$, which corresponds, by taking exponentials, to some continuous homomorphism $\xi: \mathbb{R} \rightarrow G$ (by defining $\xi(t) = e^{t(\dot{\gamma}(0) + \dot{\delta}(0))}$).

Recall that when we showed $\mathfrak{g}$ (which we were calling $T_I G$ back then) is a vector space, we showed that $\eta(t) = \gamma(t) \cdot \delta(t)$ is a continuously differentiable map $\mathbb{R} \rightarrow G$ with

$$\dot{\eta}(0) = \dot{\gamma}(0) + \dot{\delta}(0) \in T_I G = \mathfrak{g}.$$

But since G might not be abelian, η isn't necessarily a homomorphism. So we might not have that $\xi = \eta$, but $\xi(t) = e^{t\dot{\eta}(0)}$. The formula in the corollary lets us express $e^{t\dot{\eta}(0)}$ in terms of η :

$$\xi(t) = e^{t\dot{\eta}(0)} = \lim_{n \rightarrow \infty} \left(\gamma\left(\frac{t}{n}\right) \cdot \delta\left(\frac{t}{n}\right) \right)^n$$

Theorem 4.5.24

Let $G \leq GL_n(\mathbb{F})$ be a closed subgroup. There are open neighborhoods U of 0 in $\mathfrak{g} = T_I G$ and $V = \exp(U)$ of I in G such that $\exp$ restricts to a homeomorphism $\exp|_U: U \rightarrow V$.

Proof (non-examinable). First, we pick a complementary (real) vector subspace W to $\mathfrak{g}$ in $M_n(\mathbb{F})$, i.e. a subspace $W \leq M_n(\mathbb{F})$ such that $\mathfrak{g} \oplus W = M_n(\mathbb{F})$ (i.e. $\mathfrak{g}$ and W together span $M_n(\mathbb{F})$, and their intersection is $\{0\}$). Define a function $\widetilde{\exp}: \mathfrak{g} \oplus W \rightarrow GL_n(\mathbb{F})$ to send (X, Y) (where $X \in \mathfrak{g}$ and $Y \in W$) to $e^X \cdot e^Y$. Note that since X and Y might not commute, this isn't necessarily the same as $\exp$, which sends (X, Y) to e^{X+Y} . So $\widetilde{\exp}$ is a modified version of the exponential map. Note that $\exp$ and $\widetilde{\exp}$ agree on $\mathfrak{g}$, so crucially if the theorem is true for $\widetilde{\exp}$, it's true for $\exp$ too. So we'll prove the theorem with this modified version of the exponential map, $\widetilde{\exp}$.

Claim 4.5.25. The derivative of the modified exponential map at 0, $d\widetilde{\exp}_0: M_n(\mathbb{F}) \rightarrow \mathbb{F}$, is the identity.

Proof of Claim 4.5.25. Let $(X, Y) \in \mathfrak{g} \oplus W = M_n(\mathbb{F})$. Let $\gamma(t) = e^{tX} \cdot e^{tY} = \widetilde{\exp}(X, Y)$. Then,

$$\dot{\gamma}(0) = \left(\frac{d}{dt} \Big|_{t=0} e^{tX} \right) \cdot e^{0 \cdot Y} + e^{0 \cdot X} \cdot \left(\frac{d}{dt} \Big|_{t=0} e^{tY} \right) = X \cdot I + I \cdot Y = X + Y.$$

But by the chain rule, $\dot{\gamma}(0) = d\widetilde{\exp}_0(X, Y)$. So $d\widetilde{\exp}_0(X, Y) = X + Y$. $\square$

Similarly to with $\exp$, by the inverse function theorem, there's a neighborhood $\tilde{U}$ of 0 in $M_n(\mathbb{F})$ and a neighborhood $\tilde{V} = \widetilde{\exp}(\tilde{U})$ of I in $GL_n(\mathbb{F})$ such that

$$\widetilde{\exp}_{\tilde{U}}: \tilde{U} \rightarrow \tilde{V}$$

is a homeomorphism.

Claim 4.5.26. There's a neighborhood Z of 0 in $\tilde{U} \subseteq M_n(\mathbb{F})$ such that

$$\widetilde{\exp}^{-1}(G) \cap Z = \mathfrak{g} \cap Z.$$

In words, this says that for $(X, Y) \in Z \subseteq \mathfrak{g} \oplus W$, we have that $\widetilde{\exp}(X, Y) \in G$ if and only if $Y = 0$.

Proof of Claim 4.5.26. Suppose this claim weren't true. This means that for every neighborhood Z of 0 in $M_n(\mathbb{F})$, there's some $(X, Y) \in \mathfrak{g} \oplus W = M_n(\mathbb{F})$ with $Y \neq 0$ and $\widetilde{\exp}(X, Y) = e^X \cdot e^Y \in G$. This means there's a sequence of vectors $(X_i, Y_i) \in \mathfrak{g} \oplus W$ converging to 0, such that $Y_i \neq 0$ for all i , and such that $\widetilde{\exp}(X_i, Y_i) \in G$ (i.e. $e^{X_i} \cdot e^{Y_i} \in G$) for all i . $-X_i \in \mathfrak{g}$ too, so $e^{-X_i} \in G$, so $e^{-X_i} \cdot e^{X_i} \cdot e^{Y_i} \in G$ too. The idea now is to use this sequence Y_i to find some non-zero $Y \neq 0 \in W$ such that $e^{tY} \in G$ for all t . However this implies that $Y \in \mathfrak{g}$, but $\mathfrak{g} \cap W = \{0\}$, so this is a contradiction. Since each $Y_i \neq 0 \in W$ is non-zero, each $\frac{Y_i}{|Y_i|} \in S^d$ (for $d = \dim(W) - 1$), where $S^d = \{y \in W \mid |y| = 1\}$. These $\frac{Y_i}{|Y_i|}$ are the Y_i , *normalized*. By sequential compactness of S^d , after passing to some subsequence,

$$\frac{Y_i}{|Y_i|} \rightarrow Y \in W$$

for some $Y \in S^d \subseteq W$ with $|Y| = 1$, so in particular $Y \neq 0$. We'll show that $e^{tY} \in G$ for all t , implying $Y \in \mathfrak{g}$, a contradiction as $Y \neq 0 \in W$. Fix $t \in \mathbb{R}$. Choose integers m_i such that $m_i|Y_i|$; we can do this by choosing the m_i such that

$$m_i|Y_i| \leq t \leq (m_i + 1)|Y_i|$$

for all i . Now consider $e^{m_i Y_i}$. We have that $e^{m_i Y_i} = (e^{Y_i})^{m_i} \in G$ since $e^{Y_i} \in G$ for all i . But

$$m_i Y_i = (m_i|Y_i|) \frac{Y_i}{|Y_i|} \rightarrow tY$$

since the first term $m_i|Y_i| \rightarrow t$ and the second term $\frac{Y_i}{|Y_i|} \rightarrow Y$. This implies that $e^{m_i Y_i} \rightarrow e^{tY}$. So since $G \subseteq GL_n(\mathbb{F})$ is a closed subset, $e^{tY} \in G$ for all t , so $Y \in \mathfrak{g}$, a contradiction since $Y \neq 0 \in W$, thus proving the claim. $\square$

With this, we prove the theorem, with $U = \mathfrak{g} \cap Z$. By the first claim (4.5.25), $\widetilde{\exp}|_Z: Z \rightarrow \text{Im}(\widetilde{\exp}|_Z)$ is a homeomorphism to a neighborhood of I in $GL_n(\mathbb{F})$. Restricting to $U := Z \cap \mathfrak{g}$, which is an open neighborhood of 0 in $\mathfrak{g}$, this implies that

$$\widetilde{\exp}|_U: U \rightarrow \text{Im}(\widetilde{\exp}|_U)$$

is a homeomorphism. The right hand side is, by the second claim, $G \cap \text{Im}(\widetilde{\exp}|_Z)$, which is $G \cap (\text{some open neighborhood of } I)$, so it's an open neighborhood of I in G . This prove the theorem. $\square$

Corollary 4.5.27

Let $G \leq GL_n(\mathbb{F})$ be a closed subgroup. Then for all A in G , there is an open neighborhood U of A in G which is homeomorphic to an open subset of $\mathbb{R}^d$, where d is the dimension of G .

Proof. When $A = I$, this follows from the theorem, since $\mathfrak{g}$ is homeomorphic to $\mathbb{R}^d$. When $A \neq I$, multiplication by A , $\cdot A: G \rightarrow G$, is a homeomorphism, and sends an open neighborhood of I (which is homeomorphic to an open subset of $\mathbb{R}^d$) to an open neighborhood of A in G . $\square$

Corollary 4.5.28

Let $G \leq GL_n(\mathbb{F})$ be a closed subgroup. Then G is a discrete topological space if and only if its dimension is 0 .

Proof. ($\Leftarrow$): If the dimension of G is 0 , every $A \in G$ has a neighborhood homeomorphic to $\mathbb{R}^0$, which is just a point. This means that all points in G are open, so G has the discrete topology.

($\Rightarrow$): If G has the discrete topology, there's a neighborhood U of I in G , homeomorphic to an open neighborhood of $\mathbb{R}^d$, where d is the dimension of G . $U \subseteq G$ is open, and G is discrete so U is too, so $d = 0$. $\square$

Example 4.5.29

For any closed subgroup $G \leq GL_n(\mathbb{F})$, making the open neighborhood in the theorem smaller if necessary, there's an open neighborhood U of I in G , which is homeomorphic to a ball in $\mathbb{R}^d$. In particular, U is path-connected.

Example 4.5.30

Consider the topological group $(\mathbb{Q}, +)$. No open neighborhood of 0 in $\mathbb{Q}$ is path-connected. So $(\mathbb{Q}, +)$ isn't isomorphic to a closed subgroup of any $GL_n(\mathbb{F})$, i.e. $(\mathbb{Q}, +)$ isn't a matrix Lie group.

Example 4.5.31

Define

$$G := U(1) \times U(1) = \left\{ \begin{pmatrix} z & 0 \\ 0 & w \end{pmatrix} \mid z, w \in \mathbb{C}, |z| = |w| = 1 \right\}.$$

Let $\gamma: \mathbb{R} \rightarrow G$ be the continuous homomorphism sending $t \in \mathbb{R}$ to the matrix

$$\gamma(t) := \begin{pmatrix} e^{it} & 0 \\ 0 & e^{i\sqrt{2}t} \end{pmatrix}.$$

You could replace $\sqrt{2}$ with your favorite irrational number if you'd like. Then γ is injective, but no open neighborhood of $I \in \text{Im}(\gamma)$ is path-connected. So $\text{Im}(\gamma) \leq G$ is a subgroup but it's not closed. Compare this to the earlier parts of the course: images of (group/ring/module) homomorphisms are sub(groups/rings/modules), but this doesn't hold for matrix Lie groups.

Example 4.5.32

Let's compute the Lie algebra of $O(n)$, $\mathfrak{o}(n) = T_I O(n)$. First, let $A = \mathfrak{o}(n)$. Then $tA \in \mathfrak{o}(n)$ for all $t \in \mathbb{R}$ too. So $e^{tA} \in O(n)$ for all t too. This means that

$$e^{tA} \cdot (e^{tA})^T = I = e^{tA} \cdot e^{tA^T}.$$

So since the inverse of e^{tA^T} is e^{-tA^T} , we have that $e^{tA} = e^{-tA^T}$. Since $\exp$ isn't injective, we can't conclude yet that $tA = -tA^T$. For $t > 0$ small enough, tA and $-tA^T$ both lie in an open neighborhood U of 0 in $M_n(\mathbb{F})$ on which $\exp$ is in fact injective. So for $t > 0$ small enough, we do have that $tA = -tA^T$. Dividing by t , we see that $A = -A^T$, meaning A is skew-symmetric. Conversely, suppose A is skew-symmetric; we'll show that $A \in \mathfrak{o}(n)$. A and $A^T = -A$ commute, so for all $t \in \mathbb{R}$, we have

$$e^{tA} \cdot (e^{tA})^T = e^{tA} \cdot e^{-tA} = e^0 = I.$$

Thus, set $e^{tA} \in O(n)$. So $\gamma_A(t) = e^{tA}: \mathbb{R} \rightarrow O(n)$ is a continuous homomorphism, so $A \in \mathfrak{o}(n)$. Altogether, what we've shown is that $\mathfrak{o}(n) \leq M_n(\mathbb{R})$ is the set of skew-symmetric matrices.

Definition 4.5.33. Let $G \leq GL_n(\mathbb{F})$ and $H \leq GL_n(\mathbb{F})$ be closed subgroups, and let $\phi: G \rightarrow H$ be a continuous homomorphism. We define a function $\phi_*: \mathfrak{g} \rightarrow \mathfrak{h}$ as follows.

For $A \in \mathfrak{g}$, let $\gamma_A(t) = e^{tA}$. So $\gamma_A: \mathbb{R} \rightarrow G$ is a continuous homomorphism. Now $\phi \circ \gamma: \mathbb{R} \rightarrow H$ is a composition of two continuous homomorphisms, so it's a continuous homomorphism. So $\phi(\gamma(t)) = e^{tB}$ for some unique $B \in \mathfrak{h}$. We define

$$\phi_*(A) := B.$$

Another equivalent way of saying this is that $\phi_*(A)$ is defined by the equation

$$\gamma_{\phi_*(A)} = \phi \circ \gamma_A.$$

Plugging 1 into this equation, using $\gamma_{\phi_*(A)}(1) = e^{\phi_*(A)}$ and $\phi \circ \gamma_A(1) = \phi(e^A)$ implies that the two maps $\exp \circ \phi_*$ and $\phi \circ \exp$, from $\mathfrak{g}$ to H , are the same:

$$\exp \circ \phi_* = \phi \circ \exp: \mathfrak{g} \rightarrow H.$$

Lemma 4.5.34

ϕ_* is a linear map $\phi_*: \mathfrak{g} \rightarrow \mathfrak{h}$ of $\mathbb{R}$ -vector spaces.

Proof. Let $A, B \in \mathfrak{g}$, and $\lambda \in \mathbb{R}$. We first show that $\phi_*(\lambda A) = \lambda \phi_*(A)$. Recall that for any $t \in \mathbb{R}$,

$$e^{t\phi_*(\lambda A)} = \phi(e^{t\lambda A}) = e^{(t\lambda)\phi_*(A)} = e^{t(\lambda\phi_*(A))}.$$

Since this is true for all $t \in \mathbb{R}$, this implies that $\phi_*(\lambda A) = \lambda \phi_*(A)$. We now show that $\phi_*(A + B) = \phi_*(A) + \phi_*(B)$. We know that for all $t \in \mathbb{R}$,

$$\gamma_{A+B}(t) = e^{t(A+B)} = \lim_{n \rightarrow \infty} \gamma \left(\gamma_A \left(\frac{t}{n} \right) \gamma_B \left(\frac{t}{n} \right) \right)^n.$$

Here we're using the earlier lemma which said that if $\delta: \mathbb{R} \rightarrow G$ sends $0 \mapsto I$ and is continuously differentiable, then $e^{t\dot{\delta}(0)} = \lim_{n \rightarrow \infty} \delta \left(\frac{t}{n} \right)^n$, as well as the fact that if $\delta(t) = \gamma_A(t) \cdot \gamma_B(t)$, then $\dot{\delta}(0) = A + B$; this is what we used when we proved that $A + B \in \mathfrak{g}$. The same equation applied to $\phi_*(A)$ and $\phi_*(B)$ says that

$$\begin{aligned} \gamma_{\phi_*(A)+\phi_*(B)} &= \lim_{n \rightarrow \infty} \gamma \left(\gamma_{\phi_*(A)} \left(\frac{t}{n} \right) \gamma_{\phi_*(B)} \left(\frac{t}{n} \right) \right)^n \\ &= \lim_{n \rightarrow \infty} \gamma \left(\phi \left(\gamma_A \left(\frac{t}{n} \right) \right) \phi \left(\gamma_B \left(\frac{t}{n} \right) \right) \right)^n \\ &= \lim_{n \rightarrow \infty} \phi \left(\gamma \left(\gamma_A \left(\frac{t}{n} \right) \gamma_B \left(\frac{t}{n} \right) \right)^n \right) \\ &= \phi \left(\lim_{n \rightarrow \infty} \gamma \left(\gamma_A \left(\frac{t}{n} \right) \gamma_B \left(\frac{t}{n} \right) \right)^n \right) \quad \text{since } \phi \text{ is continuous.} \\ &= \phi(\gamma_{A+B}(t)) \\ &= \gamma_{\phi_*(A+B)}(t) \end{aligned}$$

□

Remark 4.5.35 — Note that if we view ϕ_* as a differential, this is obvious.

Lemma 4.5.36

ϕ_* is a linear map. Furthermore $\phi_*([M, N]) = [\phi_*(M), \phi_*(N)]$, for all M, N in $\mathfrak{g}$. In particular, ϕ_* is a homomorphism of Lie algebras.

Proof. We first show $\phi_*(tA) = t\phi_*(A)$ for $t \in \mathbb{R}$. Let $\lambda_t: \mathbb{R} \rightarrow \mathbb{R}$ be the homomorphism given by multiplication by t . Then $\gamma_{tA} = \gamma_A \circ \lambda_t$, and so

$$\gamma_{\phi_*(tA)} = \phi \circ \gamma_A \circ \lambda_t = \gamma_{\phi_*(A)} \circ \lambda_t = \gamma_{t\phi_*(A)}.$$

So $\phi_*(tA) = t\phi_*(A)$. We next show $\phi_*(A+B) = \phi_*(A) + \phi_*(B)$. We know that

$$\gamma_{A+B} = \lim_{n \rightarrow \infty} \left(\gamma_A \left(\frac{t}{n} \right) \gamma_B \left(\frac{t}{n} \right) \right)^n$$

and

$$\gamma_{\phi_*(A)\phi_*(B)}(t) = \lim_{n \rightarrow \infty} \left(\phi \left(\gamma_A \left(\frac{t}{n} \right) \right) \phi \left(\gamma_B \left(\frac{t}{n} \right) \right) \right)^n$$

which equals $\phi(\gamma_{A+B}(t))$ since we can take the ϕ outside of the brackets since it's a homomorphism, and outside the limit since it's continuous. Together this says that $\gamma_{\phi_*(A+B)} = \phi \circ \gamma_{A+B} = \gamma_{\phi_*(A)+\phi_*(B)}$.

We next show ϕ_* respects the Lie bracket. Let $A, B \in T_I G$. Then letting $\delta_s(t) = \gamma_A(s)\gamma_B(t)\gamma_A(-s) = e^{sA}e^{tB}e^{-sA}$, we have that $\dot{\delta}_s(0) = e^{sA}Be^{-sA}$, which, as s varies, defines a C^1 path in $\mathfrak{g}$ whose derivative at 0 is $[A, B]$. We define

$$\beta_s(t) = \gamma_{\phi_*(A)}(s)\gamma_{\phi_*(B)}(t)\gamma_{\phi_*(A)}(-s),$$

so the path $\dot{\beta}_s(0)$ is a C^1 path in $\mathfrak{h}$ whose derivative at 0 is $[\phi_*(A), \phi_*(B)]$. Then since $\phi \circ \delta_s = \beta_s$, $\phi_*(\dot{\delta}_s(0)) = \dot{\beta}_s(0)$ by definition of ϕ_* (not by the differentiability of ϕ). Therefore, taking derivatives at $s = 0$ implies the result:

$$\phi_*([A, B]) = [\phi_*(A), \phi_*(B)].$$

□

Lemma 4.5.37

Let $\phi: G \rightarrow H$ and $\psi: H \rightarrow K$ be continuous homomorphisms between closed subgroups of $GL_n(\mathbb{F})$. Then $\psi_* \circ \phi_* = (\psi \circ \phi)_*$.

Proof. Let A lie in $\mathfrak{g}$. Write γ_B for the homomorphism given by $t \mapsto e^{tB}$. Then $\phi_*(A)$ is determined by the equation $\gamma_{\phi_*A} = \phi \circ \gamma_A$, and $\psi_*(\phi_*(A))$ is determined by the equation $\psi_*(\phi_*(A)) = \psi \circ \gamma_{\phi_*(A)} = \psi \circ \phi \circ \gamma_A$. $(\psi \circ \phi)_*(A)$ is determined by the equation $\gamma_{(\psi \circ \phi)_*(A)} = \psi \circ \phi \circ \gamma_A$. Thus, $(\psi \circ \phi)_*(A) = \psi_*(\phi_*(A))$. □

Corollary 4.5.38

If $G, H \leq GL_n(\mathbb{F})$ are closed subgroups which are isomorphic as topological groups, then their Lie algebras $\mathfrak{g} = T_I G$ and $\mathfrak{h} = T_I H$ are isomorphic.

Proof. Let $f: G \rightarrow H$ and $g: H \rightarrow G$ with continuous homomorphisms which are inverse to each other. Then, f_* and g_* are inverse homomorphisms. □